Female Urology & Pelvic Reconstruction

Campbell-Walsh-Wein · Chapters 77, 101, 105, 108, 114, 116–119, 121, 122

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 7h 59min even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 118* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
101	Surgical Anatomy of the Female Pelvis	25 min
108	UI & Prolapse: Epidemiology & Pathophysiology	38 min
105	Evaluation of Women With UI & POP	26 min
114	Conservative Management of UI	57 min
116	Retropubic Suspension (Burch)	17 min
118	Slings: Autologous, Biologic, Synthetic, Mid-Urethral	83 min
117	Reconstructive Surgery for Pelvic Organ Prolapse	56 min
119	Mesh Complications & Their Repair	24 min
121	Urinary Tract Fistulae	59 min
122	Bladder & Female Urethral Diverticula	38 min
77	Female Sexual Function & Dysfunction	49 min
★	Putting it together: female pelvic medicine decisions	7 min
	Total (33480 words, at 70 words/min)	**7h 59min**

101 Surgical, Radiographic and Endoscopic Anatomy of the Female Pelvis

Reading time ≈ 25 min (slow pace)

1 · PERINEUM AND EXTERNAL GENITALIA

- **Vulva** = mons pubis, labia majora and minora, clitoris, vestibule. External urethral meatus lies **1 cm** anterior to the vaginal opening.
- **Skene's (paraurethral) glands**: either side of the meatus; embryological homologue of the prostate. Obstruction of periurethral glands → urethral diverticulum or cyst.
- **Bartholin's (greater vestibular) gland**: under the posterior end of each vestibular bulb — site of cyst/abscess.
- **Perineal membrane** (formerly urogenital diaphragm) spans the pubic arch; pierced by urethra and vagina.

	Deep perineal space	Superficial perineal space
Boundary	Floored by perineal membrane	Roofed by Colles fascia
Contents	Urethral sphincter, urethrovaginal sphincter, compressor urethrae, deep transverse perineal	Superficial perineal muscles, clitoris, vestibular bulbs, Bartholin's glands

- **Perineal body**: convergence of bulbospongiosus, external anal sphincter, superficial and deep transverse perineal muscles. Episiotomy is angled laterally to spare the anal sphincter.
- **Blood supply of labia majora is dual**: superficial external pudendal (femoral) above, posterior labial branches of internal pudendal (internal iliac) below — this is why the labial fibrofatty **Martius flap** is a reliable interposition flap.

2 · BONY PELVIS AND LIGAMENTS

- **Ischial spine** (≈ level S5): found in lithotomy by palpating sacrum → sacrospinous ligament laterally → spine. Key landmark for apical suspension sutures. Interspinous line divides anterior from posterior compartment.
- **Standing**, ASIS and symphysis are in one vertical plane, so the inlet faces forwards and abdominal pressure falls on bone, not on muscle/fascia.

Fixation point	Used for	Adjacent danger
Sacrospinous ligament (SSL): ischial spine → lower sacrum/coccyx	SSL fixation (apical prolapse)	Pudendal nerve/vessels near spine; sacral plexus exits just posterior
Cooper's ligament (medial superior pubic ramus)	Burch colposuspension for stress urinary incontinence (SUI)	Obturator vessels; aberrant obturator
Obturator foramen; canal superolateral	Transobturator tape (TOT) trocar	Obturator nerve/vessels → groin pain, bleeding
Anterior longitudinal ligament (ALL) over S1	Sacrocolpopexy mesh	Presacral veins, middle sacral vessels, left common iliac vein, right ureter
Uterosacral ligament	Uterosacral (apical) suspension	Ureter (lateral, closest at cervix); sacral nerve entrapment

- SSL + sacrotuberous ligament form the **greater sciatic foramen** (piriformis, sacral plexus, pudendal nerve leaving pelvis) and **lesser sciatic foramen** (pudendal nerve re-entering towards Alcock's canal; obturator internus).

3 · PELVIC FLOOR MUSCLES

- Pelvic diaphragm = **levator ani** + coccygeus. Levator ani innervated directly from **S3–S5 (mainly S4)**.

Muscle	Origin → insertion	Role
Puborectalis	Back of symphysis → loops behind anorectal junction	U-sling pulling anorectal junction towards pubis
Pubococcygeus	Posterior pubis, arcus tendineus → viscera, anococcygeal raphe	Slings vagina and urethra (pubovaginal fibres); raises urethral resistance
Iliococcygeus	Arcus tendineus levator ani, ischial spine → raphe, coccyx	Flat posterior part of levator plate
Coccygeus MCQ	Ischial spine → lower sacrum, coccyx	NOT part of levator ani; lies on the SSL

- **Levator hiatus** transmits urethra, vagina, rectum. **Levator plate** = midline fusion forming a horizontal shelf on which viscera rest.
- Resting tone supports viscera, resists rises in abdominal pressure and gives passive urinary and faecal continence. Muscle or nerve injury → lax hiatus → plate tilts → **prolapse**.

4 · ENDOPELVIC FASCIA AND VISCERAL LIGAMENTS

- Of three fascial strata, the **intermediate** (parametrium on uterus, paracolpium on vagina, pubocervical fascia) **provides most organ support**.
- **Pubourethral (pubovesical) ligaments**: pubis → bladder neck/urethra; homologue of puboprostatic; with anterior vaginal wall form a hammock under the mid-urethra; weakness → SUI.
- **Vesicopelvic fascia** → arcus tendineus; defect → lateral (paravaginal) cystocele.
- **Cardinal (Mackenrodt's)**: side wall → cervix/upper vagina; carries internal iliac branches and pelvic plexus — division at hysterectomy → bladder dysfunction.
- **Infundibulopelvic** (ovary → side wall, ovarian vessels; ureter just beneath) vs **ovarian ligament** (ovary → uterus) MCQ.

DeLancey levels of vaginal support †

Level	Segment	Support	Failure
I – suspension	Cervix, upper third	Uterosacral–cardinal complex to sacrum/side wall	Uterine or vault prolapse, enterocele
II – attachment	Middle third	Paravaginal attachment to arcus tendineus fasciae pelvis and levator fascia	Anterior (paravaginal or central) and posterior wall prolapse
III – fusion	Distal third	Fused to urethra, perineal body, levator ani, perineal membrane	Urethral hypermobility (SUI); distal rectocele, perineal descent

> **KEY POINT** Support ligaments are connective tissue condensed around vessels and nerves — so the best anchors (uterosacral, cardinal) are exactly where suturing injures the ureter, pelvic plexus and sacral nerves. → *Ch 108*

5 · URETHRA AND CONTINENCE ANATOMY

- Length **4.7–5.1 cm**; behind symphysis; suspended by posterior pubourethral ligaments; rests on anterior vaginal wall.
- Layers: infolded epithelium (transitional → squamous distally) → submucosa with **spongy vascular plexus** → longitudinal and circular smooth muscle → periurethral fascia.
- **Mucosa and submucosa generate most closure pressure and are oestrogen-dependent** (rationale for vaginal oestrogen in atrophy).

	Internal sphincter	External (striated) sphincter
Muscle	Smooth, continuation of detrusor	Striated; horseshoe-shaped at mid-urethra
Control	Involuntary (autonomic)	Voluntary (pudendal)
Site	Bladder neck, proximal urethra	Distal two-thirds

- Striated urogenital sphincter complex: sphincter urethrae, **compressor urethrae** (closes urethra against anterior vaginal wall), **urethrovaginal sphincter** (narrows hiatus).
- Bladder neck and urethral **hypermobility** is one cause of SUI. Autologous pubovaginal sling sits at the **bladder neck**; synthetic mid-urethral sling (MUS) at the **mid-urethra**.
- **Diverticulum**: infected or ruptured paraurethral gland breaching periurethral fascia.

6 · BLADDER AND URETER

Bladder surface	Relation	Injury consequence
Dome	Peritoneum	Intraperitoneal leak → formal repair (thin wall heals poorly with drainage; bowel may herniate)
Base	Anterior vaginal wall, lower uterus	Vesicovaginal or vesicouterine fistula

- **Pelvic ureter course**: crosses pelvic brim over common iliac bifurcation → lateral wall towards ischial spine → behind ovary → base of broad ligament → **under the uterine artery** ("water under the bridge") → turns medially → over anterior vaginal fornix → bladder.

ALGORITHM
Ureteric danger points: pelvic brim (infundibulopelvic ligament/ovarian vessels) → under uterine artery (hysterectomy) → beside cardinal/uterosacral ligament near cervix → anterior vaginal fornix (vaginal surgery) → near sacral promontory (right ureter, sacrocolpopexy)

7 · NERVES AT SURGICAL RISK

- **Pudendal nerve (S2–S4)**: leaves greater sciatic foramen → crosses SSL/coccygeus **close to the ischial spine** → lesser sciatic foramen → Alcock's canal. Branches: inferior rectal, perineal, dorsal nerve of clitoris. Supplies external sphincters, perineal muscles; carries urethral sensation.
- **Pudendal block** (pudendal neuralgia): inject at SSL just posterior to the ischial spine, transvaginally.
- To stay safe in SSL fixation, place sutures ≈ 2 cm medial to the ischial spine through the ligament, not around it †.

Nerve	Procedure / mechanism	Deficit
Common peroneal	Lithotomy: compression at fibular head	Foot drop
Sacral plexus (L4–S4)	SSL fixation (exits just posterior to SSL)	Posterior leg motor/sensory loss
Pudendal	SSL fixation near spine	Perineal pain, sensory loss, sphincter dysfunction
Sacral trunks S1–S4	Uterosacral suspension (deep sutures)	Buttock/posterior thigh pain
Ilioinguinal (L1), genitofemoral (L1–L2)	Hernia repair, caesarean, hysterectomy, lymphadenectomy	Mons/labial numbness or pain
Obturator (L2–L4) †	TOT trocar, obturator lymphadenectomy	Groin pain, adductor weakness
Posterior femoral cutaneous (S1–S3)	Gluteal pressure, pelvic tumour	Perineal sensory loss

KEY POINT Post-lithotomy foot drop is peripheral (fibular head), not pelvic. Post-op leg pain and swelling after long lithotomy → suspect **compartment syndrome**: surgical emergency († fasciotomy).

8 · VESSELS AND LYMPHATICS

Internal iliac	Anterior division	Posterior division
Visceral	Obliterated umbilical, superior vesical, uterine, vaginal, middle rectal	None
Parietal	Obturator, internal pudendal, inferior gluteal	Iliolumbar, lateral sacral, superior gluteal

- Common iliacs bifurcate where the **ureter crosses**. Branching is variable: obturator artery may arise from the inferior epigastric.
- **Internal iliac**: rich bilateral collaterals → ligation/embolisation for haemorrhage tolerated. **External iliac**: poor collaterals → inadvertent ligation threatens the limb (emergency).
- Ovarian veins: right → inferior vena cava (IVC); left → left renal vein.
- **Presacral venous plexus**: valveless, high pressure (worse in lithotomy); a 1–2 mm vein can bleed > **1 L/min**.

ALGORITHM
Sacrocolpopexy: expose ALL at promontory over L5/S1 disc → open peritoneum distally → suture into ALL over **S1** (strongest ALL; less discitis) → keep clear of left common iliac vein, middle sacral vessels, presacral plexus, right ureter (each ≈ 2.7 cm from mid-promontory)

Organ	First nodes
Distal ⅓ urethra, lower ⅓ vagina, vulva	Inguinal → pelvic
Proximal ⅔ urethra, upper ⅔ vagina	Iliac, obturator (pelvic)
Bladder, uterus	Perivisceral → pelvic → presacral → great vessels
Ovary MCQ	Along gonadal vessels → para-aortic (bypasses pelvis)

9 · ENDOSCOPIC ANATOMY

ALGORITHM
Distal urethra (glands, diverticular ostia, mesh, stones) → mid-urethral coaptation (external sphincter) → bladder neck coaptation → trigone and orifices (30° lens) → whole-mucosa survey

- Patulous orifice → reflux; ectopic orifice → duplicated system. In marked anterior/apical prolapse, a vaginal sponge stick reduces the prolapse to restore anatomy.
- Trigonal squamous metaplasia, cystitis cystica are benign but not reliably diagnosed by appearance → biopsy if any doubt.

10 · IMAGING

- **Translabial ultrasound**: real-time bladder-neck mobility on Valsalva, bladder wall thickness, prolapse; shows **mesh, MUS, bulking agents**. 3D adds axial view.
- **Endoanal ultrasound**: internal sphincter defect = bright break in dark muscle; external sphincter defect = dark lesion in bright muscle.
- **Magnetic resonance imaging (MRI)**: most comprehensive radiation-free modality. **T2-weighted MRI is the best test for urethral diverticulum** and to distinguish it from other vaginal cysts (Gartner, Skene, Bartholin, müllerian, epidermal inclusion).

- **MR defecography**: dynamic rest/strain with rectal contrast; prolapse graded by descent below the **pubococcygeal line** (inferior symphysis → last coccygeal joint). Use for defaecatory dysfunction or symptoms out of proportion to examination.

11 · MCQ ONE-LINERS

1. External urethral meatus lies 1 cm anterior to the vaginal opening.
2. Skene's glands are the female homologue of the prostate.
3. Compressor urethrae and urethrovaginal sphincter lie in the deep perineal space.
4. Coccygeus is not part of levator ani; it lies on the SSL.
5. Levator ani is supplied directly by S3–S5, mainly S4.
6. Martius flap works because of dual (femoral and internal iliac) labial supply.
7. Cooper's ligament: anchor for Burch colposuspension.
8. Obturator canal lies superolateral in the obturator foramen.
9. Pudendal nerve enters Alcock's canal via the lesser sciatic foramen.
0. Pudendal nerve is most vulnerable near the ischial spine during SSL fixation.
11. Sacral plexus exits immediately posterior to the SSL.
12. Post-lithotomy foot drop: common peroneal compression at fibular head.
13. Female urethra ≈ 5 cm; closure pressure mainly from oestrogen-dependent mucosa/submucosa.
14. Pubovaginal sling at bladder neck; synthetic MUS at mid-urethra.
15. Ureter passes under the uterine artery: "water under the bridge".
16. Sacrocolpopexy sutures go into ALL over S1.
17. Presacral vein of 1–2 mm can bleed more than 1 L/min.
18. Posterior division of internal iliac has no visceral branches.
19. Distal third urethra drains to inguinal nodes; proximal two-thirds to iliac/obturator.
20. T2-weighted MRI is the best test for urethral diverticulum.

108 Urinary Incontinence and Pelvic Prolapse: Epidemiology and Pathophysiology

Reading time ≈ 38 min (slow pace)

1 · DEFINITIONS AND TERMINOLOGY

- **Urinary incontinence (UI)**: complaint of involuntary loss of urine (International Continence Society (ICS) / International Urogynecological Association (IUGA) joint terminology, 2010).
- Lower urinary tract symptoms (LUTS) are **non-specific**: urgency occurs in overactive bladder (OAB), bladder outlet obstruction (BOO), pelvic organ prolapse (POP) and interstitial cystitis/painful bladder syndrome. So diagnosis triangulates **signs** (clinician-observed, e.g. leak on cough test), **symptoms** (patient report, questionnaires) and **observations** (bladder diary; urodynamic studies (UDS)).
- **Bother drives treatment**: a symptom without quality-of-life (QoL) impact may not need treatment.

Type	Complaint of loss…	Urodynamic correlate / trap
Stress UI (SUI)	On effort, cough, sneeze, raised abdominal pressure	urodynamic (UD) SUI: leak with rise in abdominal pressure **and no detrusor contraction**
Urgency UI (UUI)	Associated with urgency	**Detrusor overactivity (DO) not required**; DO incontinence (DOI) is a separate UD diagnosis
Mixed UI (MUI)	With urgency and with effort	Urge-, stress-predominant or equal
Overflow UI	Frequent small-volume loss with retention	Detrusor underactivity or obstruction; high post-void residual (PVR)
Functional UI †	Loss because of impaired mobility or cognition despite a normal lower urinary tract	Common in frail elderly and long-term care
Continuous UI	Day and night, few voids	Suspect fistula (vesicovaginal, ureterovaginal)
Nocturnal enuresis	During sleep	≠ nocturnal UUI (wakes, cannot reach toilet)

- **Urgency** = sudden compelling desire, difficult to defer (pathological) vs **urge** = normal desire to void. **Increased sensation**: early desire but voiding can be postponed.
- **OAB syndrome**: urgency ± UUI, usually with frequency and nocturia; **defined by urgency**. Frequency ≥ **8 voids/24 h** is above normal but polyuria must be excluded.

2 · EPIDEMIOLOGY OF FEMALE UI (HEADLINE)

- Under-reported: ask directly. Prevalence figures vary with definition, time frame, instrument and population — compare only within a study.
- Annual incidence (40–60 y) 1–10%; remission up to ≈12%/year, **lowest for MUI**.
- **Pregnancy**: commonest cause of incident UI in young women (≈ **40%**, mostly SUI). >30% still leak at 1 year; UI at 3 months usually persists long term. Pelvic floor muscle training (PFMT) during and after pregnancy reduces UI.
- **Hysterectomy** can cause or cure UI; vaginal/laparoscopic routes carry more UI than open.

3 · RISK FACTORS FOR FEMALE UI

Factor	Key point
Ageing	Strongest independent non-modifiable factor. Type shifts: SUI peaks 40–59 y, UUI/MUI dominate >60. UUI linked to impaired mobility; in nonagenarians UI is linked to frailty and death. Long-term care residence: >40% at admission; driven by impaired activities of daily living, dementia, stroke, Parkinson disease
Vaginal delivery	Causes later **SUI, not UUI**. Caesarean section (CS) protects only if **every** delivery is CS; number needed to treat (NNT) **13** to prevent one moderate–severe UI. Elective CS < CS after labour. Baby ≥4 kg: odds ratio (OR) ≈1.5. Forceps, episiotomy, labour length: unproven
Parity	First delivery OR ≈1.5; modest rise with each further delivery; strongest for SUI
Obesity	Body mass index (BMI) ≥30 kg/m² more than doubles risk (SUI, MUI most). Dose- and duration-dependent and **reversible** with weight loss. Little effect on SUI surgery success
Menopause / oestrogen	SUI incidence rises across the menopausal transition. **Oral (systemic) oestrogen increases UI** (incident UI nearly doubled in a placebo-controlled trial); vaginal oestrogen not linked to SUI and treats atrophy and recurrent urinary tract infection (UTI)
Race	White > Black/Asian for SUI (higher urethral closure pressure in Black women)
Smoking	Urgency, frequency, UUI (dose-dependent)
Caffeine	UUI, MUI, OAB — not SUI
Diabetes mellitus (DM)	Up to 2× prevalence; neuropathy, macroalbuminuria independent factors
Depression, multimorbidity	Bidirectional with UI; hypertension, heart failure, osteoarthritis additive

> **KEY POINT** Two families of risk: obstetric injury and loss of support → **stress** UI; ageing, neurological, metabolic and lifestyle factors → **urgency**. Hence a woman's dominant UI type moves from stress to urge with age.

Male UI (contrast)

- Much less common; **UUI > MUI > SUI**. Male SUI is almost always iatrogenic (radical prostatectomy (RP), radical pelvic surgery) or neurological. Remission up to **40%** (reversible urgency causes).
- Post-RP UI 8–60% depending on definition; better with bladder-neck, nerve and Retzius sparing and younger age. Transurethral resection of the prostate (TURP): ≈1%.

4 · PHYSIOLOGY OF CONTINENCE

Low-pressure storage

Pathway	Nerve	Role during filling
Parasympathetic	Pelvic	Suppressed → no detrusor contraction
Sympathetic	Hypogastric	β → detrusor relaxation; α → bladder-neck closure
Somatic	Pudendal (Onuf's nucleus)	Tonic external sphincter resistance
Central	Pontine micturition centre; suprapontine	Suprapontine centres tonically inhibit the pontine voiding reflex until voiding is appropriate

- Bladder factors: dome expands freely into the peritoneal cavity; umbrella cells flatten; **poor electrical coupling** of detrusor cells stops isolated stretch-induced contractions spreading; extracellular matrix (type I collagen, collagen:elastin ratio) maintains compliance — altered by inflammation, infection, radiation.

Outlet

- Three components: **urethral mucosa** (seal), **internal sphincter** (bladder neck/proximal urethra), **distal rhabdosphincter**.
- **Female**: continence zone is the **proximal and mid-urethra**; bladder neck often incompetent even in continent women. Circular striated muscle compresses the mid-urethra under tonic pudendal drive; compressor urethrae and urethrovaginal sphincter distally. Oestrogen-sensitive spongy submucosa provides up to **30%** of closure.
- **Support**: anterior vaginal wall = firm backing; pubourethral ligaments limit proximal movement; urethropelvic ligaments anchor to the arcus tendineus. Lax vaginal support with intact pubourethral ligaments creates **shear** in the continence zone.

	Male	Female
Bladder neck	Powerful (α-adrenergic, prostate)	Weak; often incompetent
Loss of one unit	Tolerated (TURP loses bladder neck; rhabdosphincter keeps continence)	Not tolerated: external sphincter deficiency → UI
Exposure	Fixed, protected	Mobile; vulnerable to external forces

> **KEY POINT** The female outlet relies on a **single** mid-urethral sphincter compressed against a mobile backing — so female continence depends on support as much as on muscle. Continence needs support, intrinsic urethral properties and muscle together.

5 · GENERAL PATHOPHYSIOLOGY: BLADDER VS OUTLET

Mechanism	Causes	Result
Loss of suprapontine inhibition	Stroke, multiple sclerosis, Parkinson disease; early DM	Neurogenic DO → UUI
Outlet obstruction	Women: after anti-incontinence surgery; men: BOO	DO → de novo UUI
Detrusor underactivity	Conus/cauda disease, peripheral neuropathy (DM, alcohol), radical hysterectomy, abdominoperineal resection, opioids incl. tramadol	Retention → **overflow UI**
Poor compliance	Pelvic radiotherapy	High detrusor leak point pressure → UI
Sphincter deficiency	Surgery (RP, urethral/anti-incontinence surgery), cord injury, α-blockers and muscle relaxants	Intrinsic sphincter deficiency (ISD) → SUI

- **DM** acts in two phases: early polyuria and UUI → later loss of sensation and contractility → poor emptying, UTI, overflow.
- Cervical/upper thoracic spinal cord injury → detrusor–sphincter dyssynergia with neurogenic DO.
- Multiple system atrophy: intrinsic sphincter destroyed; ≥20% UI after TURP — counsel.
- Post-RP UI: **ISD predominates (>70%)**; DO/compliance contribute in >30%.
- Women: each failed urethral/anti-incontinence operation adds scarring, periurethral fibrosis and ISD.
- **Labour**: ligament/fascial injury, ischaemia from compression, **pudendal denervation**, direct urogenital injury.

6 · PATHOPHYSIOLOGY OF FEMALE SUI

- Urethral support: endopelvic fascia, reinforced anteriorly by the pubourethral ligament and posteriorly by the vaginal wall.

Theory	Mechanism	Treatment implication
Pressure transmission (Enhorning)	Hypermobile proximal urethra descends out of the abdominal pressure zone; raised abdominal pressure reaches the bladder but not the urethra	Restore urethral position (retropubic suspension)
Urethral hanging	Proximal urethra hangs below a less mobile bladder neck and pulls it open (funnelling)	—
Hammock hypothesis (DeLancey)	Urethra is compressed against a firm backing — anterior vaginal wall and its lateral attachments to levator ani and arcus tendineus fasciae pelvis — plus active mid-urethral sphincter. Loss of backing → SUI	Restore backing; **repositioning not essential**
Integral theory (Petros–Ulmsten)	Mid-urethra at the pubourethral ligaments is the continence zone; pubococcygeus pulls vagina forward against them. Connective-tissue (collagen/elastin) laxity → SUI; premature stretch-receptor firing → urgency	Tension-free mid-urethral sling (MUS)
ISD (McGuire)	Failure of urethral closure itself, not of support	Sphincter-directed: pubovaginal sling, bulking

Hypermobility vs ISD

- Support does not explain everything: some women with marked hypermobility are dry; some with a fixed urethra leak severely (scarred or neurogenic urethra).
- ISD was described with the autologous fascial pubovaginal sling for severe urethral dysfunction, usually after previous surgery; classified as **type III SUI** (Blaivas) MCQ.
- **"Pipestem" urethra**: fixed, non-coapting, usually iatrogenic. Subtler ISD coexists with hypermobility and underlies most SUI (birth ischaemia, progressive pudendal damage).
- **Valsalva leak point pressure (VLPP)**: abdominal pressure causing leakage (not valid during a detrusor contraction or poor compliance). **VLPP < 60 cm H_2O = ISD**; historically directed to bulking or pubovaginal sling.

> KEY POINT **Why slings work in hypermobility but Burch fails in ISD**: almost all female SUI includes some ISD. Sphincter-directed procedures (pubovaginal sling, MUS, even bulking) therefore work in hypermobile women; Burch colposuspension, which only corrects hypermobility, is less effective with severe ISD and a fixed urethra.

7 · OAB PATHOPHYSIOLOGY (BRIEF)

- Neurogenic: loss of suprapontine inhibition or cord lesions → DO (see §5).
- Outlet obstruction (BOO; post-sling obstruction) → DO and de novo urgency.
- † Non-neurogenic hypotheses: **myogenic** (altered detrusor excitability and cell-to-cell coupling allows local contractions to spread), **urotheliogenic** (urothelial mediator release and afferent sensitisation), **neurogenic/afferent** (increased afferent signalling).
- With POP: frequency tracks anterior/apical POP; UUI tracks severity of anterior POP. Mechanisms: BOO from kinking (most important), chronic stretch → DO, cystocele pulling the bladder neck open so urine enters the proximal urethra. **OAB improves after POP repair in 60–80%**, more with anterior/apical repair; no reliable predictor of persistence.

8 · INSENSIBLE INCONTINENCE

- Usually still ISD or DOI. If standard treatment fails, look for rarer causes; UDS may miss them.

Cause	Pattern	Investigation
Urethral diverticulum	Post-void dribble	Pelvic magnetic resonance imaging (MRI)
Ectopic ureter	Continuous loss with normal voids	Upper-tract imaging

Cause	Pattern	Investigation
Overflow	Frequent small-volume loss	UDS / PVR

9 · PELVIC ORGAN PROLAPSE: DEFINITION AND STAGING

- **POP**: descent of uterus/vault and/or vaginal walls with adjacent organs, from loss of support of one or more compartments. **Name the vaginal segment** (anterior, apical, posterior), not the presumed organ (cystocele, rectocele, enterocele).
- Enterocele: peritoneal sac with bowel in the rectovaginal space.
- Most specific symptom: **visible or palpable vaginal bulge**; correlates best once the leading edge passes the hymen.
- Constipation without bulge rarely justifies posterior repair (exception: stool trapped "in a pocket" needing splinting).
- Never attribute postmenopausal bleeding or pain to POP — evaluate first. A **pessary trial** is diagnostic: if symptoms improve, surgery is likely to help.
- Ask whether sexually inactive women wish to resume — influences choice of repair.

POP-Q stage	Most distal point on maximal strain
0	Aa, Ba, Ap, Bp = −3; C/D ≤ −(tvl − 2)
1	> 1 cm above hymen (< −1)
2	Within 1 cm of hymen (−1 to +1)
3	> +1 but < +(tvl − 2)
4	Complete eversion: ≥ +(tvl − 2)

- **Pelvic Organ Prolapse Quantification (POP-Q)**: hymen = 0; negative inside, positive beyond; cm. Points: **Aa/Ap** (3 cm from hymen, range −3 to +3; Aa ≈ urethrovesical junction), **Ba/Bp** (most distal wall point), **C** (cervix/cuff), **D** (posterior fornix; omitted after hysterectomy); **gh** (genital hiatus), **pb** (perineal body), **tvl** (total vaginal length) at rest. 3 × 3 grid.
- Traps: stage set by the single most distal point; Ba/Bp = C in total eversion; measuring straight after pessary removal underestimates.
- Baden–Walker halfway system: grades 0–4 relative to hymen; poor interobserver reliability. POP-Q is the recommended (IUGA/ICS) system.
- Imaging rarely needed. Dynamic MRI/defecography for defaecatory dysfunction, enterocele or symptom–examination mismatch.

10 · POP EPIDEMIOLOGY AND RISK FACTORS

- Some loss of support in 43–90% on examination; **asymptomatic stage 1–2 is normal** and rarely needs treatment. Symptomatic POP 3–12%.
- Compartment frequency: **anterior > posterior > apical**, but apical descent is under-diagnosed — it accompanies most significant anterior POP and is universal at stage 4. **Planning a cystocele repair → assess apex and consider vault suspension.**
- Natural history: slow progression, regression common in early stages. Progression predictors: age >60, Ba ≥4 cm, gh ≥5 cm.

Risk factor	POP	UI (contrast)
Vaginal childbirth	Strongest; relative risk rises steeply with parity; forceps worse. Elective CS ≈ nulliparous; CS after labour ≈ vaginal	SUI, not UUI
Age	≈40% rise per decade; surgery peaks 7th decade	Shift to UUI/MUI
Obesity	≈1.4–2.5-fold	>2-fold, reversible

Risk factor	POP	UI (contrast)
Hysterectomy	Raises risk, most if done for POP (omitted vault suspension)	Cause or cure
Race	Hispanic highest, Black lowest	White highest
Oestrogen / menopause	Not consistently associated	Oral oestrogen ↑ UI
Other	Family history (≈5× in siblings), connective-tissue disease, chronic constipation (2.5×)	—

- Recurrence after surgery predicted by preoperative stage 3–4.

11 · PATHOPHYSIOLOGY OF POP

- Support = endopelvic fascia + perineal membrane + **levator ani** (pubococcygeus, puborectalis, iliococcygeus) + bony attachments. Levator is the primary support; uterosacral–cardinal ligaments hold organs in position over it.

DeLancey level	Segment	Attachment	Loss produces
I – suspension	Cervix, upper third	Uterosacral–cardinal complex (parametrium, paracolpium) to sacrum and side wall	Uterine or vault prolapse; enterocele
II – attachment	Middle third	Paravaginal attachment to arcus tendineus fasciae pelvis (ATFP) and levator fascia	Anterior wall POP: **lateral (paravaginal)** or **central** defect
III – fusion	Distal third	Urethra anteriorly, perineal body posteriorly, levator laterally; perineal membrane	Anterior: **urethral hypermobility**; posterior: distal rectocele, perineal descent

- ATFP: pubis (1 cm from midline) → just above ischial spine. Commonest cystocele defect at surgery: ATFP detachment from the ischial spine.
- Defects are rarely isolated; level I loss correlates strongly with anterior POP — the apex is the keystone.

ALGORITHM
POP sequence: levator trauma/denervation (labour) → loss of tone → widened urogenital hiatus → fascia/ligaments now carry the whole load → stretch and fail over time (ageing, straining, hysterectomy) → POP → hiatus widens further

- Levator defects on MRI in up to **20%** of primiparas (none in nulliparas). **Levator avulsion** → advanced and multicompartment POP.
- Levator electromyographic neuropathy in ≈ a quarter of primiparas after vaginal delivery or CS in labour; almost none after elective CS — **labour itself is the injury**.
- Connective tissue: altered collagen (type I:III ratio); Ehlers–Danlos and Marfan increase risk.

12 · POP AND UI INTERACTION

- 40–55% of women with SUI have significant anterior POP; ≈20% of OAB.
- **Occult SUI**: SUI seen only when the prolapse is reduced (up to a quarter of advanced POP).

ALGORITHM
Cystocele → urethral kink → obstruction masks SUI → POP repair relieves kink → SUI unmasked. Detect by UDS / stress test **with prolapse reduced** (pessary or speculum).

- De novo SUI after POP surgery more likely with age >66, DM, low urethral closure pressure.
- Missing symptomatic POP at sling surgery raises later prolapse surgery — examine for POP in every UI evaluation.

13 · LANDMARK TRIALS

Trial	Design	Result	Take-home
CARE (Colpopexy and Urinary Reduction Efforts)	Stress-continent women, abdominal sacrocolpopexy ± Burch	Burch reduced postoperative SUI; SUI on reduction UDS predicted it	Prophylactic Burch at sacrocolpopexy prevents de novo SUI
OPUS (Outcomes following vaginal Prolapse repair and midUrethral Sling)	Stress-continent women, vaginal POP repair ± prophylactic MUS vs sham	UI at 1 year 27% vs 43% (NNT ≈6); more bladder perforation, UTI, bleeding, retention	Counsel: fewer de novo SUI, more adverse events

14 · MCQ ONE-LINERS

1. UUI does not require DO on urodynamics.
2. UD SUI requires leakage without a detrusor contraction.
3. OAB is defined by urgency.
4. Pregnancy is the commonest cause of incident UI in young women.
5. Vaginal delivery causes SUI, not UUI.
6. CS benefit is lost after even one vaginal delivery; NNT 13.
7. BMI ≥30 more than doubles UI risk; weight loss reverses it.
8. Systemic oestrogen increases UI; vaginal oestrogen does not.
9. Male UI: UUI > MUI > SUI; remission up to 40%.
10. Post-RP incontinence is mainly ISD (>70%).
11. Female continence zone: proximal and mid-urethra.
12. Submucosal vascular plexus contributes up to 30% of urethral closure.
13. Hammock hypothesis (DeLancey): repositioning the urethra is not essential.
14. Integral theory (Petros–Ulmsten): pubourethral ligament laxity; basis of tension-free MUS.
15. ISD = type III SUI; VLPP <60 cm H_2O.
16. Burch is less effective in ISD with a fixed urethra.
17. DeLancey level I loss → apical/vault prolapse.
18. DeLancey level III anterior loss → urethral hypermobility.
19. Commonest paravaginal defect: ATFP detachment from the ischial spine.
20. Describe POP by vaginal segment, not organ.
21. POP-Q reference point: hymen = 0.
22. Aa and Ap cannot exceed +3.
23. POP-Q stage 2: leading edge within 1 cm of hymen.
24. Strongest POP risk factor: vaginal childbirth (parity).
25. Elective CS ≈ nulliparous risk; CS after labour ≈ vaginal delivery.
26. Occult SUI: unmasked only when prolapse is reduced.
27. OAB improves after POP repair in 60–80%.

105 Evaluation and Management of Women With Urinary Incontinence and Pelvic Organ Prolapse

Reading time ≈ 26 min (slow pace)

1 · DEFINITIONS AND RISK FACTORS

- **Pelvic floor disorders** (urinary incontinence [UI], faecal incontinence, pelvic organ prolapse [POP], defecatory dysfunction) affect ~1 in 4 women; UI commonest.
- **UI** (International Continence Society, ICS): complaint of involuntary loss of urine. Risk factors: age, obesity, smoking, comorbidity, postmenopausal hormone therapy; vaginal delivery → stress UI (SUI).
- **POP** (ICS): descent of anterior wall, posterior wall, uterus/cervix or vault (cuff). Commonest: **anterior > posterior > apex**. About half of parous women have anatomical POP, but disease = **symptoms attributable to it**. **Vaginal bulge** is the most specific symptom and the one most likely to resolve after repair.

Category	POP risk factors
Primary only	Older age, high body mass index (BMI), high birth weight, levator defects, low education, poverty
Recurrence only	**Younger age**, preoperative POP-Q stage 3–4
Both	Vaginal delivery, parity, levator hiatal ballooning
Protective (primary)	Caesarean delivery; **smoking** MCQ

- **ICS UI terms** (all symptoms, defined by trigger): **SUI** — effort, sneeze, cough; **UUI** — with urgency; **MUI** — both; **nocturnal enuresis** — involuntary *voiding* in sleep; **disability-associated** — aware but cannot reach toilet; **insensible** — unaware how/when; **occult SUI** — only after POP reduction; also coital, continuous, postural.

2 · HISTORY

- Women rarely volunteer symptoms → ask. Exclude reversible causes — **DIAPPERS**: Delirium, Infection, Atrophic vaginitis/urethritis, Psychological, Pharmacological, Excess urine output, Restricted mobility, Stool impaction.
- **SUI = urethral failure; UUI = bladder failure; MUI = both**. Leakage with cough/laugh/activity → SUI; preceded by urgency or "on the way" → UUI. In MUI, the more **bothersome** component is treated first.
- A negative SUI history does not exclude SUI → examination, diary or urodynamic studies (UDS) if type unclear.
- Record type, severity, pad use, trajectory, prior treatment, goals, voiding symptoms. Lower urinary tract symptoms (LUTS) may precede a neurological diagnosis → ask about numbness, tremor, balance, vision.
- **POP**: bulge, pressure; worse later in day. Symptoms begin when the prolapse reaches the hymen. **Splinting** to void/defecate suggests high-grade POP; high-grade anterior POP kinks the urethra → voiding difficulty. Posterior POP: obstructed defecation. POP is associated with overactive bladder (OAB); repair may not cure it.
- SUI coexists in ~**40%** with POP. Patients often expect prolapse repair to cure leakage → counsel.
- Past history: neurological disease, diabetes, dementia, pelvic radiation, rectal/pelvic surgery (pelvic nerve injury), prior UI/POP surgery or mesh; obstetric (prolonged second stage, forceps/vacuum, macrosomia); **local** oestrogen may improve UI, **systemic** may worsen it. Family history of POP raises risk and recurrence.

Occult SUI (ICS-defined; ~⅓ before vaginal POP repair) vs **de novo SUI** (after POP surgery, no preoperative occult SUI; not ICS-defined) → shared decision on concurrent vs staged anti-incontinence surgery.

- **Drugs** act via outlet, detrusor, urine volume or cognition: α-agonists → retention; **α-blockers → worsen SUI in women**; anticholinergics, calcium-channel blockers, opioids → poor emptying; cholinesterase inhibitors → UUI; diuretics, SGLT2 inhibitors, lithium → polyuria; glitazones → oedema → nocturia; sedatives → confusion; **ACE inhibitors → cough → SUI**.

3 · PHYSICAL EXAMINATION

- General: mobility, cognition; frailty assessment preoperatively at ≥65 y (e.g., Clinical Frailty Scale).
- Genitalia: **oestrogen status** — atrophic = pale, flat, dry, no rugae; well-oestrogenised = pink, rugated. Note caruncle, urethral mass.

SUI testing

- American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (AUA/SUFU) 2023: **objectively demonstrate SUI with a comfortably full bladder before surgery**.
- **Cough stress test (CST)**: patient coughs; leakage synchronous with cough = positive. Supine first; **stand if supine negative**. CST at half functional capacity (largest diary void) has high sensitivity and specificity; at comfortably full bladder similar sensitivity, lower specificity.
- With POP, **repeat with prolapse reduced** (anterior wall supported): a large prolapse kinks the urethra and masks SUI (occult SUI).
- **Urethral hypermobility**: **Q-tip test** — lubricated swab to the urethrovesical junction, Valsalva, angle change from horizontal **>30°** = hypermobility. Uncomfortable; now mostly teaching/trials. Vaginal swab test equivalent; **visual urethral mobility examination** (meatal rotation < or >30° on cough) is routine practice.

POP examination

- Examine lithotomy **and standing** (foot on a step), straining, to reproduce the worst prolapse.
- **Split Graves speculum**: retract posterior wall → see anterior compartment; rotate → see posterior; both blades → isolate apex; withdraw posterior blade while straining. Blade expelled with posterior wall foreshortening = **loss of level I (cardinal–uterosacral) support → vault prolapse**; blade stays = isolated rectocele/enterocele.
- Rectocele (posterior wall into vagina) ≠ rectal prolapse (mucosa out through anus). Enterocele = bowel at the vaginal apex.

Pelvic Organ Prolapse Quantification (POP-Q) system

Objective, in centimetres, relative to the **hymen (= 0)**; above negative, beyond positive. Six points on straining + three lengths.

Point	Definition	Range
Aa	Anterior wall, 3 cm proximal to hymen (fixed)	−3 to +3
Ba	Most distal point of upper anterior wall (mobile)	−3 to +tvl
C	Cervix or vaginal cuff	—
D	Posterior fornix (absent after hysterectomy)	—
Ap	Posterior wall, 3 cm proximal to hymen	−3 to +3
Bp	Most distal point of upper posterior wall	−3 to +tvl
gh	Genital hiatus: mid urethral meatus → posterior hymen	—
pb	Perineal body: posterior gh → mid anus	—
tvl	Total vaginal length, prolapse reduced	—

No prolapse: Aa, Ba, Ap, Bp all at −3. Stage is set by the single most distal point (leading edge).

Stage	POP-Q criteria	Baden–Walker grade (by eye)
0	Aa, Ba, Ap, Bp −3; C or D ≤ −(tvl − 2)	0 — normal
I	Leading edge < −1 cm	1 — upper vagina to midplane
II	Leading edge −1 to +1 cm	2 — midplane to hymen
III	> +1 cm but < +(tvl − 2)	3 — through/beyond hymen
IV	≥ +(tvl − 2) — complete eversion	4 — complete eversion

KEY POINT Stage II is the 2-cm window around the hymen. The III/IV boundary depends on tvl, not a fixed number. Worked example: tvl 8, Bp +5 → between +1 and +6 → **stage III**.

Other elements

- **Neurological** if suspected. **Bulbocavernosus reflex** (S2–S4) present in only ~**70%** of normal women → absence alone not pathological. Anal tone (S2–S4).
- **Prior mesh**: International Urogynecological Association (IUGA)/ICS terms — *prominence* (no epithelial breach) vs *separation* (breach); *extrusion* (out through vaginal epithelium) vs *perforation* (into bladder/urethra/viscus).
- **Trauma-informed care** (patient controls timing, may decline speculum); always offer a **chaperone**.

4 · SUPPLEMENTAL EVALUATION

Universal tests: UA and PVR. AUA/SUFU 2023 index patient wanting SUI surgery: **urinalysis (UA), postvoid residual (PVR) and objective demonstration of SUI** are required; cystoscopy, UDS and imaging are not routine. In non-neurogenic OAB only UA is required (PVR optional).

Any departure from the uncomplicated **index patient** earns further testing:

- Diagnostic uncertainty; SUI not demonstrated.
- Neurogenic lower urinary tract dysfunction; urgency-predominant MUI.
- Unexplained haematuria/pyuria; raised PVR; significant voiding dysfunction.
- POP-Q stage ≥3 without demonstrable SUI on reduction.
- Prior failed anti-incontinence or POP surgery.

Symptom quantification

- **Bladder diary** (frequency–volume chart, 3 days): diagnostic and therapeutic (self-monitoring) → *Ch 114*. Largest void = functional capacity.
- **Questionnaires**: International Consultation on Incontinence Questionnaire (ICIQ) modular; ICIQ-UI Short Form correlates with pad tests. Pairs: **UDI-6** (Urogenital Distress Inventory, bother) / **IIQ-7** (Incontinence Impact Questionnaire, impact); **PFDI** (Pelvic Floor Distress Inventory) / **PFIQ** (Pelvic Floor Impact Questionnaire) for POP.
- **Pad tests** (mainly research). ICS 1-hour: drink 500 mL in <15 min, then standardised activities (walking, stairs, coughing, running, bending, hand-washing) → weigh.

	1-hour test	24-hour test
Normal (sweat/discharge)	<1.4 g	<4.4 g
Mild	1–10 g	4–20 g
Moderate	11–50 g	21–74 g
Severe	>50 g	≥75 g

Pad *counts* do not correlate with 24-hour pad weight.

Dye tests: oral phenazopyridine (orange) confirms fluid is urine; **double-dye tampon test** (oral phenazopyridine + intravesical methylene blue) — orange = ureterovaginal, blue = vesicovaginal

fistula.

UA and PVR

- UA: haematuria, pyuria, glucosuria → secondary cause. Do not treat asymptomatic bacteriuria.
- **PVR**: ultrasound bladder scan preferred to catheter. Single value unreliable → repeat. No agreed cut-off; asymptomatic women almost always <100 mL. A raised PVR alters counselling before SUI surgery or antimuscarinics (retention risk).

Urodynamics

- Not needed in the index SUI patient → *Ch 107*.
- † **VALUE trial** (Value of Urodynamic Evaluation): in demonstrable SUI, office evaluation alone was non-inferior to office evaluation + UDS for 1-year outcome after surgery.

Cystoscopy

- Not routine before UI/POP surgery. Indications: haematuria, urgency/irritative LUTS, recurrent urinary tract infection (UTI), prior sling/mesh (eroded mesh/suture), pelvic radiation.

Imaging

Modality	Main indication
Voiding cystourethrogram	Reflux, diverticula, cystocele
Renal ultrasound	Hydronephrosis: chronic retention, **severe POP (ureteric kinking)**, fistula, neurogenic bladder
Translabial ultrasound	**Modality of choice for mesh** (CT/magnetic resonance imaging (MRI) poor); levator defects
MRI / dynamic MRI	**Urethral diverticulum (standard of care)**; complex/recurrent POP, symptoms out of proportion to examination

- Urethral diverticulum classic **3 Ds**: dysuria, postvoid dribbling, dyspareunia (often absent) → *Ch 122*.
- MRI **HMO system**: H-line = hiatal width; M-line = levator plate descent; O = organ beyond H-line.

5 · MANAGEMENT PATHWAY

ALGORITHM

History + examination + UA + PVR → exclude DIAPPERS, UTI, obstruction, tumour → classify and grade bother → red flags (pain, haematuria, recurrent UTI, pelvic mass, prior radiation/pelvic surgery, prolapse beyond introitus, fistula) → specialist work-up; otherwise initial management → reassess → extra tests only if they change management → surgery.

Diagnosis	Initial management
SUI	Lifestyle, behavioural therapy incl. pelvic floor muscle training (PFMT) → *Ch 114*; † pessary; surgery if these fail
UUI / OAB	Lifestyle, behavioural therapy; antimuscarinic trial → *Ch 113*
Significant PVR	Treat constipation, review drugs; catheter drainage if PVR 200–500 mL, then reassess
Symptomatic POP	† Observation, PFMT, pessary, or surgery by bother and goals

6 · LANDMARK TRIALS

Trial	Design	Result	Take-home
† VALUE	RCT, uncomplicated demonstrable SUI: office evaluation ± UDS before surgery	Office evaluation alone non-inferior	Preoperative UDS not needed in the index SUI patient

7 · MCQ ONE-LINERS

1. Commonest POP compartment: anterior.
2. Nocturnal enuresis = voiding, not leakage.
3. In MUI, treat the most bothersome component first.
4. Most specific POP symptom, most likely to resolve after repair: vaginal bulge.
5. Smoking: protective for primary POP, risk factor for UI.
6. Younger age is a risk factor for POP recurrence.
7. Occult SUI: revealed only after reduction of prolapse.
8. α-blockers worsen SUI in women; ACE inhibitors cause SUI via cough.
9. Cholinesterase inhibitors for dementia can precipitate UUI.
0. Negative supine CST → repeat standing.
11. Q-tip hypermobility: >30° from horizontal on Valsalva.
12. POP-Q reference point: hymen (0).
13. POP-Q stage II: leading edge −1 to +1 cm.
14. POP-Q stage IV: leading edge ≥ +(tvl − 2) cm.
15. Posterior speculum blade expelled on straining → level I (apical) support loss.
16. Bulbocavernosus reflex absent in ~30% of normal women — not pathological alone.
17. Required for index SUI surgical patient: UA, PVR, objective SUI demonstration.
18. 1-hour pad test: 500 mL drink; >50 g = severe.
19. Best imaging for mesh: translabial ultrasound.
20. Standard imaging for urethral diverticulum: MRI.

114 Conservative Management of Urinary Incontinence: Behavioural and Pelvic Floor Therapy, Urethral and Pelvic Devices

Reading time ≈ 57 min (slow pace)

1 · DEFINITION AND PRINCIPLES

- **Conservative treatment** of urinary incontinence (UI) and lower urinary tract symptoms (LUTS) = any therapy that is neither surgery nor drugs.
- **Behavioural treatment** improves UI and LUTS by changing the patient's behaviour or environment, or by teaching new skills. It is the mainstay because it is effective and **essentially risk free** → first-line in all guidelines for stress UI (SUI), urgency UI (UUI), mixed UI (MUI), overactive bladder (OAB) and nocturia.
- Programmes are multicomponent and individualised: education, bladder diary, pelvic floor muscle training (PFMT), stress strategies ("the knack"), urge suppression, bladder training (BT), biofeedback (BF), electrical stimulation (E-stim), scheduled voiding, fluid/caffeine/weight management. Usually delivered by nurses and advanced practice providers (APPs) after simple non-invasive assessment.
- Outcome cannot be predicted, so it is offered to anyone willing and able; even patients with dementia benefit from caregiver-led programmes.

	Outlet-directed	**Bladder-directed**
Target	Urethral closure	Bladder function via voiding habits
Skills	Pelvic floor muscle (PFM) strength and control; urge suppression	Changed voiding habits plus urge control
Examples	PFMT; behavioural training with urge suppression	BT; delayed voiding

	Patient-dependent	**Caregiver-dependent**
Examples	BT, PFMT, behavioural training	Timed voiding, habit training, prompted voiding
Needs	Intact function, ability to learn, motivation	A knowledgeable, motivated caregiver; not the patient's cognition
Suits	Cognitively and functionally intact	Cognitively/functionally impaired, nursing home

> **KEY POINT** Why first-line if surgery cures SUI more often? Because conservative therapy is risk free and at least as effective as drugs. This asymmetry of harm drives every recommendation below.

2 · ASSESSMENT BEFORE BEHAVIOURAL TREATMENT

- **Focused history**: onset, duration, progression, triggers of leakage — these design the programme. Assessment targets *triggers and habits* rather than diagnosis.
- **Bladder diary / frequency–volume chart, 3–7 days**: fluid type and amount, voids and volumes, urgency, leakage episodes and circumstances, pads.
  - Best non-invasive tool to monitor habits and treatment effect; voided volumes estimate **functional bladder capacity**.
  - **Self-monitoring effect**: the patient sees how intake and activity relate to leakage → the diary is assessment tool, outcome measure and treatment at once.
- **Questionnaire-Based Voiding Diary (QVD)**: validated recall alternative (last month), completed at the first visit in 5–7 min.
- **Education** first: anatomy of bladder, urethra, PFMs; mechanism of the patient's own leakage; improvement is gradual and depends on daily practice.

3 · LIFESTYLE MODIFICATIONS

Self-care changes, usually adjunctive. Evidence weaker than for skill-based therapy, except weight loss.

Fluid management

	Excess intake	Inadequate intake
Effect	Urgency, frequency, UI; very high intake (~3.7 L/day) worsens frequency	Dehydration (elderly), urinary tract infection (UTI), constipation, concentrated irritant urine, reduced capacity
Reason	"Flushing the kidneys", dieting, habit	Self-restriction to control symptoms
In practice	Less common	**More common**
Action	Reduce if 24-h output >~**2100 mL**	Increase to target

- **Target**: six to eight 8-oz glasses, or **30 mL/kg/day**, minimum **1500 mL**, at set times. Institute of Medicine total water 2.7 L/day for women *includes food water*.
- **Timing**, not blanket restriction: restrict before outings and make up later; stop fluids **3–4 h before bed** for nocturia.
- Nocturia from daytime oedema mobilised at night → support stockings or leg elevation in late afternoon. Loop diuretic → give mid-to-late afternoon, or switch to non-loop.
- Tip: dilute urine smells less — a persuasive argument for drinking more.
- Evidence: small RCTs show reducing intake by ~¼ cuts frequency, urgency, nocturia; large cohort showed no link between total intake and incident UI. Goal is a **normal range, not restriction**.

Caffeine and dietary irritants

- Caffeine is a **diuretic and bladder irritant**: raises detrusor pressure on filling, lowers sensory threshold, risk factor for detrusor overactivity (DO).
- ~**250 mg/day** (≈2 cups of coffee) associated with moderate–severe UI in men. Coffee is not associated with nocturia.
- Reduction reduces SUI and UUI episodes, but only a subset is caffeine-sensitive and the amount needed is unknown.
- **Practical**: 3–5-day reduction trial; if better, taper over weeks by mixing with decaffeinated drinks to avoid withdrawal headache.
- Other suspected irritants: aspartame, citrus, spicy food, tomato, carbonated and artificially sweetened drinks (associated with OAB/UI onset). Use a food diary and eliminate one item at a time. Alcohol: heavy drinking → more LUTS; evidence weak.

Bowel function

- Constipation (<3 stools/week) and straining are linked to LUTS: shared space and innervation, bladder–bowel cross-sensitisation, and straining-induced **pudendal denervation** of PFMs.
- **Faecal impaction** (nursing homes) → OAB or obstruction with overflow.
- Management: bowel history; fibre **with adequate fluid** (fibre fails without it); exercise; regular evacuation after a meal (gastrocolic reflex).

Obesity and weight loss

- Obesity ~doubles UI risk. Mechanisms: chronic raised intra-abdominal pressure; metabolic syndrome → UUI; inflammation altering collagen.
- **Loss of 5–10% body weight** sustained for 12 months substantially reduces UI episodes (mainly SUI). Bariatric surgery resolves UI in morbid obesity.
- Behavioural weight loss in overweight women = the **only grade A lifestyle recommendation**.

Smoking and diet

- Smoking: no UI recommendation possible (grade D); offer cessation for general health (grade C in men with UI).
- Diet high in fruit, vegetables and whole grains helps prevent UI in postmenopausal women (B); vitamin D reduced UUI episodes in postmenopausal Black women (B).

4 · PELVIC FLOOR MUSCLE TRAINING (PFMT)

- Cornerstone for SUI, UUI, MUI, urgency and frequency. Popularised by **Arnold Kegel (1948)** MCQ — SUI attributed to poor PFM awareness; repetitive contractions reduced it.
- **Mechanism**: (1) contraction lifts the urethra and compresses it against the pubic symphysis, preventing descent; (2) training → striated muscle **hypertrophy** → greater external urethral pressure; (3) better bladder-neck support during rises in abdominal pressure. Plus a **skill** route — timed urethral occlusion (the knack).
- Three elements: learn to contract and relax; daily regimen for strength and control; active contraction during effort.

Assessment of PFM function

- Observation: correct contraction = **perineal elevation** (inward, ventrocephalad movement); incorrect = perineal descent (straining).
- **Digital vaginal examination** (commonest): 1–2 fingers 6–8 cm in; palpate levator ani just above the hymen at 4 and 8 o'clock for strength and tenderness; assess resting tone, symmetry, maximal hold and quick "flicks".
- Scales: **Modified Oxford** (most used); **Brink score** — pressure, duration, displacement (lift of examiner's finger).
- Anal route: puborectalis palpable 2.5–4 cm from the anal verge; gripping "pull-in" around the whole finger.
- Note co-contraction (abdominals, gluteals) and **overactivity** (hypertonicity, cannot relax; may contract during voiding) vs hypotonicity (weak contraction).

Teaching PFM control

- Verbal feedback during palpation, BF or E-stim. Instructions: "tighten as if stopping gas, pull up and in, without tensing buttocks, legs or stomach"; men — "move the penis up and down without moving anything else". If the patient cannot find the muscle, use tactile cues (tapping, quick stretch).
- Commonest errors: **co-contraction** of rectus abdominis/gluteals and **Valsalva substitution** — straining worsens the condition being treated. Avoid breath-holding: count aloud.
- **Verify correct contraction before prescribing exercises** — failure to isolate the muscle is probably the commonest cause of poor outcome. A pamphlet or "stop the stream" instruction does not ensure this.

Exercise regimen

No ideal regimen; good results with **45–60** contraction–relaxation pairs/day. Sample prescription:

ALGORITHM
Empty bladder → **lying**: 5 quick (1–2 s) + 5 sustained (5 s building to 10 s) → **sitting**: same → **standing**: same (= 30 contractions) → rest as long as each contraction (work:rest **1:1 to 1:2**) → twice daily = **60/day**.

- Maximal effort per contraction — intensity drives strength gain; spread sessions to avoid fatigue; progress reclining → upright so the skill works in daily positions.

Stress strategy — "the knack"

- Volitional contraction **just before and during** cough, sneeze, lift ("squeeze before you sneeze"). Immediately reduces cough leakage; becomes automatic with practice.

- If leakage occurs, squeeze anyway — this links the squeeze to the trigger.

Evidence

- PFMT beats no treatment/placebo for SUI, UUI and MUI (cure, quality of life [QoL], leakage), immediately and long term; pure SUI does best.
- **Supervised > unsupervised**; offer the most intensive professional-led programme possible (grade A). Effect enhanced by specificity, progressive overload, confirmed contraction, the knack.
- **Age is no deterrent** — older women do as well → first-line at all ages.
- Pregnancy: structured antenatal PFMT in continent women prevents UI late in pregnancy and postpartum; supervised postpartum PFMT for UI persisting ~3 months after delivery.
- **Men after radical prostatectomy (RP)**: post-prostatectomy incontinence is almost universal early. PFMT (taught by digital rectal examination, BF or E-stim) hastens continence in the first 3 months; 8 weeks of behavioural training helped men leaking 1–17 years after RP, sustained to 12 months. Adding E-stim or BF does not improve continence, though BF is a good teaching tool.
- Post-micturition dribble: strong PFM contraction after voiding or urethral milking.
- **Neurological OAB** (multiple sclerosis, incomplete spinal cord injury, Parkinson disease, stroke): PFMT alone or combined is safe and first-line (B); transcutaneous electrical nerve stimulation is a first-line option after stroke.
- PFMT combined with drugs: adds benefit to tolterodine for UUI; Pilates/yoga are not substitutes for PFMT after prostatectomy.

Vaginal cones and other devices

- **Weighted vaginal cones**: sensation of slipping prompts a reflex PFM contraction → progressive overload. Protocol: heaviest cone retained standing → walking → increase weight; goal 2 × 20 min/day.
- Limitations: discomfort, narrow introitus; **cannot be retained with pelvic organ prolapse (POP) or wide introitus**; the vagina is not vertical, so some retain the cone without contracting. Better than no treatment; not superior to PFMT; adding cones to PFMT adds nothing.
- **Colpexin sphere**: sized polycarbonate sphere placed above the levators that elevates mild prolapse to ease contraction; no added strength over exercise alone.
- Home devices: pressure perineometers rise with both a lift and a Valsalva; **motion-sensing (accelerometer) inserts** distinguish a correct lift from straining — relevant since Valsalva substitution is the commonest error. An app-guided motion device was superior to a written/video home programme, sustained at 12 months.

5 · BEHAVIOURAL TRAINING WITH URGE SUPPRESSION

- Basis: a well-timed PFM contraction **reflexly inhibits detrusor contraction** and suppresses urgency. Originally learnt with bladder–sphincter BF; equally teachable with verbal feedback during palpation.
- Rushing to the toilet raises bladder pressure, and toilet cues act as a conditioned trigger → leakage.

ALGORITHM
Urge → do not rush, stay away from toilet → stand still or sit, relax → several quick PFM squeezes → concentrate on inhibiting, distract → urge subsides (it is a wave) → walk normally to void → once mastered, wait 5 min, then extend gradually.

- Stand-alone efficacy for UUI: mean **60–80%** reduction in incontinence; also reduces urgency, frequency, nocturia in both sexes.
- **Nocturia protocol**: on waking with urge, suppress; if it passes, sleep; if not within 1–2 min, void.
- Equivalent to or better than antimuscarinics; adding it to drugs improves outcomes (see trials).
- Men: behavioural training lowered frequency more than antimuscarinic + α-blocker; in men already on an α-blocker, adding behavioural therapy reduced nocturia more than adding an

antimuscarinic.

6 · BIOFEEDBACK (BF)

- Not a treatment itself but a **teaching technique**: instantaneous visual/auditory display of PFM activity (manometry or surface electromyography [EMG]); learning by operant conditioning. Kegel's perineometer was the first vaginal pressure BF device.
- EMG BF permits multichannel display — PFM contraction alongside relaxed abdominals/gluteals — and functional positions; pressure BF cannot distinguish lift from strain.
- On EMG: fast-twitch = ~2-s spikes; slow-twitch = ~10-s plateaus; return to baseline between.
- **Evidence**: adding BF to supervised PFMT gives **no clear extra benefit**; apparent benefit in older trials reflected extra clinician contact.
- **Use it for**: very weak PFMs, atrophy, reduced proprioception (cannot find the muscle), uncertainty of correct contraction; motivation and tracking progress.

7 · ELECTRICAL STIMULATION (E-STIM)

- Non-implanted low-grade stimulation of PFMs (vaginal/anal probe or surface electrodes) — twofold action: **PFM contraction** and **detrusor inhibition**.

	Long-term (chronic)	**Maximal**
Intensity	Below sensory threshold	Just below pain threshold
Aim	Detrusor inhibition via pudendal afferents (reflex hypogastric activation, central pelvic efferent inhibition)	Urethral closure by direct and reflex periurethral striated muscle contraction
Dose	6–12 h/day for months	15–30 min, several times a week to twice daily

Indication	**Frequency**
UUI	**5–20 Hz** (low)
SUI (incl. post-prostatectomy)	**20–50 Hz** (higher)
MUI	~20 Hz or alternating
All	Pulse 200–1000 µs, rectangular, biphasic

- Better than sham for SUI, UUI and MUI, but **adds nothing to PFMT** (± BF); PFMT preferred first-line (grade A).
- **Niche**: women who **cannot contract voluntarily** — intravaginal E-stim with concurrent effort restores the ability to contract.
- **Tibial nerve stimulation (TNS)** for UUI/OAB: better than no treatment; transcutaneous = percutaneous in efficacy with fewer adverse effects → *Ch 115*.

> **KEY POINT** E-stim is dosed by goal: sub-sensory, low-frequency, prolonged → detrusor inhibition; near-painful, higher-frequency, brief → sphincter contraction. Volitional PFMT achieves both, so E-stim earns its place only where volitional contraction is absent.

8 · BLADDER TRAINING AND SCHEDULED VOIDING

Bladder training (BT)

- = bladder drill/retraining; developed for UUI. Breaks the cycle: urgency → frequent voiding (relief) → habit → reduced functional capacity and DO → more urgency.
- Components: education, **incremental voiding schedule**, urge control (distraction, self-affirmation "I can wait", quick PFM flicks).
- Mechanism unknown; central feature = **dissociating voiding from urgency** (void by the clock, not the urge); improved cortical control and central afferent modulation proposed.

ALGORITHM
Diary → initial interval ~1 h while awake (or longest comfortable; 30 min if needed) → void on waking, at each interval and at bedtime, not between → increase by **15–30 min/week** if fewer leaks and urgency controlled → target **3–4 h** → weekly review with positive reinforcement.

- Evidence: more effective than no treatment for UUI, SUI and MUI; first-line in women (grade A); most intensive supervision possible. Insufficient evidence in men. Agency for Healthcare Research and Quality (AHRQ): no difference between BT and PFMT.
- Works in motivated frail older women who are cognitively and physically intact; BT with prompting in older adults admitted with hip fracture reduced functional UI at 6 months.

Delayed voiding (DV)

- No schedule: at each urge, suppress, then **wait 5 min** before voiding; extend gradually. Patients are surprised the urge fades → confidence. BT is triggered by the clock; DV by the urge plus a wait.

Caregiver-dependent toileting programmes — elderly and nursing homes

Programme	Patient role	Key features
Timed voiding (scheduled toileting)	Passive	Fixed intervals every 2–4 h (by day); goal dryness, no urge resistance. Also for impaired bladder sensation (neurogenic) and infrequent voiders. ~20% of frail elders become dry, 30–50% improve. 2–3-day trial predicts success.
Habit training	Passive	Schedule individualised from the diary, set shorter than usual interval and timed to **pre-empt** expected leaks; suits diuretic-induced or time-patterned UI.
Prompted voiding (PV)	Responsive	**Monitoring** (wet/dry check) → **prompting** (ask, assist) → **praise** (social reinforcement). Every 2–2.5 h. 25–40% of nursing home residents respond (daytime leaks ≤1); 3-day trial predicts response.
Functional incidental training	Responsive	PV plus mobility, transfer and strength exercise — improves function and UI.
Bladder training	Active	Lengthens interval; needs ability and motivation.

- Choice depends on cognition, mobility, voiding pattern and the need for reinforcement. Only BT aims to lengthen the interval; caregiver programmes aim only at dryness. PV requires a consistently available caregiver and a person able to cooperate or recognise the need to void.
- Ultrasound-assisted PV (prompting at measured optimal bladder volume) reduced daytime leakage more than fixed-interval PV.
- Caregiver burden is high (UI is the third most troublesome caregiving task); success depends on caregiver consistency more than patient skill.
- † Before toileting programmes in the frail elderly, exclude reversible causes (DIAPPERS: delirium, infection, atrophy, psychological, pharmacological, excess output, restricted mobility, stool impaction) and check postvoid residual (PVR) → *Ch 105*.

9 · HIGH-TONE (OVERACTIVE) PFM DYSFUNCTION

- Raised resting tone, poor relaxation → pelvic pain, bladder pain syndrome, hesitancy, straining, incomplete emptying.
- Many women void by Valsalva, which provokes a reflex PFM contraction that impedes flow. Normal voiding begins with PFM relaxation.
- **Down-training**: contract to feel tension, then contrast with relaxation; work:rest **1:4 or longer**; BF especially useful; carry relaxation into voiding.
- Voiding technique: slow down → deep breath → relax body and PFMs → wait for flow. Double voiding for incomplete emptying.

10 · MECHANICAL DEVICES

Two types: **intravaginal** (support urethra/bladder neck) and **intraurethral/meatal** (occlude). Insufficient evidence to favour any device or show superiority over other treatments.

Continence pessaries

- **Rationale**: with weak support the proximal urethra descends below the pelvic floor, out of the zone where abdominal pressure aids closure. The pessary restores it above the floor, **stabilises the bladder neck, lengthens the urethra and compresses it against the pubis** on straining — like a sling. The knob of an incontinence ring sits at the urethrovesical junction.
- **Types** (SUI): incontinence **ring** (with knob), **dish**, **Hodge** — each with or without support. Inert silicone/plastic. † For POP: support pessaries (ring, Gehrung) and space-filling pessaries (Gellhorn, cube, donut) for advanced prolapse or poor perineal support.
- † **Fitting**: largest comfortable size, one finger-breadth between pessary and vaginal wall; patient then walks, coughs, strains and voids before leaving — must not expel, cause pain or obstruct voiding.
- Worn continuously or only during exercise. Also helps urgency/frequency due to POP.
- **Contraindications**: active pelvic infection, severe ulceration, silicone/latex allergy, **likely non-compliance with follow-up**.
- **Complications**: common — discharge, odour; rare but serious (neglected long-term use) — **erosion, impaction, vesicovaginal and rectovaginal fistula**, bleeding. Self-insertion/removal difficulty limits use.
- **Care**: follow-up every **24 weeks** is acceptable (vs 12, grade C); anaesthetic cream reduces removal pain; oestrogen + Lactobacillus tablet adds no benefit. † Inspect for erosion at each review; local oestrogen is commonly used in atrophic women.
- Evidence: ~half improve; for OAB a ring pessary gave cure similar to behavioural therapy (retrospective). **ATLAS** — see trials; ICI: pessary or PFMT may be offered; pessary + PFMT better than PFMT alone for POP symptoms.

Other intravaginal devices

	Uresta	Impressa
Design	Reusable bell-shaped pessary with handle; sizes 3, 4, 5	Disposable tampon-like device: silicone core, support poles, mesh cover, applicator
Fitting	Clinician or self-fitted	Self-inserted; removed by string
Mechanism	Wide base supports urethra	Forms a tension-free suburethral support when abdominal pressure rises
Use	Inserted like a tampon	Need not be removed to void; remove for menses and intercourse

Urethral inserts and meatal devices

- None currently available in the USA.
- † Intraurethral insert: single-use plug with a retaining balloon, removed to void — occludes the urethra in pure SUI; problems are UTI, haematuria, discomfort, migration. Meatal devices: adhesive or suction caps over the meatus.

† Containment products and catheters

- † Absorbent pads and briefs when treatment fails, is declined or as an adjunct; choose by leakage volume, mobility and skin; change promptly to prevent **incontinence-associated dermatitis**. Female external (wicking/suction) catheters are an option for bed-bound women.
- † **Indwelling catheter is the last resort**: retention not manageable otherwise, sacral/perineal wounds contaminated by urine, terminal care or patient preference. Risks: catheter-associated UTI, stones, urethral erosion. Clean intermittent self-catheterisation is preferred for retention.

11 · VAGINAL OESTROGEN

- SUI with vaginal atrophy: PFMT + intravaginal oestrogen better than oestrogen alone (grade C); AHRQ: behavioural therapy + hormones better than no treatment for SUI.
- † Local (not systemic) oestrogen improves urgency, frequency and recurrent UTI in atrophic postmenopausal women and improves pessary tolerance → *Ch 113*.

12 · ADHERENCE, PREVENTION AND DELIVERY

- **Adherence caps efficacy**, since therapy is self-management. Barriers: remembering, finding time. Remedies: realistic expectations (weeks–months), scheduled follow-up, reminders (texts, timers), linking exercise to daily cues. Predictors: self-efficacy, positive intention, therapist enthusiasm. Adherence declines over time.
- **Prevention** in three predictable insults:
  - **Older women**: group behaviour programmes and bladder-health classes improve continence.
  - **Childbearing**: intensive supervised antenatal PFMT (12 weeks from 16–24 weeks' gestation) prevents late-pregnancy and early postpartum UI (grade A); postnatal programme after forceps or baby ≥**4000 g**. Antenatal PFMT does **not** prevent postnatal prolapse.
  - **Prostatectomy**: preoperative ± early postoperative PFMT shortens time to continence. Most men still receive no instruction.
- **Delivery models**: limited by access (too few trained providers), not efficacy — continence APPs within a urology Continence and Pelvic Health service; group classes (modest effect, cheap, reach mild cases); self-administered internet programmes (as effective as postal, maintained at 2 years); evidence-based apps (e.g., Tät for SUI). Most apps are of poor quality. Each model trades intensity, and so effect size, for reach.

13 · EVIDENCE GRADING — KEY ICI RECOMMENDATIONS

Grade (Oxford)	Meaning
A	Consistent level 1 evidence; effectively mandatory
B	Consistent level 2/3 studies or majority of randomised controlled trial (RCT) evidence
C	Level 4, or Delphi-processed expert opinion
D	**No recommendation possible** (not "weak")

International Consultation on Incontinence (ICI, 2023) recommendation	Grade
Supervised PFMT first-line for UI in women of all ages	A
PFMT over E-stim as first-line	A
BT first-line for women with UI	A
Behavioural weight loss for overweight women with UI	A
Antenatal supervised PFMT in continent women to prevent UI	A
PFMT reduces POP symptoms (stage reduction uncertain)	A; B
PFMT over drugs first-line (fewer adverse effects, less dropout)	B
PFMT before surgery (surgery more effective but riskier)	B
Adding BF, cones or E-stim to PFMT — not routine	B/C
Fluid adequacy and caffeine reduction	B
Smoking cessation for UI	D

- AHRQ (2018): behavioural therapy is **more likely than anticholinergics to cure UUI** (high strength) — drugs are not the superior first line for urgency incontinence.

14 · LANDMARK TRIALS

Trial	Design	Result	Take-home
First behavioural vs drug RCT	Older women with UUI: behavioural training vs titrated drug	Behavioural fewer leaks, more satisfaction	Behavioural therapy ≥ drugs for UUI
MOTIVE (Male Overactive Bladder Treatment in Veterans)	Men with OAB, no obstruction: behavioural vs oxybutynin ER	Equivalent reduction in voids	Behavioural therapy works in men
NIDDK Urinary Incontinence Treatment Network (†BE-DRI)	Urgency-predominant UI: tolterodine ER ± behavioural training	Combination better satisfaction and bother, persisting after drug stopped	Add behavioural training to drugs
ATLAS	Stress-predominant UI: pessary vs behavioural vs both	3 mo: behavioural better satisfaction; combination not better than behavioural; no difference at 12 mo	Pessary is a reasonable alternative; behavioural first
Weight-loss RCT (†PRIDE)	Overweight women: 6-month intensive weight loss vs education	Greater UI reduction with weight loss	Recommend 5–10% weight loss (grade A)
GBT (Group-Administered Behavioural Treatment)	Women with UI: one 2-h bladder health class vs none	Modestly better ICIQ-SF to 12 months	Group classes are a cheap, scalable option

15 · MCQ ONE-LINERS

1. Conservative therapy = neither surgery nor drugs.
2. Only grade A lifestyle recommendation: behavioural weight loss in overweight women.
3. 5–10% weight loss substantially reduces UI episodes.
4. Caffeine ~250 mg/day (two cups) associated with UI in men.
5. Fluid target: 30 mL/kg/day, minimum 1500 mL.
6. Stop fluids 3–4 h before bed for nocturia.
7. Commonest fluid problem in practice: over-restriction, not excess.
8. PFMT popularised by Kegel (1948); perineometer was the first BF device.
9. Correct PFM contraction: perineal elevation; incorrect: perineal descent.
0. Commonest PFMT error: Valsalva and abdominal/gluteal co-contraction.
1. Levator ani palpated at 4 and 8 o'clock.
2. PFMT work:rest 1:1–1:2; down-training 1:4.
3. Typical PFMT dose: 45–60 contractions/day.
4. "The knack": contraction just before and during a cough.
5. Supervised PFMT is grade A first-line at all ages.
6. BF adds no benefit to supervised PFMT; use for poor proprioception.
7. Vaginal cones cannot be retained with POP or wide introitus.
18. E-stim for UUI 5–20 Hz; for SUI 20–50 Hz.
19. E-stim niche: women unable to contract PFMs voluntarily.
20. Transcutaneous TNS = percutaneous TNS efficacy, fewer adverse effects.
21. Urge suppression: stop, squeeze quickly, let urge pass, then walk.
22. BT interval increased 15–30 min/week to 3–4 h.
23. BT dissociates voiding from urgency.
24. Prompted voiding: monitoring, prompting, praise.
25. Habit training = timed voiding individualised to the voiding pattern.
26. Only BT aims to lengthen the voiding interval.
27. Prompted voiding responders: 25–40% of nursing home residents.
28. AHRQ: behavioural therapy cures UUI more often than anticholinergics.
29. Grade D = no recommendation possible.
30. Continence pessary supports bladder neck and compresses urethra against pubis.
31. Serious pessary complications: erosion, impaction, vesicovaginal/rectovaginal fistula.
32. ATLAS: behavioural therapy ≥ pessary; combination no better than behavioural alone.
33. Antenatal PFMT prevents UI but not postnatal prolapse.
34. Indwelling catheter is the last resort for UI.

116 Retropubic Suspension Surgery for Incontinence in Females

Reading time ≈ 17 min (slow pace)

1 · DEFINITIONS AND MECHANISM

- **Stress urinary incontinence (SUI)**: involuntary loss of urine on effort (cough, sneeze, exertion). Others: mixed (MUI), urgency (UUI).
- Mechanisms, alone or combined: **urethral hypermobility** and **intrinsic sphincter deficiency (ISD)**.
- **Pressure transmission** (Enhorning) theory: urethra drops out of the retropubic zone on straining → leak — the basis of suspension. **Hammock hypothesis** (DeLancey): urethra compressed against a firm backstop.

> **KEY POINT** Suspension corrects **hypermobility, not the sphincter**. It lifts and stabilises the bladder neck and proximal urethra retropubically, so it presupposes a mobile urethra. A fixed, ISD urethra needs a device that compresses (sling, artificial urinary sphincter (AUS)).

2 · PATIENT SELECTION

Work-up

- Examination: stress test supine → standing with full bladder if negative; urethral mobility (Q-tip angle >30° = hypermobile); pelvic organ prolapse (POP); oestrogen status; prior abdominal incisions.
- Minimum (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction, AUA/SUFU): **post-void residual (PVR) + urinalysis**. Urodynamics not needed if SUI demonstrated; use if leak not shown. Valsalva leak point pressure (VLPP) <60 or maximal urethral closure pressure (MUCP) <20 cm H_2O suggests ISD.
- For Burch, **urethral mobility matters more than urethral function**.

Indications (Burch remains an AUA/SUFU guideline option alongside midurethral sling (MUS), autologous sling, bulking)

- SUI with hypermobility, or stress component of MUI. Not for UUI without SUI.
- **Concomitant abdominal surgery** (e.g. sacrocolpopexy).
- **Difficult vaginal access** (hip/leg contractures, vaginal stenosis).
- **Mesh aversion**.

Contraindications	Raise failure risk
Immobile vagina (cannot be lifted to Cooper's)	Baseline urgency, preoperative anticholinergic use
Immobile urethra (negative Q-tip ≈ **5-fold** failure) → sling or AUS	Obesity, menopause, age >60
ISD as sole cause; "lead-pipe" urethra (fixed, rigid, open)	Prior hysterectomy or SUI surgery; advanced POP; MUI

3 · ANATOMY

- **Space of Retzius**: avascular preperitoneal space between symphysis and bladder; floor = endopelvic fascia.
- **Cooper's (iliopectineal) ligament**: on superior pubic ramus.
- **Arcus tendineus fascia pelvis (ATFP)**: posterior symphysis → ischial spine along obturator internus; lateral vaginal detachment here = **paravaginal defect**.

4 · BURCH COLPOSUSPENSION — OPERATIVE STEPS

Principle: anterior vaginal wall and para-bladder-neck tissue sutured to Cooper's ligament; sutures elevate, and the resulting **fibrosis maintains** elevation.

ALGORITHM
Low lithotomy, prep umbilicus to perineum incl. vagina, Foley → Pfannenstiel/low midline (± Cherney) → enter Retzius, sweep fat laterally → locate bladder neck by Foley balloon + vaginal finger → dissect bladder off vagina **laterally** → expose Cooper's ligaments → sutures → cystoscopy

- **Sutures**: No. 0 or 1-0 delayed-absorbable/non-absorbable, SH needle; 2–4 each side.
- First suture at **mid-urethra level, as lateral as possible**, full thickness of vaginal wall **without entering lumen**; next suture more cephalad and more lateral at bladder neck; all medial → lateral, then through Cooper's.
- Tie **distal first**, with vaginal elevation; tie **loosely** (Tanagho modification) — **≈ 2 fingerbreadths** left between bladder neck and pubis. Over-tightening → obstruction.

KEY POINT **Danger points**: never dissect in the midline over urethra/bladder neck (Santorini plexus bleeding, urethral/bladder injury, obstruction if sutures too medial). Passing through Cooper's may cause sudden heavy bleeding — pressure, clips, sutures.

Approaches

Approach	Bottom line
Open (Pfannenstiel)	Cure **85–90% at 1 y, ≈70% at 5 y**
Laparoscopic (1991) / robotic; intra- or extraperitoneal	Same anchor; subjective cure equivalent to open to 18 months
Needle suspension (Pereyra, Stamey, Raz)	Abandoned — sutures tore through vaginal tissue

5 · OTHER RETROPUBIC PROCEDURES

- **Marshall–Marchetti–Krantz (MMK)**: two pairs of sutures from paraurethral vaginal wall at bladder neck → **pubic periosteum** beside symphysis. Unique complication: **osteitis pubis**.
- **Paravaginal repair** (Richardson): lateral vaginal sulcus reattached to ATFP/obturator fascia from symphysis to ischial spine.
- **Vagino-obturator shelf (VOS)**: Burch–paravaginal hybrid; vagina sutured to internal obturator fascia ± iliopectineal ligament.

Technique	Anchor	Holds by	Obstruction
Burch	Cooper's ligament	Sutures; high V-shaped angle	Greater
MMK	Symphyseal periosteum	Sutures; high	Greater
Paravaginal	ATFP	Scarring; flatter	Less
VOS	Obturator fascia	Scarring; flatter	Less

KEY POINT AUS in women: second-line for ISD (recurrent or neurogenic SUI). Placed at bladder neck, preferably **abdominally**; the transvaginal route raises infection and cuff erosion risk.

6 · PERIOPERATIVE CARE

- **Intraoperative cystoscopy is mandatory**: no suture in bladder/urethra or through the Foley; confirm bilateral ureteric efflux → if absent, release sutures and recheck.
- Retropubic drain if bleeding concern. Catheter optional if voiding trial passed.
- Incomplete emptying: teach clean intermittent catheterisation (CIC) (preferred over suprapubic tube) until PVR consistently **<150 mL**.

7 · OUTCOMES VERSUS SLINGS

- Burch ≈ retropubic MUS (tension-free vaginal tape, TVT) in cure; meta-analyses slightly favour MUS (more bladder perforation); Burch: longer operation, slower recovery, more voiding dysfunction.
- Autologous pubovaginal sling (PVS) more effective than Burch but more obstructive (SISTEr).
- Long-term failure ≈ 1 in 5 for both.
- With POP: stress-continent women having sacrocolpopexy benefit from prophylactic Burch (CARE); if SUI already present, MUS beats Burch.

8 · COMPLICATIONS AND MANAGEMENT

Complication	Notes / management
Suture through bladder/urethra; ureteric kinking	Cystoscopy; remove/replace suture. Missed → infection, irritative voiding, stones, fistula
Voiding dysfunction (2–27%)	Overcorrection; CIC; persisting obstruction → urethrolysis → *Ch 118*
De novo storage symptoms / OAB (8–27%)	Exclude obstruction; †standard OAB therapy
Posterior compartment prolapse (2.5–27%)	†Elevating the anterior wall opens the posterior cul-de-sac; examine posterior compartment, repair if needed
Osteitis pubis (MMK, 0.7–2.5%)	2–12 weeks: suprapubic pain to thighs, tender symphysis → rest/analgesia; osteomyelitis → antibiotics → debridement

9 · FOLLOW-UP

- Early contact (telemedicine acceptable); visit by 6 months: continence, voiding, overactive bladder (OAB) symptoms, sexual function, examination, PVR. Results decline with time; long-term = >48 months.

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SISTEr	Randomised controlled trial (RCT), 655: Burch vs autologous fascial PVS	PVS higher cure at 2 and 5 y; more urinary infection, voiding dysfunction, de novo UUI	Sling more effective, more morbid
Ward–Hilton (TVT vs colposuspension)	First RCT, 348	No difference to 5 y; TVT more bladder perforation; Burch more voiding dysfunction, later POP	MUS equals Burch, less invasive
CARE	RCT, 322 stress-continent, sacrocolpopexy ± Burch	Less postoperative SUI with Burch, sustained 2 y	Add prophylactic Burch at sacrocolpopexy

11 · MCQ ONE-LINERS

1. Burch anchors paravaginal tissue to Cooper's (iliopectineal) ligament.
2. MMK anchors to pubic symphyseal periosteum; complication: osteitis pubis.
3. Gap left between bladder neck and pubis: about two fingerbreadths.
4. Negative Q-tip test raises Burch failure about five-fold.
5. Q-tip hypermobility: angle change >30°.
6. Open Burch cure ≈85–90% at 1 y, ≈70% at 5 y.
7. Needle suspensions abandoned owing to suture pull-through.
8. Intraoperative cystoscopy is mandatory after Burch suture placement.
9. Burch-specific late complication: posterior compartment prolapse (enterocele).
10. SISTEr: autologous sling beat Burch but caused more voiding dysfunction.
11. CARE: prophylactic Burch at sacrocolpopexy reduces postoperative SUI.

2. ISD threshold: VLPP <60 or MUCP <20 cm H_2O.
3. AUS in women: second-line, preferably abdominal not transvaginal.
4. Burch sutures: first at mid-urethra level, most lateral.
5. Sutures must not enter the vaginal lumen.
6. Tanagho modification: sutures more lateral, tied loosely.
17. MMK and Burch obstruct more than paravaginal repair.
18. Paravaginal repair anchors to arcus tendineus fascia pelvis.
19. No ureteric efflux after Burch: release sutures.
20. Prefer CIC over suprapubic tube; target PVR <150 mL.

118 Slings: Autologous, Biologic, Synthetic and Mid-Urethral

Reading time ≈ 83 min (slow pace)

1 · HOW SLINGS WORK — THEORIES OF CONTINENCE

Theory	Concept	Surgical corollary
Hammock hypothesis (DeLancey)	Urethra rests on a suburethral layer attached laterally to the arcus tendineus fasciae pelvis (ATFP). In stress urinary incontinence (SUI) this layer is deficient → bladder neck/urethral hypermobility → unequal pressure transmission → leak	**Pubovaginal sling (PVS)** at the **bladder neck** restores a strong backstop for dynamic compression
Mid-urethral / Integral theory (Petros and Ulmsten)	Continence needs three components: **pubourethral ligaments, suburethral vaginal hammock, pubococcygeus**. Maximal urethral closure pressure (MUCP) is at the mid-urethra. Injury (parturition, surgery, ageing, oestrogen loss) to any one → SUI	**Mid-urethral sling (MUS)** placed **loosely** at the **mid-urethra**

- Integral theory support: incontinent women have ≈ **40% less collagen** in skin/round ligament biopsies MCQ.
- **Urethral hypermobility is a SIGN** of damaged support, **not the cause** of SUI.
- Surgical principle that follows: a PVS is placed under **minimal tension at the bladder neck** to rebuild the hammock; a MUS is placed **loosely under the mid-urethra** to stop the posterior urethral wall moving and help the walls coapt. Neither should obstruct at rest.

Historical milestones MCQ

- Autologous fascial PVS reintroduced 1978 (McGuire); modern short free rectus graft with adjustable tension (Blaivas).
- Intravaginal slingplasty 1995 → renamed **tension-free vaginal tape (TVT)**, the first retropubic MUS. Its first series defined the two complications **unique to MUS**: mesh penetration of the urinary tract and vaginal mesh exposure.
- Transobturator route (Delorme, 2001) — designed to avoid bladder, bowel and vessel injury.
- Single-incision mini-sling (SIMS): first approved 2006 (TVT-Secur, since withdrawn).

Guideline status (American Urological Association/Society for Urodynamics and Female Urology, AUA/SUFU)

- MUS may be offered to the index patient — **strong recommendation**, grade A.
- Retropubic or transobturator — either (conditional, grade A).
- SIMS — conditional, grade B.
- Must discuss mesh-specific risks **and** non-mesh alternatives before a synthetic MUS (Clinical Principle).

2 · PREOPERATIVE ASSESSMENT AND PATIENT SELECTION

AUA/SUFU minimum evaluation (index patient)

1. Focused history incl. bother (validated questionnaire).
2. Focused pelvic examination.
3. Objective demonstration of SUI with comfortably full bladder — cough stress test (CST) or Valsalva; if supine negative, repeat **standing**.
4. Post-void residual (PVR) — any method.
5. Urinalysis.

- **History**: onset, severity, storage/voiding symptoms, triggers (intercourse, exercise); **baseline urgency** (predicts poorer sling outcome); neurological disease, radiation, prior surgery, medications, bowel symptoms.
- **Examination**: pelvic organ prolapse (POP), atrophy, fistula, foreign material, urethral mobility. Q-tip (cotton-swab) test: hypermobility = **≥30°** from rest on straining.
- **Cystoscopy** is NOT part of evaluation of the index patient; do it for haematuria, suspected neoplasm/foreign body, prior pelvic surgery.
- Further tests when indicated: upper-tract imaging (haematuria, neurogenic bladder); magnetic resonance imaging (MRI) of brain/spine if a neurological lesion is suspected. Imaging is not part of the routine work-up.
- Formal assessment of **bother** is integral: incontinence affects quality of life (QoL), not survival, so the patient's bother and goals drive the decision to operate.

Additional evaluation (incl. urodynamics) should be considered when

- Diagnosis unclear or SUI not demonstrated; mismatch between symptoms and findings.
- Neurogenic lower urinary tract (LUT) dysfunction; unexplained haematuria/pyuria.
- Urgency-predominant mixed urinary incontinence (MUI), overactive bladder (OAB), elevated PVR, significant voiding dysfunction.
- High-grade POP (POP Quantification, POP-Q, stage ≥3) without SUI on reduction — test with and without prolapse reduction (pessary, speculum, swab).
- Prior anti-incontinence or POP surgery.

Urodynamic studies (UDS) — the controversy

- May be **omitted** in index patient with demonstrable SUI (conditional, grade B) — based on **ValUE**.
- Criticism of ValUE: most screened women were ineligible (uncomplicated SUI is a minority); surgeons were fellowship-trained.
- UDS are the only way to quantify **urethral function**:
  - MUCP **<20 cm H_2O** = "type 3 SUI" = **intrinsic sphincter deficiency (ISD)**.
  - Abdominal/Valsalva leak point pressure (ALPP/VLPP) **<60 cm H_2O** = ISD; **≥90** = little or no ISD.
- ISD is a **matter of degree**, not a separate disease — every woman with SUI has some ISD. It is now largely a clinical diagnosis.
- Predictive value of low MUCP/VLPP after MUS is inconsistent (worse mainly after transobturator slings). ISD matters most when combined with a **hypomobile urethra**.
- Bladder findings: **detrusor overactivity (DO)** is the most consequential — it may improve (if stress-induced: urine entering proximal urethra triggers contraction) or worsen after surgery; consider treating DO first. Low capacity/compliance also worsen outcomes. Detrusor underactivity and Valsalva voiding may predict post-operative voiding difficulty but should **not exclude** surgery.

Non-surgical options (discuss with all)

- Education, fluid management, **weight loss** (dose–response reduction in SUI), **pelvic floor muscle training (PFMT)** — recommended before surgery for most; adherence is the limiting factor.
- Pessaries/devices; periurethral bulking (minimally invasive, frequent retreatment) → *Ch 120*.

3 · CHOOSING THE SLING

Situation	Preferred option	Reasoning
Index SUI, mobile urethra	MUS (retropubic or transobturator); SIMS acceptable	Equivalent efficacy, least morbidity
ISD with mobile urethra	Retropubic MUS > transobturator	Comparative studies and meta-analyses favour retropubic

Situation	Preferred option	Reasoning
Fixed, immobile urethra	Retropubic MUS, autologous PVS, bulking, adjustable sling	AUA Expert Opinion; mobility predicts MUS success
Recurrent SUI after failed MUS	Retropubic MUS or autologous PVS	Retropubic better than transobturator for recurrence
Mesh complication, urethral injury, fistula, diverticulum, urethral reconstruction	**Autologous PVS**	No mesh near damaged urethra; adds tissue cover
Declines mesh	Autologous PVS or Burch → *Ch 116*	—
Neurogenic SUI on clean intermittent catheterisation (CIC)	Autologous PVS (may be tight)	Only once UDS confirm safe capacity/compliance
Concomitant POP surgery	MUS, PVS or Burch (any)	Conditional, grade C

4 · PUBOVAGINAL SLING

Indications

- Primary and recurrent SUI from **both hypermobility and ISD**; pure SUI and stress-predominant MUI.
- Concomitant anteroapical POP; neurogenic SUI (e.g. myelodysplasia).
- **Urethral reconstruction**: after trauma, mesh or graft perforation, iatrogenic hypospadias; interposition for urethrovaginal fistula or diverticulum repair.
- Failed retropubic suspension or failed MUS.
- **Autologous fascial PVS remains the gold standard for all forms of SUI.**

Mechanics

- Hypermobility: on straining, posterior urethral wall slides away from the anterior wall (held by pubourethral ligaments) → bladder neck opens (funnelling) → leak. ISD: urethra cannot coapt at rest.
- PVS gives **dynamic compression without obstruction** during rises in abdominal pressure. Too much tension → ischaemia, retention, erosion.
- What prevents give-way is **incorporation into the endopelvic fascia with fibrosis**, not entry into the retropubic space.
- Width **<2 cm** — wide enough to compress without forming a narrow constricting band.

Materials

Material	Behaviour	Status
Autologous — rectus fascia, fascia lata	Greatest fibroblast ingrowth and neovascularisation, minimal reaction; grafts intact years later; negligible erosion	**Gold standard**; cost: harvest time, pain, wound complications
Allograft (fresh frozen, freeze dried, solvent dehydrated, dermal)	Processing may disrupt microstructure and strength; heterogeneous, less durable results	Mostly withdrawn
Xenograft (porcine dermis, small intestinal submucosa; bovine dermis)	Weaker, most encapsulation; cure falls steeply with time; porcine inferior to autologous in randomised trials	Mostly withdrawn
Synthetic at bladder neck	Strong inflammatory/foreign-body reaction; urethral perforation **15×**, vaginal exposure **14×** other materials (esp. woven polyester)	**No longer used at bladder neck**
Vaginal wall sling (in situ)	Weak; inclusion cysts, vaginal shortening	Historical

KEY POINT **Fibrosis paradox**: more fibrosis should mean better support, but the inflammatory infiltrate that drives it also degrades the sling and destroys tissue → erosion. Autologous fascia remodels with minimal inflammation — the ideal balance.

- **Rectus vs fascia lata**: similar biocompatibility, strength and cure. Fascia lata: no abdominal wound complications, less blood loss; but repositioning, unfamiliar field, thigh pain. Prior **ventral hernia repair** → use fascia lata. Scarred rectus fascia is usually still usable.
- Thigh muscle herniation depends on graft width: narrow (1.5 cm) strips → none. Using a **Crawford fascial stripper** reduces thigh pain.

5 · AUTOLOGOUS PVS — OPERATIVE PROCEDURE

Preparation

- **Counselling**: transient and permanent voiding dysfunction, de novo urgency; teach CIC preoperatively in selected patients. Manage expectations — the sling will not fix bowel or other pelvic floor symptoms.
- Spinal or general anaesthesia, agreed by patient, surgeon and anaesthetist.
- **Antibiotics** (clean-contaminated): single dose of 1st/2nd-generation cephalosporin (cefazolin adequate); alternatives: ampicillin/sulbactam, aminoglycoside or aztreonam + metronidazole or clindamycin, clindamycin alone.
- **Venous thromboembolism**: intermittent pneumatic compression (IPC) before induction; heparin by individual risk.
- **Dorsal lithotomy**, careful padding. Nerve injury rare — lateral femoral cutaneous and femoral commonest, then common peroneal; almost all resolve (median ≈1 month).
- Prep abdomen from umbilicus and vagina (povidone-iodine; chlorhexidine ≤4% alcohol acceptable). Weighted speculum, 18 Fr Foley, Trendelenburg, ring retractor.

Rectus fascia harvest

ALGORITHM
Pfannenstiel 6–7 cm, ≈2 cm above symphysis → mark **2 × 8 cm** graft (transverse or longitudinal) → excise → close fascia with running No. 1 polydioxanone (PDS) → graft cleaned of fat on side table → No. 1 PDS (or 1-0 polypropylene) stitched across each end, left long → keep in saline

- Transverse incision: stay **≥2 cm above the symphysis** for tension-free closure. Undermining fascial edges helps closure but may weaken fascia. If closure impossible (rare) → bridge with mesh or biologic graft.

Fascia lata harvest

- Set-up: IPC below knee on harvest side, knee flexed on a bag, hip internally rotated and taped; mark greater trochanter and lateral femoral condyle.
- 3 cm longitudinal incision just above patella over iliotibial band → two parallel incisions 2 cm apart → free and transect distal end → stripper advanced proximally → divide. Graft 8 cm usual (classically 20 × 2 cm).
- Close in layers **without closing the fascia lata**; compression wrap for 8 h; early ambulation.

Vaginal dissection

- Hydrodissect with saline → **inverted-U incision** (apex ≈2 cm below meatus, arms to bladder neck located by Foley balloon) — best exposure of bladder neck and endopelvic fascia. Alternative: midline incision.
- Raise **thick** flaps above the periurethral fascia; keep U arms wide to protect flap blood supply.
- **Empty the bladder**, then perforate the **endopelvic fascia** with Metzenbaum scissors pointed to the ipsilateral shoulder, just under the ischiopubic ramus, superolaterally.
- Finger-dissect the retropubic space until the back of the symphysis is felt with little intervening tissue; abdominal and vaginal fingers should meet. No aggressive medial mobilisation (bladder

injury).

- **Reoperation**: sharp dissection hugging the pubic periosteum — safest plane.

Passage, cystoscopy, fixation

ALGORITHM
Bladder empty → Stamey/Cobb-Ragde needles (or long clamp) passed **from above**, kept on the back of the pubis, out lateral to bladder into vaginal incision → **cystoscopy (70° lens)** while needles are moved → injury: withdraw and repass **more laterally**, re-scope after every pass → confirm bilateral ureteric efflux → sutures loaded, graft centred → needles withdrawn, ends tagged above

- Anchor graft to periurethral tissue with two 4-0 polyglactin sutures; close vagina watertight (2-0 polyglactin).
- Do any vaginal POP repair **before tensioning**; **remove the weighted speculum** (it distorts tension).
- **Tension**: tie sutures over the rectus fascia leaving **two fingerbreadths** of slack — deliberately **loose**. Tighter only if lifelong CIC is planned.
- Close Scarpa and skin; Foley; optional vaginal pack (± oestrogen cream).

Postoperative care

- Liberal fluids and diet the same night. Pack and catheter out on day 1 → voiding trial; failure → home with Foley, retrial within 5 days (or CIC).
- No lifting >10 lb (≈4.5 kg) and no intercourse for **6 weeks** and until healing confirmed.

6 · PVS OUTCOMES

- Durable at long-term follow-up, even in complex patients (radiation, spinal cord injury, trauma). Cure high even with MUCP <4 cm H_2O → **severe ISD is not a contraindication**.
- Reported success varies widely with definition of success and population. No direct correlation between objective and subjective cure.
- **Recurrent SUI**: excellent cure; repeat PVS after failed sling and PVS after failed MUS (with partial mesh excision) both succeed in most.
- Commonest cause of "failure" = **storage symptoms/urgency urinary incontinence (UUI)**, not stress leakage. De novo UUI 2–22%.
- Four-quadrant modification (graft fixed to pubocervical and periurethral tissue): high cure; every failure was due to urgency or storage symptoms. Shorter (4 × 2 cm) grafts also work in ISD.
- Prior POP surgery predicted de novo urgency after PVS; "over-tight" technique, age, body mass index (BMI) and radiation did not.
- Fascia lata vs rectus: equivalent continence at 1 year; rectus has more wound complications, fascia lata less blood loss. Thigh: transient pain and numbness, occasional tendinitis.
- **Cochrane (PVS vs MUS)**: PVS probably no better, may be less effective; MUS shorter operation, fewer complications except bladder perforation, less voiding and storage dysfunction.
- After mesh excision or diverticulectomy, the damaged urethra has the poorest prognosis, but PVS still gives continence to most and relieves mesh symptoms.
- **MUI**: surgery may cure, improve or worsen urgency — unpredictable; anticholinergics alone resolve urgency in only ≈ two-thirds and cannot fix the outlet. PVS cure in MUI similar to pure SUI; stress-induced DO often improves.
- Combined PVS + anterior colporrhaphy effective in grade 3–4 cystocele with SUI.

7 · PVS COMPLICATIONS AND MANAGEMENT

Voiding dysfunction / bladder outlet obstruction (BOO)

- Incidence 2.5–35% — **more than after Burch** (SISTEr: all outlet procedures were in PVS arm). Retention >4 weeks ≈ 8%; **permanent retention ≤5%** (≈2% if not overtensioned).
- Cause: usually **excessive tension** → hypersuspension. Host factors: pre-existing voiding dysfunction, poor detrusor contractility, dysfunctional voiding, **straining to void** (the sling's dynamic resistance rises with abdominal pressure, so straining tightens the outlet).
- Presentation is variable: weak stream, positional voiding, retention; or **storage symptoms only** (urgency/UUI is commoner than retention); recurrent urinary tract infection (UTI), suprapubic pain, dysuria — even with a low PVR.
- Uncorrected or new **POP in any compartment** can kink the urethra — exclude.
- Preoperative prediction is unreliable: poor detrusor contraction, PVR >100 mL or maximum flow rate (Qmax) ≤20 mL/s were linked with prolonged CIC in some series but not others. Hence low detrusor pressure or Valsalva voiding is a reason to **counsel about CIC**, not to refuse surgery.
- Concurrent PVS with prolapse repair (even grade 3–4 POP) has little extra effect on emptying.

Evaluation

- Examination: urethral angulation, fixed urethra, short non-pliable vagina (may be normal).
- PVR (no clear cut-off); cystoscopy (perforation, hypersuspension); video-UDS selectively — the classic high-pressure/low-flow pattern is often **absent in women**.

> **KEY POINT** **The key diagnostic criterion is the temporal relationship** between surgery and onset of symptoms, not a urodynamic number. Preoperative UDS does not predict who will obstruct (SISTEr analysis).

Management — timing

ALGORITHM

Transient retention (most void within **10 days**) → catheter/CIC, timed and double voiding, PFMT, biofeedback, anticholinergic → autologous PVS obstruction often improves: wait up to **3 months** → if complete retention, earlier: **<6 weeks** loosen autologous sling under anaesthesia (downward traction on urethra with cystoscope) → **>6 weeks** or failed conservative care → sling incision or urethrolysis

- Downward loosening is **not for synthetic slings**. Urethral dilation: anecdotal, risks fibrosis.
- **Transurethral incision/resection of bladder neck fails** — the sling is extraluminal — and risks sphincter damage, incontinence, contracture.

Procedure	Features
Sling incision (midline or lateral, transvaginal)	84–100% success; shorter, less morbid → **first choice**; ≈20% recurrent SUI. Graft or vaginal-wall interposition does not reliably prevent recurrence
Transvaginal urethrolysis	Least effective for PVS — lateral dissection cannot relieve direct suburethral compression
Suprameatal urethrolysis	Better access to lateral sling wings
Retropubic urethrolysis	Best single-series results; used after failed aggressive transvaginal attempt

- Formal urethrolysis success 65–93%; recurrent SUI 0–19%. No reliable predictor of success (prior urethrolysis is the only suggested adverse factor).
- Failure: usually **inadequate lysis**; also recurrent fibrosis, DO, poor contractility, learned voiding dysfunction. **Repeat urethrolysis** relieves obstruction in ≈90%, but storage symptoms often persist.
- **Refractory storage symptoms** after urethrolysis in >half (preoperative DO may hint at this): anticholinergics/β3-agonists, **sacral neuromodulation (SNM)** (test-phase response predicts

durable benefit), onabotulinumtoxinA (100 U; ≈40% dry at 12 weeks, similar to idiopathic OAB).
- Counsel: lysis may not cure urgency and may make a dry patient leak; untreated female BOO has unknown long-term bladder effects.
- Recurrent SUI after lysis: salvage with bulking, repeat autologous PVS or MUS.

Perforation and exposure

- Rare with autologous fascia; contributors: missed needle perforation, excess tension, later urethral instrumentation. Urethral perforation → **excise involved sling + repair urethra** (± Martius flap); bladder erosion → endoscopic removal.
- Most urethral perforations by bladder-neck slings present 1–18 months after surgery (mean ≈9 months).
- Synthetic PVS exposure: discharge, pain, recurrent UTI → partial/total removal; recurrent SUI after removal 44–100% → second **autologous** PVS.

Non-urological

- Commonest systemic: pulmonary, cardiovascular, neurological, gastrointestinal. Mortality ≈ **3 per 10,000** anti-incontinence procedures.
- Harvest site: infection, seroma, **incisional hernia** (≈1%; usually needs repair).

8 · MID-URETHRAL SLING — MECHANISM AND MATERIALS

- On imaging the proximal urethra is often open at rest; on stress, shearing between anterior and posterior walls opens it.
- MUS mechanism: **secures the posterior urethral wall** (stops it moving away), ± compression of the lumen and dynamic **kinking** during stress.
- Anchorage: retropubic → endopelvic fascia; transobturator → obturator internus/externus and membrane. Mesh becomes fixed along its **whole length** — why 50–93% stay continent after midline incision.
- **Urethral mobility**: preoperative immobility predicts failure (fixed urethra ≈ ISD). Persisting postoperative mobility is **not** harmful for retropubic/transobturator slings — ideal = loose sling + mobile urethra. **Exception, SIMS**: success depends on restricting mobility → tension tighter.
- Whether MUS obstructs is disputed: some studies show unchanged flow; others lower Qmax, higher detrusor pressure at Qmax and PVR.

> **KEY POINT** **Why does a loose tape work?** At rest it does not touch the urethra, so voiding is unobstructed. On coughing, the mobile urethra descends against the tape: the posterior wall is arrested, the urethra kinks over it and the lumen closes. Hence a mobile urethra is needed — a fixed urethra never meets the tape, and tightening the tape to compensate creates obstruction.

- Tensioning in practice varies widely: few surgeons use an objective intraoperative test; most tension retropubic and transobturator tapes similarly.

Materials

- Current standard: **type 1 (Amid) macroporous, monofilament, loosely woven polypropylene**, pore **>75 µm**.
- Why >75 µm: allows leukocytes and macrophages in (infection surveillance) and fibroblast/vessel ingrowth.
- Cautionary history: UraTape (microporous with silicone core) and ObTape (<50 µm) → high vaginal exposure → withdrawn; later tapes have larger pores (e.g. 200 µm).

9 · MUS OPERATIVE PROCEDURES

- **Counselling**: the mesh is a **permanent implant**; uncommon but sometimes serious mesh-specific complications (exposure, perforation, pain); transient or permanent voiding dysfunction; de novo or worse storage symptoms; alternatives without mesh. Document it.

- Anaesthesia: local ± sedation, spinal or general. Intraoperative cough test under local was meant to set tension, but outcomes are the same under spinal → **cough-test tensioning is unnecessary**.
- Antibiotics, thromboprophylaxis, positioning as for PVS (a systematic review found little effect of prophylaxis on infection or exposure, but trials were small — still give a single dose).
- Brief thigh flexion (≥70°) helps trocar passage.

Retropubic (TVT) — bottom-up

ALGORITHM
Local infiltration (retropubic and paraurethral) → two suprapubic stabs ≈2 cm from midline → 1.5 cm midline vaginal incision starting 1.5 cm from meatus → minimal lateral tunnels to the endopelvic fascia (**not perforated**, unlike PVS) → rigid catheter guide deflects bladder to the opposite side → trocar through endopelvic fascia, Retzius, rectus, out suprapubically, **hugging the back of the pubis** → repeat → cystoscopy → tension → remove sheath → trim

- Kit: two 5 mm trocars; 40 × 1.1 cm polypropylene tape in plastic sheath.
- Local anaesthesia: ≈5 mL to vaginal and suprapubic sites; 20 mL behind the pubis to the urogenital diaphragm; 10 mL each side of urethra.
- Keep the trocar tip in **metal-to-bone contact** with the back of the pubis with slow graded pressure — this keeps it extravesical and extraperitoneal. Trendelenburg in top-down helps move bowel cephalad.
- Top-down variant: trocar from suprapubic skin onto the opposite index finger; no catheter guide. Bottom-up gives better subjective cure and less voiding dysfunction, perforation and exposure.
- **Tensioning**: clamp or metal sound between tape and urethra while removing sheath — tape lies **loosely**; function is not mainly compression.

Transobturator — outside-in (TOT)

- Lithotomy with hips **hyperflexed (120°)**; vaginal dissection carried laterally to the ischiopubic ramus.
- Skin puncture in the labiocrural (genitofemoral) fold at the level of the adductor longus tendon notch (thumb on the pinched tendon).
- Trocar pierces obturator membrane (felt "give") → turned medially around the ramus onto index finger → out of vaginal incision. Check for vaginal sulcus "button-hole".
- Tension: clamp passes easily between sling and urethra.

Transobturator — inside-out (TVT-O)

- Exit stabs 2 cm above urethral level and 2 cm lateral to labial folds. Obturator membrane opened with scissors; winged guide at 45° to sagittal plane → helical passer rotated out through thigh. Remove both sheaths together without tension.
- In-to-out vs out-to-in: same cure; inside-out → more voiding dysfunction, less vaginal perforation; outside-in → possibly more urinary tract injury.

Single-incision mini-sling (SIMS)

- Vaginal incision only; device-specific inserter anchors each end into **obturator internus** (hammock) — no skin exit.
- Tension so a hemostat tip passes, i.e. **tighter than standard MUS**; learning curve — place close to urethra, wide and deep tunnels (less perforation and exposure).

Feature	Retropubic (TVT)	Transobturator (TOT/TVT-O)	SIMS
Path / anchor	Retzius, rectus fascia; U-shape	Obturator foramen; flatter hammock	Obturator internus only
Cure (index SUI)	Equal to transobturator up to 5 y; best long-term data (>17 y)	Equal; slightly more repeat surgery long term	Non-inferior to transobturator (TVT-Secur excluded); retropubic better

Feature	Retropubic (TVT)	Transobturator (TOT/TVT-O)	SIMS
ISD / recurrent SUI	**Preferred**	Worse with low VLPP	Limited data
Bladder perforation	**4.5%**	**0.6%**	Rare (but erosion higher with TVT-Secur)
Vascular/visceral injury, bleeding	More	Less	Rare
Voiding dysfunction	More	Less	Less
Pain	Suprapubic 2.9%	**Groin 6.4%** vs 1.3% retropubic	Least postoperative pain
Vaginal exposure	≈2%	≈2%	Similar once TVT-Secur excluded
Cystoscopy	Mandatory (70°)	Recommended; examine urethra	Essential (30° rigid scope suffices)

10 · INTRAOPERATIVE CYSTOSCOPY

- Purpose: detect trocar injury of bladder or urethra before the mesh is left in.
- Requirements: **70° lens** and a **fully distended bladder** — tangential injuries are missed otherwise; inspect the **urethra** too (transobturator and SIMS).
- For PVS, also confirm ureteric efflux.

Finding	Action
Bladder trocar injury	Withdraw, re-pass more laterally, re-scope, **continue**; catheter overnight to 72 h (benign; no long-term sequelae)
Urethral injury	**Do NOT place mesh** (AUA Clinical Principle); repair, catheter drainage, **staged** anti-incontinence surgery (or autologous sling)

11 · MUS OUTCOMES

- Retropubic: most studied anti-incontinence operation; follow-up to 17–20 years; objective cure stays high, subjective cure lower and instrument-dependent. Success typically <85% on strict definitions.
- Cochrane retropubic vs transobturator (81 trials): **similar subjective and objective cure** short, medium and long term; trade-offs as in the table.
- Inside-out vs outside-in transobturator: similar subjective cure; choose by surgeon experience.
- TVT vs colposuspension: equal cure at 5 years; de novo urgency low after both (<2% vs <5%); later POP commoner after Burch.
- TOMUS at 5 years: satisfaction stayed high and similar (79% retropubic, 85% transobturator) despite falling objective success — again, satisfaction ≠ strict cure.
- Long-term series pattern: objective cure usually exceeds subjective cure; Patient Global Impression of Improvement (PGI-I) is highest; **de novo urgency rises with follow-up** (to ≈⅓ at 10–20 years); retention stays low (0–7%).
- Satisfaction does not need complete dryness: a one-step fall in leakage frequency satisfies most, at any age.
- SIMS (Cochrane): worse than retropubic overall, but this was driven by **TVT-Secur (withdrawn)**; other SIMS ≈ transobturator with less pain and shorter operation. Efficacy may fade over time.
- SIMS learning curve (hammock type): failure fell from 20% to 8% by siting the tape closer to the urethra; vaginal perforation abolished by wider tunnels; exposure reduced by deeper tunnels.
- Long-term SIMS data are sparse; the key long-term endpoint is **reoperation** (mostly for SUI rather than complications).

Special populations

Group	Key points
MUI	SUI cure high; urgency/UUI resolves in ≈60–75% (best stress-predominant, worst urge-predominant); may not persist. Persistent DO linked with lower Qmax (partial obstruction). Risk factors for de novo urgency: age, obesity, nocturia, preoperative DO, low capacity. Urge-predominant → often combination therapy. Persistent DO gradient: transobturator < retropubic < PVS; de novo urgency lower after transobturator in one comparison. Worse or new storage lower urinary tract symptoms (LUTS) may be BOO rather than OAB
ISD	Lower cure (≈50% with urodynamic ISD); immobile urethra is a stronger predictor than low VLPP. Low VLPP not a contraindication to retropubic MUS. Transobturator: 12× odds of persistent SUI with low VLPP
Recurrent SUI	No level 1 evidence (Cochrane: insufficient data for any strategy); repeat MUS is the commonest choice; retropubic better than transobturator in the larger, more recent analysis; cure ≈81–90% overall but lower than primary (≈62% vs 86% in one series, partly because repeat patients have more ISD) with more de novo urgency; same technique; more bladder perforation after prior retropubic surgery; immobile urethra fails more
Concomitant POP	Concomitant sling reduces postoperative SUI (overt or occult) and further surgery, but more voiding dysfunction and adverse events. Staged approach: ≈30% of SUI resolves after POP repair alone. Tensioning hard in occult SUI. True "prophylactic" sling only after thorough counselling (OPUS); in one review only the retropubic MUS subgroup benefited
Elderly	More DO, UUI, ISD, poor contractility, prior surgery (fixed urethra), oestrogen deficiency (poor healing), comorbidity. Safe; similar or lower cure; more de novo urgency, voiding dysfunction, medical morbidity; fewer bladder injuries. Mobility decides: Q-tip >30° → 90% continent; <30° → 42%; <10° → 80% stay incontinent. Registry: cure ≈64% at ≥75 y vs 89% at 55–64 y, yet most were satisfied. **Frailty, not age**, predicts readmission and longer stay. Do not exclude by age (AUA Expert Opinion)
Obese	Raised intra-abdominal pressure drives SUI. Success >80% but lower than non-obese (objective cure ≈87% vs 66% at 5 y in one trial analysis); obesity is a risk factor for TVT failure; less OAB improvement with BMI ≥30; more postoperative pain after inside-out transobturator; worse in BMI >35; more wound infection; fewer bladder punctures; no excess retention

12 · MUS COMPLICATIONS AND MANAGEMENT

> KEY POINT **Concurrent vs staged sling at POP repair**. Concurrent: one operation and less postoperative SUI (risk ≈39% → 8–19% in meta-analysis), but more obstruction, storage symptoms and mesh events, and tension is hard to judge when SUI is only occult. Staged: avoids an unnecessary sling (some SUI resolves after POP repair) but commits a third of women to a second operation, usually within 12 months. Decide with the patient after testing for occult SUI with the prolapse reduced.

Trocar injury, bleeding, infection

- Bladder trocar injury ≈ 2–4%, almost all retropubic; risk ↑ with prior retropubic surgery, thin habitus — see §10.
- Major vascular (≈0.07%) and bowel (≈0.04%) injuries are rare but include deaths in device-reporting databases.
- †Suspect visceral or vascular injury when there is unexplained hypotension, falling haemoglobin, abdominal pain, ileus or sepsis after a retropubic sling → urgent imaging and surgical exploration.
- Bleeding/haematoma 2–3%, more retropubic (≈4% bled ≥300 mL needing intervention in one series); on routine ultrasound ≈1 in 8 had a retropubic haematoma >5 cm, almost all resolving by 1 month — so retropubic haematoma is common and usually **asymptomatic → observe**, compress; evacuation rarely.
- **UTI**: commonest complication (≈3–4% at 30 days; ≈1 in 4 within a year, 4% de novo recurrent); peak in first month (day 7); risks: age (strongest), diabetes, BMI >40, prolonged catheterisation;

less after transobturator; UTI marks a worse course (more reoperation, de novo UUI).
- Vaginal wound infection 0.7–2%; rare necrotising fasciitis, abscess, **adductor myositis** (leg pain, gait difficulty) — high suspicion; may present years later with mesh exposure.

Groin / thigh pain

- Transobturator → groin/thigh pain (16–26% early in some trials, 6.4% pooled; may persist longer); retropubic → suprapubic pain (≈2% persistent at 5 y).
- Mostly transient (haematoma or neuropraxia): non-steroidal anti-inflammatory drugs, rest, physiotherapy; persistent → trigger-point steroid/local anaesthetic injection.
- Early groin pain in up to a quarter falls to ≈4% persistent at 12 months with conservative care — counsel accordingly.
- Refractory → mesh excision; **complete removal** gives more pain resolution than partial. Retropubic arm: vaginal + abdominal incisions; transobturator: vaginal + medial thigh.

Voiding dysfunction / retention

- Cause: tape **too tight** or **too proximal** (obstructing slings are usually found near the bladder neck), or POP.
- More after retropubic (TOMUS: surgery for voiding dysfunction 2.7% vs 0%). No reliable preoperative predictor.
- Evaluate: **temporal relation**; examine for anterior/apical POP (can obstruct a previously normal sling ≥10 years later); PVR; cystoscopy; UDS selectively (obstruction often not shown).

ALGORITHM
Retention/high PVR → CIC or catheter (most resolve within 2 weeks) → persistent obstruction → **transvaginal midline sling incision at 2–4 weeks** (single small vertical incision, minimal dissection) → >3 months: tape fixed along its course → excise lateral portion (to pubis/ischiopubic ramus) → rarely formal urethrolysis

- **Urethral dilation is potentially harmful — do not perform** (scarring, mesh perforation). Early transvaginal loosening is reasonable.
- Most remain continent after incision (70–100%). **Early** release gives better voiding recovery and **less** recurrent SUI than delayed section; delay is the main predictor of recurrent SUI.
- Conservative care works for most retention: in registry data almost all resumed normal voiding within weeks; few needed incision. But women who "recover" often keep urgency and frequency — a hint of persisting partial obstruction.
- Voiding inefficiently but volitionally: timed voiding, biofeedback, PFMT, selective drugs before surgery.

KEY POINT **Why the different timing from PVS?** Autologous fascia obstruction often settles, so one waits up to 3 months. Synthetic mesh becomes fixed by tissue ingrowth, rarely loosens, and late release needs wider dissection and causes more recurrent SUI — hence release within ≈4 weeks.

Storage symptoms / de novo urgency

- De novo OAB ≈ 6–12% in reviews (up to 15–30%); similar across all MUS types — no route protects.
- Predictors: age, DO, low capacity, high detrusor pressures.
- New or worse storage symptoms with clear temporal link → **partial BOO tops the differential**; progressive symptoms over weeks suggest obstruction. **Cystoscopy mandatory** (mesh in urethra/bladder); UDS selectively.
- Treatment: behavioural, PFMT, antimuscarinic or β3-agonist (antimuscarinics work less well than in idiopathic OAB) → sling revision vs third-line (SNM, tibial nerve stimulation, onabotulinumtoxinA — as effective as in idiopathic OAB once tape malposition excluded).

Mesh exposure and perforation

- Vaginal exposure ≈2% (retropubic ≈ transobturator); †presents with discharge, bleeding, dyspareunia or partner discomfort. Urinary tract perforation is a late event (distinct from trocar injury) → *Ch 119*.
- †Small vaginal exposure: topical oestrogen ± excision of exposed mesh and closure; larger/persistent exposure: excision. Urethral or bladder mesh: excision (transvaginal or endoscopic/abdominal) with repair; avoid immediate new mesh.

Sexual function

- De novo dyspareunia ≈3–14%; worse sexual function tracks with de novo urgency and incomplete emptying more than mesh. Sling removal may relieve dyspareunia.
- Many improve, chiefly by abolishing **coital incontinence**.
- In the TVT–colposuspension trial dyspareunia fell from 34% to 13% at 5 years; MUS and Burch affect sexual satisfaction similarly.

13 · ADJUSTABLE SLINGS

- **Remeex**: suburethral polypropylene mesh on sutures to a **varitensor** (screw and reel) above rectus fascia; tension adjusted postoperatively with a percutaneous manipulator.
- Indication: **complex SUI** — ISD, fixed urethra, reoperation. Cure ≈85–90% in series; complications 32–51% (mostly Clavien II–III), bladder injury 7–14%, readjustment 14–43%. Women with detrusor underactivity may stay dry but in retention — a quarter asked for removal. Higher BMI and pad weight predict worse results. Balance success against repeated adjustments. No level 1 evidence.

14 · ASSESSING THE FAILED OR TROUBLESOME SLING

ALGORITHM
Symptom after sling → define: persistent/recurrent SUI vs storage symptoms vs voiding symptoms vs pain/discharge → history with **timeline** → examination (mesh exposure, POP, urethral mobility, tenderness over tape) → PVR, urinalysis/culture → **cystoscopy** (mesh in urethra/bladder, hypersuspension) → UDS selectively (stress leak, DO, obstruction) → tailored treatment

- Recurrent SUI with mobile urethra → repeat retropubic MUS or autologous PVS; fixed urethra/ISD → autologous PVS, bulking, adjustable sling.
- Storage symptoms with temporal link → treat obstruction first; otherwise OAB pathway.
- Pain or exposure → excise mesh (complete excision when pain is the indication) → *Ch 119*.

15 · MESH CONTROVERSY AND REGULATORY STATUS

- MUS uptake drove a rise in SUI surgery. **2010 US Food and Drug Administration (FDA)** (United States Food and Drug Administration) safety communication on surgical mesh for POP and SUI, based on reports to the **MAUDE** (Manufacturer and User Facility Device Experience) database — many involving mesh perforation and exposure.
- Professional bodies continue to endorse MUS for SUI, with documented counselling about permanence, unique complications, and non-mesh alternatives (autologous PVS, Burch, bulking).
- †The 2019 FDA order stopping sale of transvaginal mesh concerned **POP** repair, not MUS. UK (2018) paused MUS use pending safeguards (registries, specialist centres).
- Detailed mesh complications, legal and regulatory issues → *Ch 119*.

16 · LANDMARK TRIALS

Trial	Design	Result	Take-home
ValUE	Randomised controlled trial (RCT), non-inferiority, 630, uncomplicated SUI: UDS vs office evaluation	Office evaluation non-inferior at 1 y	UDS may be omitted in index patient
SISTEr	RCT, 655: autologous rectus PVS vs Burch	PVS higher SUI cure (47% vs 38% at 2 y); more voiding dysfunction, UTI, UUI; all BOO surgery in PVS arm; advantage persisted at 5 y	PVS more effective but more obstructive
Ward–Hilton	RCT: TVT vs open colposuspension	Equal cure to 5 y; TVT more intraoperative bladder injury; Burch more postoperative morbidity, POP	MUS replaced Burch as standard
TOMUS	RCT, 597: retropubic vs transobturator MUS	Objective cure 80.8% vs 77.7% (1 y, equivalent); voiding dysfunction surgery 2.7% vs 0%; †more groin/neurological symptoms with transobturator; 5 y success 51% vs 43% (equivalence not met)	Either route acceptable; retropubic slightly more durable, more voiding dysfunction
OPUS	RCT (327 analysed): vaginal POP repair ± prophylactic retropubic MUS (sham) in stress-continent women	Urinary incontinence at 12 months 27% vs 43%; number needed to treat 6.3; more bladder perforation, UTI, bleeding	Prophylactic sling reduces SUI but adds risk — counsel

17 · MCQ ONE-LINERS

1. PVS sits at the bladder neck; MUS at the mid-urethra.
2. Integral theory: pubourethral ligaments, suburethral hammock, pubococcygeus.
3. Urethral hypermobility is a sign, not cause, of SUI.
4. Gold standard sling for all SUI forms: autologous fascial PVS.
5. Rectus fascia graft size: 2 × 8 cm.
6. Transverse rectus harvest stays ≥2 cm above symphysis.
7. Prior ventral hernia repair → use fascia lata.
8. Fascia lata is left unclosed at harvest.
9. PVS sutures tied over rectus with two-fingerbreadth slack.
10. Endopelvic fascia is perforated in PVS but not MUS.
11. Synthetic bladder-neck sling: 15× urethral perforation risk.
12. MUS mesh: type 1, macroporous (>75 µm) monofilament polypropylene.
13. ISD: MUCP <20 or ALPP <60 cm H_2O.
14. ValUE: UDS unnecessary before surgery for uncomplicated SUI.
15. SISTEr: PVS beats Burch; more voiding dysfunction.
16. TOMUS: retropubic = transobturator cure; more voiding dysfunction retropubic.
17. Bladder perforation: retropubic 4.5% vs transobturator 0.6%.
18. Groin pain: transobturator 6.4% vs retropubic 1.3%.
19. Bladder trocar injury: re-pass and continue.
20. Urethral injury at MUS: abandon mesh, stage repair.
21. Cystoscopy needs 70° lens and full bladder.
22. SIMS excess failure was due to withdrawn TVT-Secur.
23. SIMS needs tighter tension; restricted mobility predicts cure.
24. ISD or recurrent SUI: retropubic preferred over transobturator.
25. Fixed urethra predicts MUS failure better than low VLPP.
26. OPUS: prophylactic sling, number needed to treat 6.3.
27. Post-PVS obstruction: wait up to 3 months unless retention.
28. Post-MUS obstruction: incise sling at 2–4 weeks.
29. Transurethral bladder neck incision fails for sling obstruction (extraluminal sling).
30. Transvaginal urethrolysis least effective after PVS.
31. Urethral dilation after MUS is harmful.
32. Obstructing MUS usually lies too proximal.
33. Diagnosis of sling obstruction: temporal relationship of symptoms.
34. Commonest complication after MUS: urinary tract infection.
35. Remeex adjustable sling uses a suprapubic varitensor.

117 Vaginal and Abdominal Reconstructive Surgery for Pelvic Organ Prolapse

Reading time ≈ 56 min (slow pace)

1 · GOALS AND PRINCIPLES

- Pelvic organ prolapse (POP) is a quality-of-life (QoL) condition: operate on the interaction of grade, symptoms and bother — not on anatomy alone. Declining surgery is a legitimate informed choice.
- Individualise to physiological age, comorbidity, previous surgery, physical and sexual activity.
- **The first operation is the best chance**: afterwards normal planes are lost and recurrence rises with each re-attempt.
- **Success = absence of symptoms** (no bulge, few de novo symptoms), which drives satisfaction more than anatomical cure. Satisfaction tracks achievement of the patient's own stated goals — ask her.
- Routine events (catheter at discharge, pain, constipation, urgency, urinary tract infection [UTI]) are perceived as complications. Preoperative education improves Patient Global Impression of Improvement (PGI-I) without changing objective cure — preparation changes perception, not anatomy.
- **Consent sequence**: alternatives → purpose and limits → benefits → risks → hospital course → home course → coping with a catheter.

Category	Principle	Use
Obliterative	Close the vagina	Comorbid, not sexually active; contraindicated if intercourse desired
Restorative	Repair discrete fascial defects with native support	Good native tissue
Compensatory	Reinforce with synthetic, biological or autologous graft	Weak tissue, failed previous repair

- **Route**: vaginal = fewer complications, shorter operation, stay and recovery; abdominal = **greater durability**. Open vs laparoscopic: equal complications, faster recovery minimally invasive.
- POP is usually multicompartmental → combined repairs ± concurrent anti-incontinence procedure.

2 · CHOICE OF PROCEDURE BY COMPARTMENT

Compartment	Pelvic Organ Prolapse Quantification (POP-Q) point	Organ
Anterior	Aa (distal) / Ba (proximal)	Urethrocele / cystocele
Middle	C / D	Cervix or cuff / enterocele (uterosacral scar)
Posterior	Ap (distal) / Bp (proximal)	Rectocele / enterocele; plus perineal body

A points always distal, B proximal. Point D exists only with a cervix.

Site	Vaginal	Abdominal
Aa — urethra	Anterior repair; bladder neck suspension; sling	Retropubic urethropexy
Ba — bladder	Anterior repair; paravaginal repair; colpocleisis; wedge colpectomy	Paravaginal repair; sacrocolpopexy
C — cervix/cuff	Uterosacral suspension; iliococcygeus; sacrospinous fixation; Manchester; hysteropexy; vaginal hysterectomy; colpocleisis	Hysterectomy; uterosacral suspension; sacrocolpopexy; uterine suspension

Site	Vaginal	Abdominal
D — cul-de-sac	**McCall** culdoplasty	**Halban**, **Moschcowitz** culdoplasty
Ap — posterior	Posterior repair; site-specific repair	Colpoperineopexy

KEY POINT Only paravaginal repair and uterosacral suspension are done by both routes. McCall = vaginal; Halban (linear sutures) and Moschcowitz (purse-string — may angulate the ureter) = abdominal.

3 · ANTERIOR COMPARTMENT

Defects

- **Central**: midline attenuation of pubocervical fascia → midline bulge. **Lateral (paravaginal)**: detachment of pubocervical fascia from the arcus tendineus fasciae pelvis (ATFP).
- Most are combined central + lateral ± apical; isolated colporrhaphy is not recommended when other defects exist. **Anterior repair does not suspend the apex** — coexisting apical prolapse needs its own repair.

Anterior colporrhaphy (Kelly plication) — steps

ALGORITHM
Dorsal lithotomy, padded, hips/knees ~90° → catheter → hydrodissection ± vasoconstrictor → midline incision apex → bladder neck → dissect vaginal wall off pubocervical fascia (white, shiny surface) midline then laterally to ATFP → reduce bladder → interrupted 2-0/3-0 delayed-absorbable plication sutures bladder neck → apex, tied while assistant reduces → trim epithelium judiciously → running absorbable closure → IV dye + **cystoscopy** → pack.

- With **midurethral sling (MUS)**: do NOT extend incision to urethra; separate midurethral incision. With **pubovaginal sling**: single incision apex → midurethra for bladder-neck placement.
- Deep sutures may enter bladder/ureter or kink the distal ureter → completion cystoscopy is mandatory.
- Abandoned for SUI: American Urological Association (AUA) meta-analysis showed failure approaching **40%**; now a prolapse operation only.

Results and augmentation

- Recurrence figures (old series 0–20%, modern up to 40%) diverge mainly by **definition of failure**: one randomised dataset gives 30–46% failure at "POP-Q stage ≥2" but 9–23% at "beyond the hymen". Objective recurrence often does not reduce satisfaction.
- Absorbable (Vicryl) mesh inlay: modest one-year gain; no suturing technique superior; biological grafts add little.
- Polypropylene mesh vs native tissue: native tissue had more awareness of prolapse, re-operation and anterior recurrence — but synthetic grafts had far more graft complications. This asymmetry plus the Food and Drug Administration (FDA) 522 post-market studies → **transvaginal synthetic POP mesh withdrawn globally** (US 2019) → *Ch 119*. QoL, re-operation and dyspareunia were not improved by mesh.
- A concomitant sling roughly **halves anterior recurrence** (−55%) — the largest single gain, bigger than any graft.

Complications and management

Complication	Point / management
De novo SUI	Rises with cystocele stage: stage 2 **41%**, early 3 **53%**, advanced 3–4 **66%**; point Ba predicts
Detrusor overactivity	De novo 5–7%; pre-existing resolves in up to 63%

Complication	Point / management
Retention	Usually from concomitant sling; also plication/neural. Brief clean intermittent catheterisation (CIC) → persistent after sling: incision/urethrolysis; hypocontractile: chronic CIC → neuromodulation → suprapubic tube
Bleeding (>350 mL)	Vaginal vascularity and venous complex under pubis; bleeding = wrong plane; worse in redo
Bladder injury	Drain bladder before perforating endopelvic fascia; two-layer absorbable closure; irradiated/tenuous → labial fat pad (Martius) interposition
Ureteric obstruction	See below
Others	Vesicovaginal fistula, vaginal shortening, inclusion cyst, dyspareunia, infection

> KEY POINT **Confirming ureteric efflux**: indigo carmine (blue, IV), fluorescein 10% (green, IV), phenazopyridine (orange, oral pre-op). No efflux → ureteric catheter/retrograde pyelogram or remove plication sutures. A patent ureter may not efflux if under-hydrated — but never close without proving patency.

Paravaginal repair

- Goal: reattach pubocervical fascia to ATFP (White, vaginal; Richardson, abdominal). Combine with colporrhaphy for combined defects.
- **Vaginal**: midline incision → dissect laterally → perforate endopelvic fascia into retropubic space → palpate ATFP (ischial spine → pubic ramus) → **5–7** non-absorbable sutures 1 cm apart (Capio device) → through lateral pubocervical fascia → tie only after all placed → cystoscopy → close.
- **Abdominal** (open/lap/robotic): retropubic space → non-dominant hand elevates lateral vaginal sulcus → **4–6** non-absorbable sutures ATFP → lateral vaginal wall, avoiding paravaginal veins → cystoscopy.
- Lateral recurrence low (2–8%) but central recurrence ~**22%** regardless → treat as two independent failures. Abdominal route more durable at long follow-up.
- **Neither colporrhaphy nor paravaginal repair treats SUI.**
- Vaginal paravaginal repair is harder with more serious complications: haemorrhage (transfusion 12%), retropubic haematoma, abscess, bilateral ureteric obstruction, lithotomy neuropathy.

4 · OCCULT SUI AND THE CONCOMITANT SLING

- High-stage anterior prolapse kinks/compresses the urethra, masking poor urethral function. SUI appearing after repair = **stress incontinence on prolapse reduction** (formerly "occult SUI"); reported in 8–66% of continent women.
- Test every high-stage prolapse **with the prolapse reduced**: pack, rectal swabs, speculum blade or pessary. None standardised or proven superior; low sensitivity, high specificity; cannot reliably predict postoperative SUI.
- **Concomitant** argument: diagnosis is non-reproducible, so SUI may be missed. **Staged** argument: sling obstruction requiring intervention (8.5%) ≈ intervention for post-op SUI without sling (8.3%).
- **Prediction models**: Outcomes following vaginal Prolapse repair and mid-Urethral Sling (OPUS) calculator — 7 predictors, accuracy 0.72 vs 0.54 for stress test alone; better than experts; for women without SUI symptoms. CUPIDO: urodynamics adds nothing in asymptomatic women.
- Concomitant Surgery and Urodynamics in Genital Prolapse and Stress Incontinence (CUPIDO) model (9 predictors): urodynamics adds no predictive value in asymptomatic women; not generalisable to abdominal surgery. A six-predictor model (age, diabetes, subjective incontinence, reduction stress test, type of prolapse surgery, concomitant sling) performs similarly.
- Women with subjective SUI having native-tissue repair or colpocleisis do not exceed 50% risk of bothersome post-op SUI → sling may reasonably be omitted.

- Women with SUI symptoms but negative urodynamic and reduction tests still carry ~30% risk of later intervention — an intermediate group to counsel.

> **KEY POINT** **Why the sling may also help the prolapse**: a suburethral sling adds distal anterior support, so it improves durability of the cystocele repair, not just continence. **Why not give everyone a sling**: OPUS showed more bladder perforation, UTI, bleeding and incomplete emptying, and patient-reported pelvic floor symptom scores did not differ — the benefit is real but must be weighed per patient.

ALGORITHM
Advanced anterior POP → reduction stress test ± urodynamics → symptomatic, occult or urodynamic SUI → offer concomitant anti-incontinence procedure → continent and test negative → individual risk–benefit (OPUS calculator); **selective, not routine**.

5 · APICAL COMPARTMENT

The apex is the cornerstone of support; failing to secure it raises recurrence exponentially. Correct by vaginal, abdominal or minimally invasive route, with or without uterine preservation.

Uterosacral ligament suspension (USLS)

- Preserves the natural vaginal axis. Ligament breaks at discrete points, allowing site-specific repair.
- **Anatomy**: S1–S3 (±S4) → cervix and posterior vagina; sacral, intermediate, cervical parts. **Ureter–ligament distance narrows from 4 cm (sacrum) to 0.9 cm (cervix)**; ureteric sheath makes **kinking** (not ligation) the mechanism of obstruction. Sutures near sacrum entrap sacral plexus (S1–S4) → buttock pain to posterior thigh; S3 injury rises >4.6 cm cephalad of the ischial spine.
- **Safe zone**: intermediate portion, ~1 cm posterior along the tensioned ligament at the level of the ischial spine; pass needles **lateral → medial**.

ALGORITHM
Intraperitoneal (after vaginal hysterectomy): stay sutures at 3 and 9 o'clock cuff dimples → excise enterocele sac → pack bowel → palpate ischial spines; ligaments posterior–medial → tent with Allis at 5 and 7 o'clock → **2–4** non-absorbable sutures per side, each proximal to the last → anterior repair/sling now → sutures through apical pubocervical and rectovaginal fascia (highest medially) → tie → **cystoscopy after tying, before trimming** → no efflux: remove most lateral ipsilateral suture first, then in sequence.

- Confirm the ureter lies lateral before placing sutures — its relation to the ligament edge shifts with the degree of prolapse; place temporary ureteric stents to aid palpation if in doubt.
- Advantage over SSLF: less risk to pudendal and gluteal vessels, and a midline, physiological vaginal axis.
- Tying draws pubocervical and rectovaginal fascia together: resuspends apex **and** closes cul-de-sac.
- Extraperitoneal variant: Breisky–Navratil retractor holds bladder and ureter anteriorly. Abdominal variant: 3 permanent sutures per side at hysterectomy (Shull).
- Unrecognised ureteric obstruction may need suture removal or, rarely, ureteroneocystostomy; other serious events include small bowel injury and pelvic abscess. Patients may note de novo dyspareunia, while irritative and obstructive voiding and SUI usually improve.
- Results: apical success ~98%, anterior ~81% → recurrence is anterior by default; ureteric kinking/injury **1–11%**. Absorbable suture non-inferior to permanent. Premenopausal status raises recurrence.
- **OPTIMAL**: USLS = sacrospinous fixation (success ~65% at 2 years by strict composite); perioperative pelvic floor muscle training no benefit.

Sacrospinous ligament fixation (SSLF)

- Richter; popularised by Nichols. Extraperitoneal, strong anchor, no laparotomy, preserves length; posterior (perirectal) or anterior (paravesical) access.

- **Anatomy**: ligament 7–8 cm, ischial spine → sacrum beneath coccygeus. Pudendal nerve/vessels pass around the spine (Alcock's canal); inferior gluteal vessels behind; hypogastric plexus above; haemorrhoidal vessels medial.
- **Safe zone**: **1.5–3.0 cm medial to the ischial spine** (1–2 fingerbreadths), **into** the ligament, limited depth, no entry above its upper border. Too lateral → pudendal bundle; too medial → thin fanned ligament (failure) + intraligamentous sacral nerves.

ALGORITHM
Mark apex and reduce to intended ligament → midline incision → blunt entry to perirectal (or paravesical) space, rectum retracted medially → palpate spine and ligament → Breisky–Navratil/Heaney ± lighted retractors → 2 sutures (Capio or Miya hook/Deschamps) → other repairs → sutures out to apex; pulley stitch if permanent → close vagina halfway → tie apex flush to ligament (**no suture bridge**) → cystoscopy before trimming → close.

- Unilateral right-sided fixation is usual (easier for a right-handed surgeon); bilateral or anterior approach gives a more midline apex and lower anterior recurrence (~7–9%) but little outcome evidence of superiority.
- Results: overall cure ~85%, apical recurrence ~5%; stage 4 prolapse predicts recurrence. Absence of any defect at 6 weeks → only 3% need further surgery — early healing predicts late success.
- Unilateral (right): 2 sutures 2 cm apart; bilateral: one per side. **Pulley stitch** keeps the permanent knot outside the vaginal lumen.
- **Signature drawback**: vaginal axis deflected posteriorly and to the right → **anterior recurrence** (~18% vs apical ~5%). Concomitant anterior repair helps only when Ba > +1 with SSLF (not with USLS). Some narrowing/shortening complaints.
- Complications: ipsilateral gluteal pain **15%** — usually resolves in 2–3 months (observe; local anaesthetic nerve injection if needed); pudendal vessel/nerve injury (too near spine); rectal injury (repair primarily); haemorrhage.

> **KEY POINT** Both ligaments are safest in their **middle third** and dangerous at their bony ends: USLS intermediate portion; SSLF 1.5–3 cm medial to the spine.

Iliococcygeus suspension

When uterosacral ligaments are deficient. Bilateral fixation to iliococcygeus fascia **1 cm distal to the ischial spine**, keeps a near-normal axis; anterior recurrence commonest.

Sacrocolpopexy (SCP) — open, laparoscopic, robotic

- **Indications**: failed vaginal repair; young, active, sexually active women; isolated vault/uterine prolapse or enterocele; patient wanting the most durable repair. Maximises functional vaginal length without axis distortion; SUI and other defects can be repaired at the same sitting.
- Two essentials: **type I macroporous monofilament polypropylene** mesh and secure fixation to promontory and vagina. Fascia lata graft: recurrence ~1/3 vs <1/10 with polypropylene.

ALGORITHM
Low lithotomy; vaginal stent/end-to-end anastomosis (EEA) sizer → bowel packed, sigmoid to left → incise peritoneum over promontory, extend along **right** side of rectum (right ureter close) → expose anterior longitudinal ligament → (Halban culdoplasty if large enterocele) → dissect bladder off anterior vagina (~6–8 cm) and rectum off posterior vagina (to rectal reflection; to perineal body = colpoperineopexy) in the areolar plane above the fibromuscularis → Y-mesh sutured to both walls full-thickness **without entering the lumen** → tension → 2–3 permanent monofilament sutures to promontory at **S1** → trim → **retroperitonealise** → anti-incontinence procedure if indicated → cystoscopy → examine for ancillary vaginal repair.

- **Sacral fixation level**: S1/promontory. Too low (S3–4) → presacral venous plexus; too high → middle sacral vessels and the **left common iliac vein** (often medial to the artery).

- **Tension**: open — two-finger gap between graft and rectum; minimally invasive — extend vagina maximally with sizer, then release; avoid hypersuspension (pain, de novo SUI).
- **Robotic/laparoscopic set-up**: anti-slip pad, low lithotomy in Allen stirrups, arms tucked, Foley and orogastric tube → umbilical camera port → 2–3 robotic ports ≥9 cm apart + assistant port on the patient's right → steep Trendelenburg; side- or parallel-dock to keep vaginal, bladder and rectal access. Laparoscopy: 0° scope, extracorporeal or barbed sutures. Steps otherwise identical; the robot's extra time is suture tying, not docking.
- Vaginal obturator (EEA sizer, Lucite stent) elevates and distends the vagina; dissecting in the loose areolar plane prevents accidental lumen entry.
- Vaginal entry: irrigate, two-layer closure; wrong plane ↑ later mesh exposure. Absorbable mesh-to-vagina sutures equivalent.
- Tacks/sutures can cause lumbosacral spondylodiscitis/osteomyelitis (mostly sutures).
- **Route**: objective success 87–100% and similar for open, laparoscopic and robotic; robotic advantage only ~90 mL less blood loss; laparoscopy shorter; robot costlier. Learning curve: laparoscopic 30–40, robotic ~50 cases.
- Failures after SCP are mainly **posterior/distal**; essentially no apical retreatment. Isolated SCP leaves cystocele/rectocele in up to one third; ~20% need vaginal repair — counsel.

SCP complication	Rate / management
Haemorrhage (presacral)	~4%; fix at S1 visible vessels, pre-emptive bipolar; † pressure, sutures, bone wax or sterile thumbtacks for presacral venous bleeding
Cystotomy / enterotomy / ureter	3% / 1.6% / 1% — cystoscopy at end; bladder injury commonest robotic visceral injury
Ileus, small bowel obstruction	3.6% / 1.1%
Mesh exposure	Quote **~10% by 7 years** (extended CARE); mean onset 14 months — bleeding, discharge; more with microporous mesh; transvaginal excision ± partial colpocleisis, or abdominal excision → *Ch 119*

- Mesh exposure is heralded by persistent pain, discharge or infection; it is commoner with microporous materials (polytetrafluoroethylene/Gore-Tex) because bacteria enter pores that macrophages cannot — hence macroporous monofilament mesh only.
- **Hysterectomy at SCP**: total hysterectomy ↑ mesh exposure → prefer **supracervical hysterectomy** (cervix as barrier) or meticulous two-layer cuff closure.

Choosing the apical operation

- Cochrane: vaginal apical repair vs SCP — more awareness of prolapse (RR 2.1), repeat surgery (2.3), recurrence (1.9), SUI (1.9), dyspareunia (2.5) after vaginal surgery. Yet vaginal repair has high success, low re-operation, and suits many older women.
- No difference between vaginal native-tissue apical repairs (USLS = SSLF).

ALGORITHM
Elderly/comorbid, not sexually active → colpocleisis → fit, primary prolapse, wants vaginal route → USLS or SSLF ± vaginal hysterectomy or hysteropexy → young/active, recurrent, short vagina, high stage → SCP (minimally invasive preferred).

6 · HYSTERECTOMY VS UTERINE PRESERVATION

- † Hysterectomy alone does not cure prolapse — the apex must always be resuspended.
- **Sacrospinous hysteropexy** is non-inferior to vaginal hysterectomy + USLS, and in newer data superior to it and to uterosacral hysteropexy. Vaginal hysterectomy with vault suspension is also effective for uterine prolapse.
- With SCP, keeping the cervix (supracervical or uterine-sparing) lowers mesh exposure.

- † Preservation needs a normal uterus: exclude abnormal bleeding, endometrial/cervical pathology and high cancer risk (e.g. Lynch syndrome); ensure cervical screening.

7 · OBLITERATIVE SURGERY — COLPOCLEISIS

- For counselled older women who do not want coital function. Short operation, regional or local anaesthesia, minimal morbidity, very low recurrence.
- **Total colpocleisis**: post-hysterectomy; whole vagina denuded (to ~2–4 cm from meatus); purse-string or transverse rows of 2-0 absorbable sutures invert the vault; enterocele sac need not be opened.
- **Partial (Le Fort)**: uterus in situ; anterior and posterior rectangles excised (3 cm from meatus to 3 cm from cervix), **lateral channels** left for cervical/uterine drainage; transverse rows from leading edge until above levator plate.
- Both: **high perineorrhaphy/levator myorrhaphy** narrows the introitus; success depends on the amount of vaginal tissue sutured together; cystoscopy.
- Uterus left: Papanicolaou smear, pelvic ultrasound ± endometrial biopsy first. Hysterectomy removes cancer and **pyometra** risk (channel obstruction) but ↑ complications (OR ~1.9).
- **Urinary**: occult SUI unmasked and urethral traction → spare ≥**1.5 cm** of distal anterior wall; screen for SUI and offer concomitant sling if SUI/occult SUI. Frail women: retention risk and difficulty with CIC → consider suprapubic tube.
- Outcome: success ~90–98%, satisfaction >90%, coital regret up to **13%**; body image and pelvic floor symptoms improve. These women are older and comorbid (43% >80 years), so shared decision-making and medical optimisation matter more than the operation itself.
- **Why no need to open the enterocele**: the layered closure obliterates the potential space into which bowel could herniate; an irreducible sac is incorporated with purse-string sutures with the uterosacral remnants.

8 · POSTERIOR COMPARTMENT

- Symptoms: bulge/bleeding; defecatory dysfunction (splinting/digitation, trapping, urgency — **not faecal incontinence**); dyspareunia.
- Distinguish **outlet obstruction** (support defects — surgical) from **motility disorders** (slow transit — diet/medication) and **anismus** (non-relaxing puborectalis — biofeedback, pelvic floor physiotherapy). Treat the latter first.
- **Predicts benefit**: bulge, need to splint, incomplete emptying on defecography, large rectocele. **Not predictive**: constipation, feeling of incomplete emptying. Counsel: **constipation often persists**.
- **Transvaginal beats transanal** repair (recurrence 7% vs 66%).

Operations

- **Midline fascial plication (posterior colporrhaphy)**: repair anterior/apical first → hydrodissection → midline incision → dissect rectovaginal fascia off epithelium (scissor tips toward epithelium) → purse-string at hernia base for large defects → midline 2-0 absorbable plication, avoiding wide lateral bites (ridges, narrowing) → incorporate perineal body → trim, close.
- **Site-specific repair**: finger in rectum identifies discrete defects (commonest **transverse**, septum detached from perineal body) → close individually.
- **Perineorrhaphy**: triangular excision → horizontal mattress approximation of perineal body; smooth contour (ridges → dyspareunia).
- **Levator ani plication is abandoned** — dyspareunia up to 50%; it creates a rigid shelf (the old posterior colpoperineorrhaphy) and also worsened bowel symptoms.
- **Outcomes**: fascial plication relieves bulge, straining, splinting and obstructed defecation (dyspareunia often improves); satisfaction correlates with relief of defecatory symptoms rather than POP-Q. Site-specific repair gives similar function but higher recurrence in advanced prolapse.

About two-thirds of women who digitated may resume it by one year — functional rectal decompensation is not corrected by surgery.

- International Consultation on Incontinence (ICI): midline plication without levatorplasty = procedure of choice (lower recurrence than site-specific); Cochrane: evidence insufficient.
- **Grafts** (synthetic or biological) not supported: no anatomical gain, more complications (RR 1.8); synthetic → dyspareunia and visceral erosion.

9 · COMPLICATIONS ACROSS POP SURGERY — SUMMARY

Problem	Prevention / management
Ureteric injury/kinking (USLS highest, 1–11%)	Lateral → medial bites at correct level; cystoscopy with dye after tying, before trimming; remove offending (most lateral) suture; stent/retrograde study; reimplant if needed
Bladder/rectal injury	Correct plane, drained bladder, rectal finger; layered closure; interposition if poor tissue
Mesh exposure	Macroporous type I mesh, full-thickness flaps, avoid lumen entry, supracervical hysterectomy; oestrogen/excision → *Ch 119*
Recurrence	Address every compartment; apex is key; anterior after USLS/SSLF, posterior after SCP; most asymptomatic
Neuropathic buttock pain	Middle-third suture placement; observe ≤3 months; local anaesthetic nerve injection
De novo SUI / retention	Preoperative reduction testing; selective sling; CIC; sling incision

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
CARE (Colpopexy and Urinary Reduction Efforts)	randomised controlled trial (RCT): stage 2–4 POP, abdominal SCP ± Burch, continent women	Post-op SUI 24% vs 44% at 3 months	Prophylactic continence surgery reduces de novo SUI
E-CARE	7-year extension	SUI 62% vs 77%; mesh exposure ~10%; composite failure 34–48%	Benefit = delay; mesh exposure rises with time
OPUS	RCT: vaginal POP repair ± MUS vs sham, no SUI symptoms	Incontinence 27% vs 43% at 12 months; more bladder perforation, UTI, bleeding, retention	Sling prevents SUI but adds harm — selective use
OPTIMAL	RCT: USLS vs SSLF ± perioperative PFMT	Equal success (~65%) at 2 years; PFMT no effect	Native-tissue apical repairs equivalent
Bilateral SSLF vs abdominal SCP	RCT	Earlier failure and 3× cystocele re-operation after SSLF	Abdominal route more durable
Transvaginal vs transanal rectocele repair	RCT	Recurrence 7% vs 66%	Repair rectocele vaginally

11 · MCQ ONE-LINERS

1. Strongest determinant of satisfaction after POP surgery: absence of bulge symptoms, not anatomical cure.
2. Abdominal POP repair: more durable; vaginal: less morbidity, faster recovery.
3. Point D is recorded only when the cervix is present.
4. McCall culdoplasty is vaginal; Halban and Moschcowitz are abdominal.
5. Moschcowitz purse-string culdoplasty may angulate and obstruct the ureter.

6. Lateral (paravaginal) defect = pubocervical fascia detached from the arcus tendineus fasciae pelvis.
7. Anterior colporrhaphy and paravaginal repair do not treat SUI.
8. Paravaginal repair sutures: vaginal 5–7, abdominal 4–6, non-absorbable, 1 cm apart.
9. De novo SUI after cystocele repair rises with stage: 41% → 53% → 66%.
0. Concomitant sling reduces cystocele recurrence by about 55%.
1. OPUS: sling at vaginal POP repair cut 12-month incontinence from 43% to 27%.
2. CARE: Burch at sacrocolpopexy halved early de novo SUI (44% → 24%).
3. Uterosacral–ureter distance: 4 cm at sacrum, 0.9 cm at cervix.
4. Ureteric obstruction after USLS is due to kinking via the ureteric sheath.
5. Do cystoscopy after tying suspension sutures, before trimming them.
6. No efflux after USLS: remove the most lateral ipsilateral suture first.
7. SSLF sutures: 1.5–3 cm medial to the ischial spine, into the ligament.
8. Commonest recurrence after SSLF: anterior compartment (posterior-right axis deviation).
9. Gluteal pain after SSLF ≈15%, usually resolves within 3 months.
20. OPTIMAL: USLS and SSLF equivalent at 2 years.
21. Iliococcygeus fixation point: 1 cm distal to the ischial spine.
22. Sacrocolpopexy mesh: type I macroporous monofilament polypropylene, fixed at S1.
23. Vessel at risk above promontory: left common iliac vein; below S1: presacral plexus.
24. Mesh exposure after sacrocolpopexy ≈10% at 7 years (E-CARE).
25. Supracervical hysterectomy preferred with sacrocolpopexy to reduce mesh exposure.
26. Commonest failure site after sacrocolpopexy: posterior compartment.
27. Le Fort partial colpocleisis keeps lateral channels to drain the uterus.
28. Serious risk of obstructed Le Fort channels: pyometra.
29. Colpocleisis: spare ≥1.5 cm distal anterior wall to avoid urethral traction SUI.
30. Levator plication at rectocele repair abandoned because of dyspareunia.
31. Commonest rectovaginal fascial defect: transverse, detached from perineal body.
32. Constipation usually persists after anatomically successful rectocele repair.
33. Transvaginal synthetic POP mesh withdrawn by FDA in 2019.

119 Genitourinary Complications of Mesh and Their Repair

Reading time ≈ 24 min (slow pace)

1 · MESH, TERMINOLOGY AND REGULATION

- Mesh is used for male and female stress urinary incontinence (SUI) and pelvic organ prolapse (POP). Harms: pain; exposure/extrusion; injury to adjacent nerves, muscle, viscera, vessels.
- Pelvic floor ≠ abdominal wall: close to bladder, urethra, vagina and rectum; large physiological stretch; abundant microbiota.

Amid type	Pore	Example	Behaviour
I	Macroporous >75 µm	Knitted monofilament polypropylene	Tissue ingrowth; **only recommended mesh**
II	Microporous ~10 µm	Expanded polytetrafluoroethylene (PTFE; Gore-Tex)	Encapsulated; >30% full-length slings excised
III	Macropores in microporous base	Mersilene, Mycromesh	Extrusion ~20%
IV	Submicronic	Silicone-coated	Extrusion/sinus up to 40%

KEY POINT **Why pore size matters**: bacteria (~1 µm) enter small pores that macrophages and lymphocytes (~50 µm) cannot — infection is sealed from immunity. Large pores (~1500 µm knitted polypropylene) allow fibrocollagenous integration. Woven polypropylene (Marlex) is stiff and extrudes. No excess cancer or autoimmune disease with mesh.

IUGA/ICS term	Meaning
Erosion	**Do not use**
Exposure	Mesh visible through separated/non-healing vaginal epithelium
Extrusion	Mesh passing gradually out of a tissue (vagina, bladder), not via the incision
Perforation	Abnormal opening into a hollow viscus
Contraction / prominence	Shrinkage / wrinkling, folding without epithelial defect

(IUGA = International Urogynecological Association; ICS = International Continence Society.)

† **Category–Time–Site (CTS) coding**: Category 1–3 vaginal (1 no epithelial separation; 2 exposure ≤1 cm; 3 >1 cm or extrusion), 4 urinary tract, 5 rectum/bowel, 6 skin/musculoskeletal, 7 systemic compromise; suffix A asymptomatic, B symptomatic, C infection, D abscess. Time T1 ≤48 h, T2 48 h–2 months, T3 2–12 months, T4 >12 months. Site S1 vaginal suture line, S2 vagina away from suture line, S3 trocar path, S4 other skin/musculoskeletal, S5 intra-abdominal.

- **Food and Drug Administration (FDA)**: class I low, II intermediate, III high risk (needs premarket approval). Mesh for male SUI, female SUI and **transabdominal POP = class II**. Transvaginal POP mesh: warnings 2008/2011 → reclassified **class III (2016)** → sales stopped **2019**. The ban does **not** apply to slings or sacrocolpopexy.
- Multi-incision midurethral sling (MUS) and sacrocolpopexy: safety/efficacy established; single-incision mini-slings: not adequately demonstrated.
- Consent: tell every patient mesh will be used and give written information; use a standard consent for mesh removal; have a policy for returning explanted mesh.

2 · RISK FACTORS AND PREVENTION

- **Contraindications to vaginal mesh**: simultaneous urethrovaginal fistula repair, urethral injury/mesh perforation or diverticulectomy.

- Relative: prior fistula/diverticulum surgery, radiation, scarring, poor/oestrogen-deficient tissue, smoking, obesity, extremes of age. Men: prior radiotherapy, urethrotomy.
- POP mesh: high mesh volume (two compartments, mesh + sling), **shallow vaginal dissection**, concurrent or prior hysterectomy, diabetes, sexual activity.
- **Intraoperative urethral injury → abandon mesh**, repair, place sling later.

3 · EVALUATION OF SUSPECTED MESH COMPLICATION

ALGORITHM
History + **prior operative notes** (implant and excisions) → speculum and pelvic examination (exposure, bands, point tenderness, levator hypertonicity) → cystoscopy/urethroscopy if haematuria, recurrent urinary tract infection (UTI), storage lower urinary tract symptoms (LUTS) or voiding pain → urodynamics (UDS) if voiding dysfunction → imaging: translabial ultrasound locates sling; **MRI** shows sacrocolpopexy mesh (invisible on CT); CT to exclude haematoma.

Time since surgery never excludes a complication — slings can expose decades later.

4 · MIDURETHRAL SLING COMPLICATIONS

Vaginal exposure/extrusion

- Causes: delayed infection, haematoma separating the incision, excess tension, mesh placed too superficially. Present with bleeding, discharge, dyspareunia, hispareunia (partner pain), UTI, pain.
- Why exposure needs examination: any discharge, bleeding or partner pain in a woman with mesh may be exposure, even years later; asymptomatic small exposures may be found incidentally.
- Small early exposures may re-epithelialise with **topical oestrogen** (e.g. 1 g cream nightly 2 weeks then 2–3×/week; or estradiol tablet 10 µg).
- **Excision**: hydrodissect (1% lidocaine with adrenaline) → midline/horizontal incision, or **inverted-U flap** if not palpable → don't cut mesh on entry → locate (sound with metal instrument, ultrasound; may lie distal or beyond bladder neck) → expose full width lateral to urethra → pass instrument behind, divide ("divide and conquer") → excise ≥**1 cm** around defect → avoid blind lateral dissection (vulval haematoma, especially transobturator) → close.

Pain

- Risk: young age, fibromyalgia, pre-existing pelvic pain. Causes: shrinkage, tension, obturator neuralgia, muscle hypertonia.
- **Immediate severe pain** (trocar injury): image to exclude haematoma, then consider **early removal** — easier before ingrowth.
- Delayed focal tenderness: pelvic floor physical therapy (PFPT), trigger point injection; atrophy → vaginal oestrogen. Persistent point tenderness → excise painful segment (60–90% improve). Diffuse late pain → treat central sensitisation; excision may fail.
- For pain, remove **as much vaginal mesh as possible** (81% pain-free).
- Persistent **groin pain** after transobturator sling → combined vaginal + thigh incision through adductor brevis to obturator externus; orthopaedic help.

Bladder perforation

- Unrecognised trocar puncture (dome/lateral wall); risk: low body mass index, prior surgery. Cystoscope with a **full bladder**; if trocar seen, withdraw and repass laterally.
- Presents with storage LUTS, haematuria, UTI, stones on mesh, vesicovaginal fistula.
- **Endoscopic** (holmium laser, scissors, resectoscope; treat stones first): low morbidity but mesh remains in the wall — watch for re-extrusion.
- **Open/robotic transvesical**: follow arm from rectus fascia to bladder, excise 2 cm beyond defect, two-layer closure, catheter 2–3 weeks until cystogram normal; minimal re-extrusion; autologous sling if SUI.

Urethral/bladder-neck perforation

- From dissection/trocar injury, over-tensioning or **urethral dilation** (avoid — divide an obstructing sling instead). Voiding pain, retention, haematuria, UTI.
- Endoscopic excision leaves mesh in the wall. Definitive: inverted-U flap → divide uninjured arm → peel mesh off urethra → minimal urethrotomy → layered closure ± **Martius flap** (fistula, poor tissue) → non-overlapping suture lines; stricture → buccal urethroplasty.
- After excision: SUI **42%** (reduced by simultaneous autologous fascial sling); urgency common.

Voiding dysfunction after sling

- Obstruction → midline **sling division**: retention resolves in 95%, symptoms in 75%. Urgency incontinence persists in up to 76%; de novo in up to 45%.
- Haematomas after excision: expectant ± antibiotics; open drainage if expanding or haemodynamic compromise.
- Persistent symptoms: UDS distinguishes obstruction, detrusor underactivity, dysfunctional voiding; repeat with wider excision and urethrolysis.
- Recurrent SUI after division/excision 20–74%; bulking, repeat MUS or fascial sling.

> KEY POINT **How much to remove**: retention — midline division; exposure — ≥1 cm each side; pain — maximal vaginal removal, rarely arms. Remove enough to fix the problem; **never remove asymptomatic mesh**; more removal = more bleeding, recurrence of POP and SUI.

5 · TRANSVAGINAL POP MESH

- Native repair vs mesh (largest randomised controlled trial, n = 1358): no difference at 6 years in symptoms, quality of life or re-operation (12% vs 10%); 8% of mesh arm needed excision.
- Kits (e.g. Perigee, Apogee) used trocar arms through obturator or iliococcygeus — obtain the original operative note, as arm routes differ by product.
- Exposure 10–12%, removal 5–8%, up to 8 years later; **contraction** ~12% (shortening, tightness, dyspareunia); prominence/tender bands; bladder perforation 0.7–3.5%, rectal 0.15%, fistula 0.2%. Most have multiple symptoms.
- **Conservative** if exposure <1 cm, minimal symptoms, no infection: observation (no risk with macroporous mesh) or oestrogen; silver nitrate. Only **size** predicts success. Office trimming: 73% later need theatre.
- Pain without exposure: PFPT, analgesia, trigger point/nerve block first; excision unpredictable.
- **Excision**: examination under anaesthesia, cystoscopy/proctoscopy → hydrodissection → thick full-thickness flaps → find edges → clamp behind mesh and divide (most bladder/rectal injuries here) → dissect viscus off, cut arms under tension (bleeding; avoid chasing arms — haematoma) → native colporrhaphy → close.
- Sequelae: cystocele recurrence 19%, repeat excision 20%, vesicovaginal fistula 3%; multicompartment excision raises transfusion (1% → 9%). Chronic pain syndromes predict persistent pain; 45% abstain sexually.

6 · SACROCOLPOPEXY MESH

- Exposure probability ~**10.5% at 7 years**. Risks: total hysterectomy (23% vs 5%), stage 3–4, multiple procedures. Cervix-sparing has low risk.
- Presents with discharge, granulation, dyspareunia, palpable mesh; cuff abscess (fever, foul discharge, weeks–months); perforation into bladder/bowel (likely unrecognised cystotomy/enterotomy); **discitis**.
- Mesh is mostly inaccessible to vaginal examination → cross-sectional imaging.
- **Exposed suture**: cut below knot in office + oestrogen. Small mesh exposure: oestrogen 6–8 weeks (low success). Vaginal excision often fails (up to 3 attempts); partial colpocleisis (traction suture, ≥1 cm epithelial flap to well-ingrown mesh, excise maximally, invert remnant) succeeds in ~50%

but shortens the vagina — only for women not sexually active. Simple trimming fails because mesh burden is high at the apex.

- **Abdominal removal** for failed vaginal approach, intractable pain, infection, discitis, ureteric complication: **remove entire infected mesh** including sacral sutures; partial vaginectomy/trachelectomy often needed (74%); bleeding 200–1000 mL, bowel injury up to 29%. **Why complete removal**: residual infected mesh causes sinuses, abscess and fistula, and a second abdominal operation is even harder. Abdominal route also allows repair of ureteric obstruction or ureterovaginal fistula.

7 · MALE SLINGS

- Fewer complications: clean perineal wound, deep multilayer cover, thick ventral bulbar urethra.
- **Advance XP** (retrourethral, repositioning): retention 14–21%, usually days; incise one arm if persistent. Neobladder patients high risk.
- **Virtue** (compressive): perineal/scrotal pain 14–17%, chronic 0–7%.
- Bone-anchored InVance: divide or remove (anchors rarely) before artificial urinary sphincter (AUS).
- **Urethral extrusion**: retention, pain, haematuria — urethroscopy in any persistent retention; excise lateral to urethra, primary repair or ventral buccal graft; avoid transection.

8 · MCQ ONE-LINERS

1. Only recommended pelvic implant mesh: type I macroporous monofilament knitted polypropylene.
2. Type I pore size: >75 µm.
3. "Erosion" should not be used for mesh complications.
4. Exposure = mesh visible through unhealed vaginal epithelium.
5. Perforation = mesh into a hollow viscus.
6. Transvaginal POP mesh: class III 2016; withdrawn 2019.
7. Slings and sacrocolpopexy mesh remain class II.
8. Urethral injury during sling dissection: abandon mesh placement.
9. Best imaging for sacrocolpopexy mesh: MRI.
0. Cystoscopy for trocar perforation requires a full bladder.
1. Small vaginal exposure: observation or topical oestrogen.
2. Exposure excision margin: at least 1 cm around defect.
13. Only predictor of conservative success: exposure size.
14. SUI after intraurethral mesh excision: 42%.
15. Sling division resolves retention in 95%.
16. Avoid urethral dilation for post-sling obstruction.
17. Infected sacrocolpopexy mesh: remove entirely.
18. Total hysterectomy at sacrocolpopexy raises exposure (23% vs 5%).
19. Exposed sacrocolpopexy suture: cut below knot in office.
20. Never remove asymptomatic mesh.
21. Complication unique to POP mesh: contraction (shrinkage), ~12%.
22. Groin pain after transobturator sling: excise via thigh through adductor brevis.
23. Male sling extrusion usually follows unrecognised urethral perforation without intraoperative cystoscopy.
24. Highest retention risk after male retrourethral sling: neobladder patients.
25. Mesh dissection most dangerous step: passing clamp behind mesh (blind).

121 Urinary Tract Fistulae

Reading time ≈ 59 min (slow pace)

1 · DEFINITION AND UNIVERSAL PRINCIPLES

- **Urinary tract fistula**: abnormal epithelial- or mesothelial-lined connection between the urinary tract (kidney, ureter, bladder, urethra) and another cavity (bowel, vessel, genital tract) or skin.
- Causes: iatrogenic (commonest in the developed world), malignancy, inflammation, infection, ischaemia, trauma, parturition.
- A fistula persists because **urine keeps flowing through a tract that cannot heal**; continuous flow lets the tract epithelialise, which prevents spontaneous closure. Every principle either diverts that flow or restores healing capacity.
- The **first repair has the best chance** — optimise patient and surgeon factors before it.

Management (whole course) — Table 121.1	Operation (any route) — Table 121.2
Optimise nutrition	Evaluate concomitant injury; rule out malignancy
Control infection (less inflammation)	Adequate exposure; remove all foreign material
Continuous urinary drainage (catheter/stent)	Mobilise for tension-free closure; vascularised edges
Eliminate obstruction (remove or bypass)	Multiple, watertight, **non-overlapping** layers
Consider malignancy — **always biopsy** if any history	Strongly consider tissue interposition; proper drainage after

> **KEY POINT** Trap: a fistula may be the only sign of recurrent cancer — biopsy the tract in anyone with prior malignancy or radiation.

2 · VESICOVAGINAL FISTULA (VVF) — AETIOLOGY

VVF is the commonest acquired urinary fistula (**≥75%** of genitourinary [GU] fistulae).

	Developing world — obstetric (>90%)	Developed world — iatrogenic
Cause	Obstructed labour, no access to operative obstetrics; young age, small pelvis, poor nutrition	Unrecognised surgical injury; **≥75%** after hysterectomy
Mechanism	Fetal head impacted for days → pressure ischaemia → necrosis → large devascularised defect	Dissection, cautery, clamp crush, or bladder wall caught in cuff closure
Extent	Level of arrest decides; trigone, urethra, ureter often involved; "obstructed labour injury complex" (renal, rectal, orthopaedic, neurological, psychosocial)	Usually solitary, **supratrigonal at the vaginal cuff**

- Common pathway: devascularisation → ischaemia → necrosis → breakdown; urine then flows from the high-pressure bladder to the low-pressure vagina, keeping the tract open.
- **Hysterectomy**: VVF ≈ 0.8/1000 overall; lowest vaginal, highest laparoscopic (highest 30-day rate after laparoscopically assisted vaginal hysterectomy [LAVH]). Risk rises with anatomical distortion (endometriosis, fibroids, prior surgery/radiation, malignancy, bleeding) and concomitant prolapse/incontinence/mesh surgery. Most VVF follow **unrecognised** injury; immediate recognition and repair prevents fistula.
- **Universal cystoscopy at hysterectomy**: increases intraoperative detection, but effect on delayed injury is inconsistent; adds time and urinary tract infection (UTI). American Urogynecologic Society (AUGS): routine with any prolapse/anti-incontinence procedure; American Association of Gynecologic Laparoscopists (AAGL): routine after laparoscopic/robotic hysterectomy; American College of Obstetricians and Gynecologists (ACOG): no recommendation after benign hysterectomy. It does not replace careful dissection.
- **Mesh**: foreign body sustains inflammation and keeps the tract open.

- **Radiation**: obliterative endarteritis + lamina propria fibrosis → necrosis → ulceration → fistula. 0.6–2% after cervical cancer radiotherapy; typically **1.5–2 years** later (up to 30); dose-dependent, worse in smokers.
- **Bevacizumab** (anti-vascular endothelial growth factor [VEGF]) raises fistula risk (hypoxia). Primary/recurrent cancer itself; rare: pessary, intrauterine device, infection, lasers, endometriosis.

3 · CLASSIFICATION OF VVF

>25 systems, mostly obstetric; World Health Organization (WHO) uses simple vs complex.

Axis	Goh	Waaldijk
Basis	Distance of distal edge from external urethral meatus	Closing (continence) mechanism and urethral involvement
Types	1: >3.5 cm · 2: 2.5–3.5 · 3: 1.5–2.5 · 4: <1.5 cm	I: closing mechanism spared · IIA: no (sub)total urethral involvement · IIB: (sub)total urethral involvement (A/B = without/with circumferential defect) · III: miscellaneous (ureteric)
Size	a <1.5 · b 1.5–3 · c >3 cm	Small <2, medium 2–3, large 4–5, extensive ≥6 cm
Tissue	i none/mild fibrosis · ii moderate/severe fibrosis · iii special (radiation, ureteric, circumferential, previous repair)	—

> **KEY POINT** Trap: Goh type 1 is farthest from the meatus (best), type 4 closest (worst). Closer to the meatus = worse closure and worse continence. Goh predicts closure better. In practice always describe size, location (trigonal vs supratrigonal), tissue loss, ureteric and urethral involvement.

4 · DIAGNOSIS OF VVF

Clinical

- Classic: **continuous leakage day and night**, independent of position, often without urge; large tract → no desire to void (bladder never fills). Small VVF → positional leak only. Also recurrent UTI, haematuria, "vaginal discharge".
- **Timing points to mechanism**: missed/failed cystotomy → first few days; thermal or crush injury → **7–10 days**; radiation → years.
- Red flags (fever, flank pain, ileus) → urinoma/ureteric injury → manage immediately.
- Examination: speculum for site (vaginal apex commonest post-hysterectomy); vaginal fluid creatinine ≈ urine creatinine confirms urine. If no defect seen: methylene blue backfill → tampon and walk 15–60 min. Biopsy friable tissue if prior cancer/radiation; examination under anaesthesia if stenosed. Vaginal oestrogen for atrophy. Judge tissue quality and likely route.
- Optimise: smoking cessation, glycaemic control, nutrition; review operative notes.

Dye tests

- **Single dye**: backfill ~300 mL methylene blue → tampon → ambulate. Proximal stain → VVF; distal → urethrovaginal fistula or stress urinary incontinence (SUI); dye from cervical os → vesicouterine fistula.
- **Double dye test**: oral phenazopyridine 100–200 mg (urine orange in 1 h) + bladder blue → tampon.

Tampon stain	Source	Meaning
Orange	Kidney	Ureterovaginal fistula (UVF)
Blue	Bladder	VVF
Both	Both	Coexisting VVF + UVF

Endoscopy

- **Cystoscopy is integral** (sensitivity ~92%): size, number, relation to ureteric orifices and bladder neck, mesh, capacity, biopsy. Small tracts: probe suspicious erythema/granulation with a guidewire or 5-Fr open-ended catheter.
- Cystoscopy under anaesthesia allows retrograde pyelography and stenting. **Vaginoscopy** shows the vaginal side, allows directed biopsy, overcomes radiation stenosis.

Imaging and urodynamics

- Cystogram/voiding cystourethrogram (VCUG): immediate vaginal filling, best on lateral view; shows concomitant urethrovaginal fistula. Magnetic resonance imaging (MRI, T2) for difficult cases.
- **Upper tract imaging is mandatory** — ureteric injury coexists in up to **12%** of VVF. Computed tomography urogram (CTU) with delayed phase; MR urography if contrast allergy/renal failure. **Partial ureteric obstruction with a VVF = UVF until proven otherwise.** Unclear CTU → retrograde pyelogram + stent; fails → percutaneous nephrostomy (PCN).
- Urodynamics (UDS)/video urodynamics (VUDS) rarely needed for diagnosis; plan concomitant procedures: 10–15% have SUI (autologous sling); radiation-reduced capacity/compliance (augmentation/diversion). Occlude a large fistula with a Foley balloon or vaginal pack to fill the bladder.

> **KEY POINT** The work-up answers three questions in order: is the fluid urine (creatinine), where does it come from (bladder, ureter, urethra, uterus — dye tests, endoscopy), and what else is damaged — the last is why upper tract imaging is compulsory even when the VVF is obvious. A PCN placed preoperatively also allows antegrade ureteroscopy during repair to locate the ureter in scar.

5 · VVF — PROGNOSTIC FACTORS AND TIMING

- No consensus definition of success; anatomical closure ≠ continence.
- **Size**: <3 mm best chance of spontaneous closure; <10 mm may close with drainage. Larger (≈ >3 cm; ≥1.5 cm for residual incontinence in obstetric fistula) predicts surgical failure.
- **Location**: most post-hysterectomy VVF are supratrigonal, midline. Within **1 cm** of a ureter → ureteric stent (identification, protection from oedema). Ureteric involvement → reimplant → abdominal route. Bladder neck involvement → reconstruction ± pubovaginal sling. Apical VVF (8–10 cm from introitus) is beyond Martius reach → peritoneal/omental flap.

Setting	Timing of repair	Rationale
Obstetric (traditional)	≥3 months, catheter meanwhile	Inflammation settles, necrosis sloughs
Iatrogenic, clean tissue	Early acceptable; European Association of Urology (EAU): no minimum delay if no infection or recent radiation	Comparable closure; less distress
Radiation-induced	≥**12 months** after radiation ends	Late radiation injury still evolving

Radiation is one of the strongest predictors of failure and brings ureteric stricture, poor compliance and intrinsic sphincter deficiency — assess the whole lower tract; consider diversion.

6 · VVF — CONSERVATIVE AND ENDOSCOPIC TREATMENT

- **Catheter drainage** cures ~15%. Favourable: small (<1 cm), catheter within **3 weeks** of injury, fresh injury. Unfavourable: large, ≥6 weeks (tract epithelialised), radiation or malignancy (never heal — vascular damage). EAU: if attempted, trial ≥**12 weeks**.
- **Fulguration** (O'Conor and Sokol) + 2 weeks' drainage for tracts <3.5 mm — added value uncertain.

- **Sealants** (fibrin glue/plug, collagen, BioGlue, cyanoacrylate): tiny series; **not recommended by EAU**; possible for <1 cm VVF in poor surgical candidates. Problems: catheter adherence, spasm, stone on glue.

7 · VVF — SURGICAL REPAIR

Preparation and counselling

- Sterile urine; biopsy if malignancy history; nutrition. Counsel: vaginal shortening/dyspareunia (esp. Latzko), prolonged catheter, spasms, persistent incontinence despite closure, failure.
- Tract excision and interposition are not proven superior in simple VVF but are valued when recurrence risk is high (irradiated, recurrent, poor tissue).

Choosing the route (Table 121.4)

	Abdominal (open / laparoscopic / robotic)	**Vaginal**
Location	Supratrigonal; narrow vagina; apex without descensus	Trigonal/distal; accessible apex
Ureters	Reimplant needed or orifice close	Ureters distant
Adjuncts	Augmentation, reimplant, diversion, bladder flap	Urethral reconstruction, pubovaginal sling
Flaps	Omentum, peritoneum, epiploic appendage, rectus	Martius, peritoneal, gracilis
Sexual function	Unchanged	Risk of shortening, dyspareunia
Other	Cannot be placed in lithotomy	Dense abdominal adhesions; no general anaesthesia; shorter time and stay

- No randomised controlled trials (RCTs); first-attempt closure ≈ **90%** either route. Transvaginal: less blood loss, shorter stay and pain — some call it the gold standard unless contraindicated. Best route = the one the surgeon is most confident with.
- Proposed abdominal indications: large (≥4 cm primary, ≥2 cm recurrent), urethral/bladder-neck or ureteric involvement, ancillary abdominal procedures, non-capacious vagina, radiation.

Transvaginal flap-splitting (layered) repair — Table 121.5

ALGORITHM
Lithotomy → cystoscopy (stent ureters if close; guidewire through tract) → ± suprapubic catheter + urethral catheter → episiotomy if narrow → Foley ≥8 Fr through fistula for traction → hydrodissect → **J- or U-incision** circumscribing tract with 1–3 mm cuff → raise anterior and posterior vaginal flaps widely → layer 1: 3-0/4-0 delayed absorbable (bladder wall + perivesical tissue) → layer 2: imbricating, **perpendicular** to layer 1 → watertight test 200–300 mL dyed fluid → ± Martius/peritoneal flap → advance opposite vaginal flap ≥2 cm beyond (layer 3) → cystoscopy for efflux

- **Postoperative drainage is the most important part**: urethral ± suprapubic catheter ≥10–14 days; ± cystogram before removal. Two catheters guard against blockage.
- Logic: tagging stitch, perpendicular layers and flap advancement all prevent overlapping suture lines; drainage protects the repair from the pressure that created the fistula.

Latzko procedure — high partial colpocleisis

- Denude vaginal epithelium **1–2 cm** circumferentially around the tract (bladder not opened, tract not excised) → close vaginal walls in 2+ layers of superficial absorbable sutures, imbricating the defect → epithelial closure. ± Martius (denude wider).

- Short, minimal blood loss, no ureteric dissection; ≥90% in non-irradiated post-hysterectomy VVF. Favoured in **radiation** or when orifices are close. Drawback: **vaginal shortening/dyspareunia**.
- **Cuff excision**: circumferential incision beyond apical scar → funnel towards bladder → excise all scar → layered closure; very high cure in non-irradiated post-hysterectomy VVF.

Abdominal repair — O'Conor (suprapubic extraperitoneal bivalving), Table 121.7

ALGORITHM
Low lithotomy, ureteric catheters → low midline, extraperitoneal bladder mobilisation → sagittal cystotomy at dome → **bivalve** posteriorly down to and around the fistula → excise tract → dissect bladder off vagina ≥2–3 cm beyond → close vagina → **omentum/peritoneum** fixed beyond vaginal line → ancillary procedures → close bladder in 1–2 layers → suprapubic ± urethral catheter, fill test → drainage 10–14 days, cystogram

- Failure causes: carcinoma ± radiation, wound infection, inadequate mobilisation. More de novo urgency urinary incontinence (UUI) than vaginal repair (long bladder suture lines).

	O'Conor	Transvesical (limited cystotomy)	Extravesical
Cystotomy	Dome to fistula	Limited	None
Orifices seen	Yes	Yes	No — ureteric risk
Strength	Best plane identification	Less bladder irritation	Fewer suture lines, less LUTS
Weakness	More lower urinary tract symptoms (LUTS)	Restricted mobilisation, limited interposition	Scarred plane; infratrigonal fistula hard

Postoperative care (all routes)

- Continuous catheter drainage 10–14 days (urethral ± suprapubic); ± cystogram before removal.
- Bladder spasms are expected and self-limiting — liberal anticholinergics or β3-agonists.
- Optional pelvic drain (check drain creatinine for urinoma); stents out at 4–6 weeks.
- †Pelvic rest (nothing per vagina) for ~6 weeks.

Laparoscopic and robotic repair

- Same closure as open with less blood loss, stay and analgesia; robot eases intracorporeal suturing. Contraindication: dense adhesions.
- **Robotic O'Conor**: cystoscopy, catheter through tract, ureteric stents, vaginal stent/sponge → vesicovaginal dissection as for sacrocolpopexy (backfill 150–300 mL) → open dome, extend cystotomy to fistula to find the fused plane → dissect 1–3 cm beyond → close vagina **transversely** → interpose omental J-flap/epiploic appendage/peritoneum → close bladder **vertically** (lines cross, not overlie) → leak test → catheter 10–14 days, cystogram; stents out 4–6 weeks.
- Extravesical (no cystotomy) suits supratrigonal VVF; transvesical suits low-lying fistulae. Indocyanine green (ICG) assesses edge vascularity.

8 · INTERPOSITION FLAPS

Rationale: reinforces repair, fills dead space, imports blood and lymphatic supply. Indicated in **complex VVF** — radiation, recurrent, ischaemic obstetric, poor tissue. An RCT in simple VVF showed **no benefit of Martius** and frequent donor-site pain/numbness.

- **Martius (labial fibrofatty) flap**: supply — internal pudendal (inferior), external pudendal (superior), obturator (lateral). Vertical labial incision 4–7 cm → mobilise pad lateral to bulbocavernosus down to Colles fascia → for VVF keep the **superior** pedicle (inferior for rectovaginal) → tunnel ~2 cm under vaginal wall → fix tension-free over repair → labial closure with drain 48 h. Reaches mid-distal VVF, bladder neck, urethra — **not the apex**.
- **Vaginal peritoneal flap**: posterior dissection to the pouch of Douglas reflection, advanced over an apical repair.

- **Omentum** (gastroepiploic arteries, lymphatic, robust J-flap) — radiation, recurrent, large VVF. Abdominal peritoneal flap or sigmoid epiploic appendages if omentum is short. Rectus abdominis, gracilis, Singapore flap — last resort/large tissue loss.

9 · OUTCOMES AFTER VVF REPAIR AND DIVERSION

- Residual urgency/UUI common, sometimes SUI. Residual incontinence after obstetric repair: Goh 3–4, >1 repair, positive cough stress or pad test; it drives quality of life (QoL) and depression. International Continence Society (ICS): pelvic floor physiotherapy for all obstetric fistula and persistent symptoms.
- Retention uncommon (vertical bladder closure). Sexual function improves overall but pain/dyspareunia persist; small vaginal calibre and fistula ≥3 cm predict dysfunction.
- **Diversion** when repair is unfeasible (multiple failures, severe radiation, massive tissue loss): conduit/continent reservoir; unfit → ureteric occlusion + nephrostomy; resource-poor → ureterosigmoidostomy.

10 · URETEROVAGINAL FISTULA (UVF)

- Usually unrecognised injury to the **distal ureter** at (often uneventful) abdominal hysterectomy — ureter passes under the uterine artery. Up to **25%** of VVF have a coexisting UVF. Delayed recognition raises fistula risk ~4-fold.
- Injury sites: pelvic brim (infundibulopelvic ligament), under uterine artery, lateral to uterosacral, anterolateral fornix near ureterovesical junction. Most within distal 6 cm.
- Prevention: anatomy. Preoperative intravenous pyelogram (IVP) does not help; prophylactic stents — selective (e.g. caesarean hysterectomy for placenta accreta).

	VVF	UVF
Voiding	Reduced/absent	**Normal voiding plus constant leak** (other ureter fills bladder)
Onset	Days to 7–10 days	Within ~4 weeks, often fever, flank pain, ileus
Upper tract	Usually normal	Hydronephrosis/obstruction common
First line	Catheter (selected), repair	Ureteric stent, then reimplant

- Diagnosis: double dye test (orange), CTU, retrograde pyelogram (abrupt cut-off or extravasation → attempt stent); exclude VVF with cystogram/cystoscopy.
- **Stent first, as early as possible** (retrograde or antegrade); success much higher within 2 weeks; leave 4–6 weeks then reimage.
- Stent fails → **ureteroneocystostomy** ± psoas hitch / Boari flap (preserve adventitia); ileal ureter or nephropexy for length. Early vs ≥6-week repair: equal success. ≥90% success; laparoscopic/robotic equal to open.
- Follow-up: ultrasound after stent removal; stricture can appear up to 30 months.

ALGORITHM
Suspect UVF → exclude VVF + confirm with CTU/retrograde pyelogram → stent → remove 4–6 weeks → reimage → resolved, or persistent/stent fails → ureteroneocystostomy

KEY POINT UVF drains past a normally filling bladder (wet but voids normally); because the ureter heals over a stent if reached early, the interval from detection to stenting largely decides whether the patient needs an operation.

11 · VESICOUTERINE FISTULA (VUF)

- Rising with caesarean rates; vast majority after **low transverse caesarean section** (bladder caught in hysterotomy closure or occult bladder injury apposed to uterine incision).
- Site: uterine body above internal os ↔ posterior bladder. Competent cervix keeps urine in uterus → **continence preserved**.

- **Youssef syndrome**: cyclical haematuria (menouria), amenorrhoea, first-trimester miscarriage, continence. Incontinence only if tract is below internal os or cervix incompetent. Differential of menouria: bladder endometriosis.
- Diagnosis: dye from cervical os; cystoscopy (posterior supratrigonal defect; scope may enter uterus — both tubal ostia seen); cystogram, hysterosalpingography, MRI; image upper tract.
- Spontaneous closure rare (~5%). Wait 3–4 months after caesarean (uterine involution). Transabdominal (increasingly minimally invasive) separation of bladder and uterus, layered closure with omentum/peritoneum interposed (uterus-sparing) or hysterectomy if family complete. Martius rarely reaches. After uterus-sparing repair: contraception until healed, deliver by caesarean.

12 · URETHROVAGINAL FISTULA

- Developing world: obstructed labour (often with VVF). Developed world: **urethral diverticulectomy** (20–25% of cases), slings, anterior repair/mesh, bulking, catheter pressure necrosis.
- Symptoms depend on position relative to sphincter: distal → often asymptomatic (split stream, vaginal voiding); proximal → continuous leak or SUI. Hispareunia after mesh.
- Diagnosis: speculum, dye test, lateral VCUG, **flexible** urethroscopy (better distension than rigid). ~20% have a separate VVF.
- Repair: transvaginal inverted-U/J flap, minimal tract excision, two perpendicular layers over a urethral sound (4-0), **Martius flap** commonly, then advance vaginal flap; buccal graft for large defects; tubularised bladder flap for near-total urethral loss. Remove all mesh in field.
- SUI: concomitant **autologous** pubovaginal sling — never synthetic (erosion of poorly vascularised urethra) — or staged.
- Postoperative: suprapubic ≥14 days, lateral VCUG before removal, nothing per vagina 6 weeks. Closure 90–95% but continence only 50–80%; stricture possible.

13 · RECTOURETHRAL FISTULA (RUF)

- Mostly iatrogenic after prostate cancer treatment: radical prostatectomy (RP), external beam radiotherapy ± brachytherapy (combined worst), cryotherapy, high-intensity focused ultrasound (HIFU; salvage HIFU highest). Rectal injury at RP: open/laparoscopic > robotic.
- Symptoms: **faecaluria, pneumaturia**, recurrent UTI, urine per rectum.

Martini	Category
Stage (size)	I ≤1.5 cm · II >1.5 cm · III any size with sphincter damage
Grade (tissue)	0 surgical injury only · 1 prior radiation/energy · 2 salvage RP or salvage ablation

- Work-up: digital rectal examination, cystoscopy + sigmoidoscopy with **biopsy**, VCUG/retrograde urethrogram, upper tract imaging, MRI if complex; assess sphincter (severe SUI may need later artificial urinary sphincter [AUS], higher erosion).
- Conservative: no sepsis/faecaluria → catheter ± colostomy, low-residue diet; simple fistula may close in ~12 weeks. Irradiated/complex rarely close.
- Surgery: **diverting colostomy usually first**.

Approach	Essentials	Pros / cons
Transperineal	Separate urethral and rectal closure; **gracilis flap** interposed (or dartos, buccal graft)	High closure, good for irradiated; urinary incontinence common
York-Mason (transsphincteric, posterior)	Prone jackknife; divide anal sphincter in tagged layers; open rectum; excise tract; close urethra, rectal muscle, rectal mucosa separately	Unscarred field, preserves continence/sexual nerves; poor bulbar exposure, no flap except rectal advancement; faecal incontinence risk

Approach	Essentials	Pros / cons
Abdominal (open / robotic)	Pouch of Douglas → separate rectum and urinary tract; omental/peritoneal flap; ± combined perineal gracilis	Large/irradiated; morbidity (open)

Irradiated/ablated RUF: mostly abdominal or combined with vascularised flap; fallback is cystoprostatectomy + ileal conduit. Tissue (grade), not size, governs repair.

14 · VESICOENTERIC FISTULA (VEF)

- Colovesical commonest; male : female **3 : 1** (uterus shields bladder; prior hysterectomy common in affected women).
- Causes: **diverticulitis** (commonest) > **Crohn disease** (ileovesical) > colorectal cancer (10–15% of colovesical fistulae have neoplasm), bladder/prostate cancer, radiation, appendicitis. †Typical site: sigmoid to left posterolateral bladder/dome.
- Symptoms: **pneumaturia** (50–70%), faecaluria, recurrent polymicrobial UTI. **Gouverneur syndrome**: suprapubic pain, frequency, dysuria, tenesmus.

Test	Role
Poppy seed test	Highest detection of the presence of fistula; cheap; no location
CT (preferred imaging)	Air in bladder, focal wall thickening, adherent bowel, contrast in bladder; shows cause
Cystoscopy	Nonspecific erythema/oedema; variable detection
Colonoscopy	Poor at finding tract; **excludes cancer**
MRI	Crohn disease, anatomy

- Conservative: minimal symptoms, benign (esp. Crohn — anti-inflammatory/immunomodulators, nutrition). Spontaneous closure <2%.
- Surgery: **resect the diseased bowel segment with the fistula** — single-stage with primary anastomosis (± loop ileostomy), increasingly laparoscopic; staged (Hartmann) for abscess, gross contamination, advanced cancer, radiation. Bladder side: small defect may be left or oversewn; **partial cystectomy** if malignancy. Interpose omentum.

> **KEY POINT** VEF is a bowel disease presenting in the bladder — cure lies in removing the diseased bowel; the bladder usually heals itself.

Renocolic fistula: spontaneous ones mostly renal origin (xanthogranulomatous pyelonephritis [XGP], chronic infection) → nephrectomy ± bowel resection; small iatrogenic (colonic injury at percutaneous nephrolithotomy [PCNL]) → conservative (antibiotics, stent or nephrostomy, nil by mouth).

15 · UROCUTANEOUS FISTULAE

- **Vesicocutaneous**: commonest after long-term suprapubic tube; also open prostatectomy, radiation. Persistence: epithelialised tract, infection, foreign body, tumour, **bladder outlet obstruction (BOO)**. Treat: drain abscess, catheter drainage, relieve BOO; fistulectomy/partial cystectomy if non-healing.
- **Urethrocutaneous fistula (UCF)**: commonest complication of hypospadias repair (up to 20%). Wait ≥**6 months**; exclude distal stricture. <1 cm → local closure; ≥1 cm or redo → vascularised regional flaps; **tunica vaginalis** cover better than dartos; massive loss → gracilis; neurogenic with pressure ulcers → ileal conduit.
- **Nephrocutaneous**: stones/staghorn, XGP, prior surgery, tuberculosis (≥10%); kidney non-functioning in ~90% → **nephrectomy**; contrast CT for diagnosis.

16 · PUBOVESICAL FISTULA (PVF)

- Tract from bladder/prostate/urethra to pubic symphysis → **osteomyelitis**. Typical: prostate radiotherapy + repeated endoscopic outlet procedures (transurethral resection of prostate [TURP], bladder neck incision, stricture treatment); median ~10 years after radiation; also after AUS.
- Symptom: **pubic pain on walking**; abscess, urosepsis, cutaneous sinus.
- **MRI is most sensitive and specific**. CT-guided bone biopsy for culture (Candida, Enterococcus, Pseudomonas; urine culture unreliable).
- Conservative usually fails. Surgery: debride/resect infected bone, excise fistula, divert urine, fill with rectus or omental flap; 4–6 weeks antibiotics. Primary repair only if sphincter functional, bladder normal, no distal obstruction; otherwise **anterior exenteration (cystoprostatectomy) + diversion**.

17 · UROVASCULAR FISTULAE

- **Arteriocalyceal fistula**: after PCNL, transplant, trauma; often with pseudoaneurysm or arteriovenous fistula → **angioembolisation**.
- **Ureteroarterial fistula (UAF)**: emergency. Secondary (85%) — long-term ureteric stents, pelvic surgery, radiation, cancer (cervix, bladder, colorectal), vascular grafts; common iliac artery commonest. Presentation: **haematuria (often intermittent)**, ~20% unstable.
- Diagnosis: **angiography** (sensitivity rises with provocation — stent removal under angiography — done only with endovascular treatment ready); cystoscopy shows pulsatile bleeding from one orifice; CT angiography insensitive. Negative test does not exclude.
- Treatment: **endovascular stent graft** ± internal iliac embolisation; open repair (bypass, ligation, nephrectomy) for infection/abscess/enteric involvement.

18 · RENAL URINE LEAK AND ANASTOMOTIC LEAK

- **Post-trauma**: leak after 1–7% of renal injuries (~quarter of grade IV–V). Multiphase CT with 10-minute delayed phase; non-opacified distal ureter → suspect ureteropelvic junction avulsion → retrograde pyelogram. Non-operative; most resolve; persistent leak/urinoma → stent or nephrostomy.
- **After partial nephrectomy** (0.5–20%; low after robotic): pelvicalyceal contact (central tumour) predicts. Drain creatinine, CT. Drain alone ± stent/nephrostomy; refractory → glue, embolisation.
- **Vesicourethral anastomotic leak after RP** — Patil grade I extraperitoneal <6 cm, II extraperitoneal ≥6 cm, III intraperitoneal. Prolonged catheter (traction, reposition) → passive drain/withdraw drain → stents/nephrostomies → reoperation last.

19 · FISTULA AFTER GENDER-AFFIRMING SURGERY

- Transmasculine (phalloplasty, metoidioplasty): urethrocutaneous fistula and stricture common (within ~3 months); **primary colpectomy** roughly halves fistula rate. Small → catheter; persistent → excision/regrafting.
- Transfeminine (vaginoplasty): urethroneovaginal or vesiconeovaginal fistula rare (bladder injury); suprapubic diversion → repair with interposition.

20 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Martius flap in simple obstetric VVF	RCT, repair ± Martius	No difference in closure; ~⅓ donor-site pain/numbness	Reserve Martius for complex fistula
Prophylactic ureteric stents in gynaecological surgery	RCT (+ meta-analysis)	No/small reduction in injury (number needed to treat 224)	Selective, not routine, stenting

21 · MCQ ONE-LINERS

1. Commonest acquired urinary fistula: VVF; developed world cause — hysterectomy.
2. Obstetric VVF mechanism: prolonged fetal head impaction → pressure necrosis.
3. Post-hysterectomy VVF typical site: vaginal cuff, supratrigonal.
4. Thermal/crush injury VVF presents at 7–10 days.
5. Double dye test: orange tampon = ureterovaginal; blue = VVF.
6. Phenazopyridine colours urine orange.
7. Coexisting ureteric injury in VVF: up to 12% — upper tract imaging mandatory.
8. Goh type 4: distal edge <1.5 cm from meatus — worst prognosis.
9. Waaldijk classification is based on closing mechanism involvement.
10. Catheter drainage cures ~15%; best if <1 cm and within 3 weeks.
11. EAU: conservative catheter trial ≥12 weeks; sealants not recommended.
12. Radiation VVF: delay repair ≥12 months after radiation.
13. Latzko = high partial colpocleisis; main drawback vaginal shortening.
14. O'Conor repair: transvesical bivalving to fistula, extraperitoneal.
15. Martius flap retained pedicle for VVF: superior (external pudendal).
16. Martius cannot reach apical VVF — use peritoneal or omental flap.
17. Omental flap supply: right and left gastroepiploic arteries.
18. Most important part of VVF repair: uninterrupted postoperative bladder drainage.
19. UVF: normal voiding with continuous leakage.
20. First-line UVF treatment: early ureteric stenting.
21. Youssef syndrome: menouria, amenorrhoea, continence — after caesarean section.
22. Commonest developed-world cause of urethrovaginal fistula: diverticulectomy.
23. Never place synthetic sling with urethrovaginal fistula repair.
24. York-Mason approach: posterior transsphincteric, prone jackknife.
25. Transperineal RUF repair uses gracilis interposition.
26. Martini grade 1–2 RUF: prior radiation/energy or salvage therapy.
27. Commonest cause of colovesical fistula: diverticulitis.
28. Most sensitive test for presence of VEF: poppy seed test.
29. Preferred imaging for VEF: CT (air in bladder).
30. Gouverneur syndrome: suprapubic pain, frequency, dysuria, tenesmus.
31. PVF: pubic pain on walking after radiation and outlet surgery; MRI best.
32. Ureteroarterial fistula commonest artery: common iliac; treat with stent graft.
33. Commonest complication of hypospadias repair: urethrocutaneous fistula.
34. Nephrocutaneous fistula: kidney non-functioning in ~90% — nephrectomy.

122 Bladder and Female Urethral Diverticula

Reading time ≈ 38 min (slow pace)

1 · BLADDER DIVERTICULA — DEFINITION AND PATHOLOGY

- **Bladder diverticulum**: herniation of urothelium through the muscularis propria, forming a thin-walled sac connected to the lumen by a neck (ostium).
- Wall: mucosa, lamina propria, scattered non-functional muscle fibres, adventitia; an outer fibrous pseudocapsule gives a surgical plane.
- **No functional muscle → the sac cannot contract** → poor emptying, high postvoid residual (PVR), stasis → infection, stones, malignancy.
- Neck is ringed by circular middle-detrusor fibres → narrow, sphincter-like ostium (the target of endoscopic neck resection).
- ~90% in adults; male : female ≈ **9 : 1**.
- Cellules → saccules → diverticula: a continuum of rising voiding pressure (no agreed size cut-offs).

Feature	Congenital	Acquired (secondary)
Age	Childhood (peak <10 years)	Men >60 years
Number	Usually solitary	Often multiple
Site	Posterolateral to ureteric orifice, ~90% near ureterovesical junction (UVJ); vesicoureteral reflux (VUR) common	Ureterovesical hiatus commonest, but anywhere
Cause	Congenital detrusor weakness at UVJ	**Bladder outlet obstruction (BOO)** or neurogenic dysfunction — almost always present
Bladder wall	Smooth, no trabeculation	Trabeculated
Presentation	Acute urinary tract infection (UTI)	Incidental, during work-up of lower urinary tract symptoms (LUTS), haematuria, UTI
Malignancy	Not increased	Increased

- Congenital: up to 60% have a syndrome, neuropathic bladder or BOO — **Menkes** (copper, kinky hair), **Williams**, **Ehlers-Danlos**, fetal alcohol; also prune-belly, posterior urethral valves (PUV). Chromosomal testing not needed for a single simple lesion.
- Acquired: ~**70%** with benign prostatic hyperplasia (BPH); also bladder neck obstruction, stricture. In women look for dysfunctional voiding, prolapse, primary bladder neck obstruction, post-sling obstruction.
- **Hutch diverticulum**: paraureteric, in neuropathic bladder with VUR (paraplegics). Iatrogenic: at cystotomy or ureteric reimplant site.

> **KEY POINT** Congenital = weak wall at normal pressure; acquired = normal wall at high pressure — so an acquired diverticulum always sends the search back to the outlet.

2 · BLADDER DIVERTICULA — EVALUATION

- Symptoms nonspecific (stasis or mass effect): incomplete emptying, double voiding; rare — iliac vein compression, diverticulum within inguinal hernia. Persistent pyuria despite antibiotics may itself justify excision.
- Work-up: history, digital rectal examination, prostate-specific antigen (PSA), urinalysis, culture, **cytology**, endoscopy and imaging of upper and lower tract; search for BOO or occult neuropathy.
- **Voiding cystourethrogram (VCUG)**: anteroposterior, oblique, lateral views; size, VUR (up to 13% of congenital), and whether the sac empties. **Paradoxic enlargement**: during voiding contrast flows from high-pressure bladder into low-pressure sac, which refills the bladder afterwards.

- Computed tomography (CT)/magnetic resonance imaging (MRI): number, anatomy, masses (filling defects), relation to ureters and rectum; needed if the neck is obstructed and contrast cannot enter.
- Upper tract: hydroureteronephrosis (HUN) in ~7% adults (up to 30% upper tract abnormality in children). Causes: ureteric compression, the underlying urodynamic problem, VUR. Pelvic ureter usually deviated **medially**. A ureter entering a diverticulum loses muscular backing → short tunnel → reflux.
- **Video urodynamics (VUDS)**: confirms BOO. **Pressure-sink effect**: detrusor decompresses into the sac, so contractility appears falsely low; raised PVR may just be diverticular urine.
- **Cystoscopy**: inspect the whole sac (flexible scope), cytology from sac, biopsy with extreme care — thin wall perforates, seeding tumour into perivesical fat. A tumour can be missed on CT.
- VUR with a congenital diverticulum: reimplantation is now **selective** — for renal scarring or deteriorating function — rather than routine.
- An incidental diverticulum in an adult always prompts evaluation for BOO or occult neuropathy, because failure to treat the cause before or at surgery leads to recurrence and postoperative voiding problems; treating the cause may itself make excision unnecessary.

> **KEY POINT** The diagnostic work-up has two jobs: find the cause (outlet or neurogenic) and exclude cancer inside a sac that imaging, endoscopy and urodynamics each see imperfectly — so the tests are combined rather than any one relied upon.

3 · MALIGNANCY IN BLADDER DIVERTICULA

- Stasis (carcinogen contact) + chronic inflammation. Tumour prevalence 0.8–10%; peak 65–75 years.
- Histology: urothelial carcinoma **70–80%**, squamous cell carcinoma (SCC) 20–25%.
- **No muscularis propria** → early transmural/extravesical spread, risk of perforation and seeding at transurethral resection (TUR), and loss of the layer against which invasion is staged. Clinical stage is the key prognostic factor; upstaging common.
- Treatment (no consensus): **radical cystectomy** — same survival as non-diverticular tumours stage-for-stage; partial cystectomy/diverticulectomy for selected low-grade, low-stage tumours; open excision without prior TUR favoured by some; TUR + intravesical therapy role undefined.

> **KEY POINT** Treat a diverticular tumour as potentially invasive from the outset — the missing muscle layer explains early spread, perforation risk and inaccurate staging.

4 · BLADDER DIVERTICULA — MANAGEMENT

ALGORITHM
Diverticulum → history, cystoscopy, cytology, VCUG → suspicious lesion? biopsy → cancer therapy → no lesion: upper tract imaging → HUN or symptoms/complications → urodynamics → treat lower urinary tract abnormality (BOO) → reassess → persistent HUN/symptoms due to diverticulum → good candidate: diverticulectomy; poor candidate: clean intermittent catheterisation (CIC) → asymptomatic, uncomplicated: surveillance

- **Indications for diverticulectomy**: refractory related LUTS, chronic relapsing UTI, **stones**, **carcinoma or premalignant change**, upper tract deterioration (obstruction or VUR). **Size alone is not an indication.**
- **Treat the outlet first or simultaneously** — otherwise recurrence. With small diverticula, relieving BOO alone may allow low-pressure emptying (staged: TURP → reassess emptying by VCUG/ultrasound → surveillance if satisfactory). Concomitant diverticulectomy + prostatectomy is also common.
- **Observation**: minimal symptoms, no complications; counsel about cancer risk; surveillance with symptoms, cytology, cystoscopy.
- **CIC/indwelling catheter**: poor emptying after BOO relief in unfit/unwilling patients.

- **Endoscopic** (elderly, unfit, or at TURP): incise/resect neck down to muscle → wide ostium empties better; ± fulguration of entire lining to obliterate. Risk: retention from "venting" into sac; poor in large diverticula.
- **Diverticulectomy**: open, laparoscopic or robotic → *Ch 124*. †Principles: ipsilateral ureteric catheter, combined intravesical/extravesical dissection in the pseudocapsule plane, excise sac, close neck in layers, reimplant ureter if it enters the sac.
- Children with syndromic multiple diverticula: observe (connective tissue disorder impairs healing, high recurrence).

Surveillance (no fixed schedule): periodic symptom review, urine cytology and cystoscopy; re-enter the algorithm if haematuria, UTI, rising LUTS or HUN develop. Natural history is unknown and malignant change from stasis is possible, so unexcised diverticula need close follow-up.

5 · FEMALE URETHRAL DIVERTICULUM (UD) — ANATOMY

- **UD**: urine-filled periurethral sac within the pelvic fascia, communicating with the urethra by an ostium.
- Urethra 3–4 cm, suspended by the urethropelvic ligament (abdominal leaf = endopelvic fascia; vaginal leaf = **periurethral fascia**); UD lie between these leaves.
- **Periurethral glands**: posterolateral in submucosa, most prominent over the distal two-thirds; ducts drain into the **distal third**. Skene glands are the largest, most distal, opening lateral to the meatus.
- Striated sphincter: horseshoe-shaped, most prominent mid-urethra, deficient on vaginal side — UD overlaps the sphincter, so continence must be planned.
- Proximal urethra: urothelium, iliac nodes; distal: stratified squamous epithelium, inguinal nodes.

6 · UD — AETIOLOGY AND PATHOLOGY

- **Young–Wahle sequence**: gland infection in distal two-thirds → repeated abscess → expanding mass disrupts the muscular envelope → herniates into periurethral fascia → dissects between fascial leaves, usually **ventrally** (anterior vaginal wall mass), sometimes laterally/dorsally → ruptures into lumen (ostium).
- Vast majority acquired, in adult women; congenital rare.
- **Periurethral gland infection/obstruction theory (Routh — most accepted)**: gland infection (Escherichia coli, gonococcus) → duct obstruction → suburethral abscess → rupture into lumen → filling, stasis, reinfection, epithelialisation → permanent UD. Predicts the posterolateral mid-distal ostium (≥90%) and ventral vaginal mass.
- Against: ~one-third are proximal (few ducts); 20–30% nulliparous (against childbirth theory). Bulking agents mimic UD; mesh slings do not cause UD.
- Prevalence 1–6% in selected series (higher in symptomatic/tertiary populations); found in proportion to how hard they are sought.
- Lining: urothelial, squamous, columnar or none; ~10% of specimens show metaplasia, dysplasia or carcinoma.
- **Stones** in 4–10% (stasis) — removed with specimen, do not change approach.
- Configuration: simple ~73%, saddlebag/horseshoe ~17%, circumferential ~10%; mid-urethral ~50%, proximal and distal ~25% each; multiple ~6–19%.

Carcinoma in UD	Primary female urethral carcinoma
Adenocarcinoma commonest, then urothelial, then SCC	SCC commonest

- Imaging cannot reliably detect small cancers; biopsy whenever non-excisional therapy is chosen. High local recurrence with local excision alone; options — anterior exenteration/cystourethrectomy ± radiotherapy, or excision + radiotherapy/chemotherapy. Nephrogenic adenoma mimics adenocarcinoma.

7 · UD — PRESENTATION

- Third to seventh decades. **"Three Ds" — dysuria, dyspareunia, (postvoid) dribbling**: historical; all three in only ~**5%**.
- Commonest: irritative LUTS, pain, recurrent UTI. Also tender anterior vaginal wall mass (~⅓), incontinence (~⅓), haematuria, discharge, retention; 10–20% asymptomatic.
- Compressing the mass may express pus per meatus. **Size does not correlate with symptoms**. Misdiagnosis for years (interstitial cystitis, vulvodynia) — about half have a palpable mass, so examine.
- Pregnancy: conservative (antibiotics, aspiration) until after delivery; repair later.

8 · UD — EVALUATION

- Examination: mass 1–3 cm inside introitus over mid/proximal urethra; meatal distortion suggests Skene lesion; **hard mass → stone or cancer**. Assess atrophy (preoperative topical oestrogen), introitus, stress test, prolapse.
- Urinalysis, culture (sterile culture does not exclude infection), cytology if suspicious.
- **Cystourethroscopy** (flexible or short-beak female scope): ostium at **4 and 8 o'clock** mid-urethra; compress the sac to see pus; found in 15–89% — failure must not stop work-up.
- **Urodynamics/VUDS**: up to 50% show stress urinary incontinence (SUI); distinguishes true SUI (needs anti-incontinence procedure) from paradoxic dribbling from the sac (cured by excision). Pressure profile may show biphasic pattern.

Imaging	Voiding needed	Notes
MRI (T2 high signal, T1 low; surface or endoluminal coil)	No	**Best** (sensitivity ~97%); maps complexity and extent, no radiation; mimics (bulking agents), may miss small cancers, cost
Double-balloon positive pressure urethrography (PPU, Trattner catheter)	No	Excellent (~95%); catheter rarely available
Ultrasound (transvaginal/translabial)	No	Anechoic cyst; ~82%; less surgical detail
VCUG	Yes	~74%; non-diagnostic without voiding/postvoid film; underestimates size
Cystoscopy	—	Finds ostium in <half

Bulking agent vs UD on MRI: bulking agent — multiple, upper urethra, no fluid–fluid level or ostium, lower T2 signal than urine.

> **KEY POINT** UD symptoms are those of common lower urinary tract complaints and do not track lesion size, so diagnosis depends on suspicion and examination; imaging must both confirm and **map** the lesion — especially extension towards the bladder neck, which raises the risk of bladder/ureteric injury and postoperative incontinence.

9 · DIFFERENTIAL DIAGNOSIS OF PERIURETHRAL MASSES

Lesion	Clue	Treatment
UD	Mid/proximal; communicates; meatus normal	Excision, reconstruction
Vaginal leiomyoma	Firm, mobile, nontender, solid; oestrogen-dependent	Transvaginal enucleation
Skene gland cyst/abscess	Lateral to meatus, distorts it; non-communicating	Aspiration, marsupialisation, excision (curative)
Gartner duct cyst	Anterolateral wall; may receive ectopic upper-pole ureter — image upper tract	Observe, sclerotherapy, excision

Lesion	Clue	Treatment
Vaginal wall cyst	Müllerian commonest; histology only	Excise if symptomatic
Urethral mucosal prolapse	Circumferential red "doughnut"; prepubertal girls, postmenopausal	Topical oestrogen; circumferential excision
Caruncle	Exophytic, red, †usually posterior meatal lip; postmenopausal	Oestrogen; excise if atypical/refractory
Bulking agent	History of injection; non-communicating	Excise if symptomatic

Skene gland adenocarcinoma may raise PSA (prostate homologue).

> **KEY POINT** Two questions sort almost every periurethral mass: does it communicate with the urethra, and where does it sit relative to the meatus? Only UD communicate; lesions at or distorting the meatus are Skene, prolapse or caruncle; firm mobile solid masses are leiomyomas.

10 · CLASSIFICATION OF UD

System	Basis	Use
L/N/S/C3 (Leach)	**L**ocation, **N**umber, **S**ize, **C**onfiguration, **C**ommunication site, **C**ontinence	Standard description; not validated
Ginsburg–Genadry	Location	Distal → marsupialisation; proximal → excision
Leng–McGuire	Periurethral fascia preserved or not	Absent fascia = **pseudodiverticulum** → needs flap/graft

11 · UD AND STRESS URINARY INCONTINENCE

- Significant de novo SUI after diverticulectomy alone: **7–16%**. Risk: size ≥30 mm, proximal, circumferential, short urethra.
- ~50% of preoperative SUI improves after excision alone → concomitant surgery only for **bothersome** SUI; mild → excise, reassess (most de novo SUI settles within 12 months).
- **Never synthetic mid-urethral mesh with UD surgery** (erosion, infection) — use **autologous pubovaginal fascial sling**.

12 · SURGICAL REPAIR OF UD

- Indications: symptoms (dysuria, dribbling, recurrent UTI, dyspareunia, pain), stones, suspicion of cancer. Asymptomatic/unfit: suppressive antibiotics + postvoid digital stripping, counselling on cancer risk.
- **Marsupialisation (Spence-Duckett)**: only distal-third UD, not sexually active — risks SUI, vaginal voiding, dyspareunia.
- Other historical options (endoscopic unroofing, fulguration, obliteration with oxidised cellulose) are largely abandoned; excision with reconstruction is standard. Laparoscopic/robotic approaches are reported for anterior or proximal UD; a suprameatal approach for purely anterior UD.
- Preparation: sterile urine, topical oestrogen, VUDS, counselling (symptoms may persist).

No.	Principles of transvaginal diverticulectomy (Box 122.2)
1	Well-vascularised anterior vaginal wall flap
2	Preserve periurethral fascia as a separate layer
3	Identify and excise the ostium
4	Remove entire sac (epithelium)
5	Watertight urethral closure
6	Multilayer, non-overlapping absorbable closure

No.	Principles of transvaginal diverticulectomy (Box 122.2)
7	Close dead space
8	Preserve or create continence

ALGORITHM
Lithotomy, ring retractor, urethral Foley (suprapubic tube placed at the **start**, never at the end — it would cross the fresh urethral repair) → **inverted-U incision** (base distal, limbs to bladder neck, widening proximally) → raise vaginal flap off periurethral fascia → incise fascia **transversely**, raise proximal and distal fascial flaps → dissect sac to ostium, excise entire wall including inflamed urethral edge → close urethra watertight over catheter (as small as 12 Fr) → close fascia **perpendicular** to urethral line, obliterating dead space → ± **Martius flap** (poor tissue, attenuated fascia, inflammation) → advance vaginal flap (3 layers; 4 with Martius) → antibiotic vaginal pack

- Pseudodiverticulum (no fascia): Martius flap or fascial graft. Circumferential UD may need segmental urethral resection.
- **Post-op**: antibiotics 24 h, pack out, antispasmodics; **pericatheter VCUG at 14–21 days** → no leak: remove catheter; leak: repeat weekly (most settle).

13 · COMPLICATIONS

Complication	Incidence	Notes / management
Recurrent UD	**10–22%** (pooled ~10%)	Incomplete excision, poor closure, dead space; risk — circumferential, complex, proximal, multiple; redo harder
Incontinence	1.7–16%	Conservative 12 months, then autologous sling
Urethrovaginal fistula	0.9–8.3%	Distal to sphincter: often asymptomatic, may not need repair; proximal: repair ± Martius
Urethral stricture	0–5%	—
Recurrent UTI	up to 31%	Investigate for recurrence
Others	—	Distal urethral necrosis, bladder/ureteric injury, vaginal scarring (dyspareunia)

Fistula prevention: meticulous haemostasis, avoid infection, preserve periurethral fascia, well-vascularised vaginal flap, multilayer non-overlapping closure. Excision of a **recurrent** UD is less likely to relieve symptoms and carries higher fistula risk (scar, obscured planes).

Persistent symptoms after technically successful excision are common; if they persist, image for recurrent or new UD or stricture.

> **KEY POINT** Every principle — complete sac removal, watertight closure, fascia closed off-axis, dead space obliterated — exists to prevent the three failures: recurrence, fistula, incontinence.

Feature	Bladder diverticulum	Female UD
Origin	Mucosal herniation through detrusor (weak wall or high pressure)	Infected, obstructed periurethral gland rupturing into lumen
Cause to treat	BOO / neurogenic dysfunction	Infection; coexisting SUI
Commonest cancer	Urothelial (then SCC)	Adenocarcinoma
Key study	Cystoscopy + cytology; VCUG for emptying	MRI
Operate for	Complications, not size	Symptoms or complications
Operation	Diverticulectomy + outlet treatment	Transvaginal excision, layered reconstruction

14 · MCQ ONE-LINERS

1. Bladder diverticulum lacks muscularis propria — cannot empty.
2. Male : female ratio of bladder diverticula ≈ 9 : 1.
3. Acquired diverticula: BOO in ~70% (BPH).
4. Congenital diverticulum site: posterolateral to ureteric orifice (paraureteric).
5. Hutch diverticulum: paraureteric in neuropathic bladder with VUR.
6. Syndromes: Menkes, Williams, Ehlers-Danlos.
7. Paradoxic enlargement of diverticulum seen on voiding VCUG.
8. Pressure-sink effect falsely lowers apparent detrusor contractility.
9. Commonest tumour in bladder diverticulum: urothelial carcinoma; second SCC.
0. Size alone is not an indication for diverticulectomy.
1. Treat outlet obstruction before or with diverticulectomy.
2. Endoscopic neck resection risk: retention by venting.
3. Commonest theory of UD: periurethral gland infection (Routh).
4. Periurethral glands drain into distal third of urethra.
5. UD ostium: posterolateral, 4 and 8 o'clock, mid-urethra.
16. All "three Ds" present in only ~5%.
17. Best imaging for UD: MRI (T2 hyperintense).
18. VCUG without voiding images is non-diagnostic.
19. Commonest carcinoma in UD: adenocarcinoma.
20. Stones in UD: 4–10%.
21. L/N/S/C3: location, number, size, configuration, communication, continence.
22. Absent periurethral fascia = pseudodiverticulum.
23. Diverticulectomy incision: inverted U.
24. Fascia closed perpendicular to urethral suture line.
25. Suprapubic tube placed at start of diverticulectomy, not the end.
26. No synthetic sling with UD repair — autologous fascial sling.
27. De novo significant SUI after diverticulectomy: 7–16%.
28. Recurrence after diverticulectomy: 10–22%; risk — circumferential.
29. Pericatheter VCUG at 14–21 days before catheter removal.
30. Skene gland cyst distorts meatus and does not communicate.
31. Gartner duct cyst may receive ectopic ureter — image upper tract.

77 Sexual Function and Dysfunction in the Female

Reading time ≈ 49 min (slow pace)

1 · FUNCTIONAL ANATOMY

The **distal vagina, clitoris and urethra** share blood supply and innervation and respond to arousal as one unit, the **clitoral complex**. This is why urethral and anterior vaginal wall surgery cannot be separated from sexual function.

- **Clitoris**: homologous to the penis but **separate from the urethra**. Two corpora cavernosa (incomplete pectiniform septum) continue as crura attached to the ischiopubic rami. The glans is very sensitive but **not erectile**. The dorsal nerves lie **superficially on the dorsum**, outside the tunica albuginea, divided by the deep suspensory ligament. This puts them at risk in hood or clitoral surgery.
- **Erectile tissue** (clitoral body, crura, "vestibular" bulbs) engorges and expands. **Non-erectile vascular tissue** (labia minora, glans, periurethral tissue, vaginal walls) has more blood flow during arousal but does not engorge. The urethra lies in the vaginal wall on one side and is surrounded by erectile tissue on all others.
- **Skene glands** are the prostate homologue (PSA-positive). **Bartholin glands** are the bulbourethral homologue. MCQ The G-spot has no agreed anatomical basis.
- Labium majus sensation: anterior third — ilioinguinal (L1); posterior two-thirds — perineal (S3); lateral — posterior cutaneous nerve of thigh (S2). The glans drains **directly to deep inguinal nodes**. The deep dorsal vein does not run with the arteries.

2 · PHYSIOLOGY OF THE SEXUAL RESPONSE

ALGORITHM
Touch on the clitoral complex/vulva → pudendal (somatic) afferents → **S2–S4** → parasympathetic efferents (nervi erigentes → cavernous nerves) → arteriolar dilatation → engorgement + lubrication

The female sexual reflex is unusual: **somatic afferents, visceral efferents**, and it can be voluntarily augmented or suppressed. **Orgasm** is a cerebral event with **sympathetic** output (T12–L1 → hypogastric nerves) causing uterine contraction, plus pudendal reflex contraction of the pelvic striated muscles. Resolution = sympathetic vasoconstriction and decongestion.

> **KEY POINT** Pudendal injury removes the **trigger**. Pelvic plexus or cavernous nerve injury removes the **response**.

ALGORITHM
Neuronal nitric oxide synthase → nitric oxide (NO) → guanylate cyclase → cyclic guanosine monophosphate (cGMP) → ↓ Ca^{2+} → smooth muscle relaxation → engorgement. Phosphodiesterase type 5 (PDE5) breaks down cGMP; arginase uses up the NOS substrate. Vasoactive intestinal polypeptide (VIP) works with NO.

Desire = a balance between **excitation** (dopamine, noradrenaline, melanocortins, oxytocin, testosterone, oestrogen) and **inhibition** (serotonin, opioids, prolactin, progesterone). The licensed drugs for low desire act on this balance.

	Oestrogen	**Testosterone**
Role	Maintains the **substrate**: epithelium, collagen, lubrication, acidic pH; keeps NOS and VIP working	Drives **desire** and responsiveness. Testosterone bound to sex hormone–binding globulin (SHBG) is inactive
Deficiency	Thin, dry vagina with fibrosis → dyspareunia. Risk rises when oestradiol is **≤50 pg/mL**	↓ Libido, arousal, sensation, orgasm

	Oestrogen	Testosterone
Risk of treatment	—	Virilisation: early and reversible (acne, hirsutism) or late and irreversible (voice change, alopecia, clitoromegaly)

Model	Stages	Key idea
Masters & Johnson (1966), linear	Excitement → plateau → orgasm → resolution	Physiological; **no desire phase**
Kaplan (1977), linear	Desire → arousal → orgasm	Adds desire as the first step
Whipple & Brash-McGreer, circular	Seduction → sensations → surrender → reflection	Reflection feeds the next seduction
Basson (2002), circular	Neutrality → receptivity to stimuli → arousal → **responsive desire** → satisfaction → back to the start	Spontaneous desire is **not needed**; intimacy is the motive

> **KEY POINT** Many women do not follow a linear sequence. Under Basson's model, lack of *spontaneous* desire (common after menopause) is not a dysfunction. The same idea led DSM-5 to merge desire and arousal into one diagnosis.

3 · CLASSIFICATION

The Diagnostic and Statistical Manual of Mental Disorders (DSM-5) requires **≥6 months** of symptoms and **clinically significant distress**. A woman with low desire who is not distressed has no disorder.

Domain	DSM-IV-TR	DSM-5
Desire + arousal	Hypoactive sexual desire disorder (HSDD); female sexual arousal disorder (FSAD); sexual aversion disorder	**Female sexual interest/arousal disorder (FSIAD)**: **≥3 of 6** — ↓ interest; ↓ fantasies; ↓ initiation or receptivity; ↓ pleasure, ↓ response to cues, ↓ sensation (in 75–100% of encounters). Aversion dropped
Orgasm	Female orgasmic disorder (FOD) after normal excitement	FOD: delay, infrequency, absence **or ↓ intensity** in **75–100%** of encounters. Normal-excitement requirement removed
Pain	Dyspareunia; vaginismus	**Genito-pelvic pain/penetration disorder (GPPPD)**, ≥1 of: penetration difficulty; pain; fear of pain; pelvic floor muscle (PFM) tensing

- The **International Consultation on Sexual Medicine (ICSM, 2016)** keeps HSDD and FSAD **separate**. It adds **persistent genital arousal disorder (PGAD)** and orgasmic illness syndrome (malaise after sex). The **International Society for the Study of Women's Sexual Health (ISSWSH)** splits FSAD into cognitive arousal disorder and **genital arousal disorder**; the genital type falls within urology. It is **not** diagnosed if the stimulation does not match the woman's preferences.
- Female sexual dysfunction (FSD): **~40%** of women have a lifetime sexual problem, but only **~8–12%** have a distressing disorder. **HSDD is the most common**, FOD second.

4 · CAUSES (BIOPSYCHOSOCIAL)

- **Biological**: diabetes mellitus (DM), hypertension, coronary disease, thyroid disease, hyperprolactinaemia, renal failure, spinal cord injury (SCI), multiple sclerosis (MS), vulval dermatoses, incontinence, pregnancy, breastfeeding, menopause.
- **Psychological**: depression (bidirectional — screen every HSDD patient), anxiety, body image, childhood abuse. **Sociocultural**: restrictive upbringing. **Interpersonal**: conflict, **partner's erectile dysfunction (ED) or premature ejaculation**.
- **Drugs**: selective serotonin reuptake inhibitors (SSRIs), serotonin–noradrenaline reuptake inhibitors (SNRIs), antipsychotics (↑ prolactin); β-blockers, angiotensin-converting enzyme (ACE)

inhibitors, calcium channel blockers, thiazides, spironolactone (antiandrogen); antiandrogens, gonadotrophin-releasing hormone (GnRH) agonists; opiates, alcohol. MCQ Combined hormonal contraceptives work through two mechanisms (↓ ovarian testosterone and ↑ SHBG); oral oestrogens, tamoxifen and thyroxine also ↑ SHBG.

- **Genitourinary syndrome of menopause (GSM)**: dryness, burning, dyspareunia, urgency, dysuria, recurrent urinary tract infection (UTI). Dyspareunia is often **vestibular** because atrophy is worse there. Dryness → pain → secondary low desire, so **treat the tissue first**.
- **Cancer treatment**: surgical or chemotherapy-induced castration, anti-oestrogens (†aromatase inhibitors cause severe GSM), pelvic radiotherapy (vaginal stenosis), and nerve injury during surgery.

5 · EVALUATION

Women rarely raise sexual problems, so the **clinician must ask**. Use gender-neutral questions and do not assume sex means penile–vaginal intercourse.

ALGORITHM
PLISSIT: **P**ermission (normalise) → **L**imited **I**nformation (response cycle, ageing, drugs) → **S**pecific **S**uggestions (lubricants, **vaginal oestrogen**, positions, vibrators) → **I**ntensive **T**herapy = **referral**

- **Brief Sexual Symptom Checklist for Women (BSSC-W)**: triage tool recommended by the American College of Obstetricians and Gynecologists (ACOG), but **not validated**.
- **Decreased Sexual Desire Screener (DSDS)**: **validated**, 5 items. Yes to items 1–4 (desire was good, has fallen, is bothersome, she wants it back) + no contributing cause = **generalised acquired HSDD**.
- **Female Sexual Function Index (FSFI)**: 19 items, 6 domains (desire, arousal, lubrication, orgasm, satisfaction, pain); the most widely used. Female Sexual Distress Scale (FSDS-R) measures **distress**. **Pelvic Organ Prolapse/Urinary Incontinence Sexual Questionnaire (PISQ-12/IR)** is the **only validated** tool for prolapse and incontinence. For HSDD: Profile of Female Sexual Function (PFSF) after menopause, Sexual Interest and Desire Inventory (SIDI) before.

History: lifelong or acquired; generalised or situational; and the **order in which problems appeared**, which separates primary from secondary (e.g. low desire → poor arousal → no lubrication → pain). Also drugs, surgery, lower urinary tract symptoms (LUTS), prolapse and the **partner's** function.

Examination (essential when there is pain):

- Atrophy: pale, flat labia, loss of rugae, narrow introitus. Vaginal **pH ≥4.5** without infection or semen = atrophy.
- **Cotton-swab (Q-tip) test** of the vestibule. **Retract the clitoral hood** to look for adhesions.
- Dermatoses: lichen sclerosus (white plaques, loss of architecture → introital stenosis); lichen planus (erosions, **Wickham striae** are pathognomonic → vaginal stenosis).
- Palpate the PFM for tone, relaxation and tenderness. **Overactive** muscle fails to relax and causes dyspareunia; **underactive** muscle causes incontinence and prolapse. Fornix tenderness suggests endometriosis; ischial spine tenderness suggests pudendal neuropathy. Do a cough stress test. Grade PFM tenderness by Lukban–Whitmore: 0 none → 4 severe pain that prevents contraction.

Laboratory tests are rarely needed. Check HbA1c and lipids if there are risk factors; prolactin and thyroid tests only if signs point to them. **Do not measure testosterone at random**: there is no female normal range, assays are unreliable at female levels, and no level predicts HSDD. Physiological tests (photoplethysmography, Doppler) are for research only.

6 · HYPOACTIVE SEXUAL DESIRE DISORDER

- ISSWSH definition: ≥6 months of loss of motivation (spontaneous or responsive) or loss of desire to take part in sex, **with distress**, not due to a drug, illness, pain, abuse or partner problems.

Exclude relationship conflict first. On the DSDS, a coexisting arousal or orgasm disorder does not exclude HSDD; the clinician decides which disorder is primary.

- Desire falls with age, but distress falls too, so HSDD peaks in **midlife**. HSDD is defined by distress, not by a hormone level.

ALGORITHM
Screen (DSDS) → history + focused examination → **correct modifiable factors** (drugs, depression, GSM, pain, partner's ED) → education / sex or couples therapy → **generalised acquired** HSDD that persists → drug therapy → review at 8 weeks. **Refer**: situational HSDD not responding to counselling, psychiatric illness, sexual trauma

	Flibanserin	**Bremelanotide**
Mechanism	Central **5-HT1A agonist / 5-HT2A antagonist**: ↓ serotonin, ↑ dopamine and noradrenaline (removes the brake)	Central **melanocortin agonist** → dopamine release (presses the accelerator)
Licence	US Food and Drug Administration (FDA)-approved only for **premenopausal** acquired generalised HSDD	
Dose	**100 mg at bedtime**, daily; about half respond by 8 weeks	**1.75 mg subcutaneous** ≥45 min before sex; ≤1 per 24 h, **≤8/month**
Side effects	Dizziness, somnolence, nausea (hence bedtime dosing)	Nausea, flushing, headache, **hyperpigmentation**
Cautions	**Alcohol** (hypotension, syncope); restricted prescribing (REMS); †avoid strong CYP3A4 inhibitors	Uncontrolled hypertension or cardiovascular disease

- **Testosterone** (off-label): transdermal testosterone improves desire, arousal, orgasm and satisfaction in **postmenopausal HSDD**. Oestrogen + androgen works better than oestrogen alone. No cardiovascular signal on meta-analysis, but long-term safety data are lacking.
- †2019 Global Consensus: use transdermal testosterone to reach **premenopausal physiological levels**. Check the level at baseline and at 3–6 weeks to exclude supraphysiological levels. **Stop at 6 months** if there is no benefit. Monitor for virilisation.
- Bupropion (off-label) is an alternative. Hormone levels do not define HSDD: low oestradiol is associated with low desire, but many women with HSDD have normal levels.

7 · AROUSAL DISORDER AND PGAD

- **Subjective and genital arousal correlate poorly.** Peripheral vasodilators raise measured blood flow without the woman feeling aroused. Start by asking what arousal means to her. If arousal fails only with her current partner, the problem is in the relationship.
- Psychological treatment: mindfulness within sex-focused cognitive behavioural therapy (CBT). Depression and anxiety predict a poor response.
- Drugs: **transdermal testosterone** has the best evidence. PDE5 inhibitors only when the neurovascular supply is impaired (type 1 DM, SCI, MS, SSRI-induced dysfunction). **Bupropion 150 mg twice daily** is the drug of choice for SSRI-induced dysfunction. Topical alprostadil does not work. L-arginine + yohimbine raised vaginal blood flow on photoplethysmography without raising subjective arousal, which shows the discordance. Sublingual apomorphine (a dopamine agonist) has only small studies.

PGAD (Leiblum–Nathan criteria): genital arousal lasting hours to days, **not relieved by orgasm**, without desire, intrusive and distressing — the reverse of HSDD. Associations: Tarlov cysts, pelvic varices, **stopping an SSRI**. Treatment: CBT, duloxetine, pregabalin; surgery for a lumbosacral annular tear.

8 · FEMALE ORGASMIC DISORDER

- A **diagnosis of exclusion that requires distress**. First exclude drugs, disease, stress and **inadequate stimulation**. If the partner's ED or premature ejaculation deprives her of stimulation, FOD cannot be diagnosed.
- **Absence of orgasm during coitus without clitoral stimulation is normal variation**, not FOD.
- Lifelong FOD usually reflects inexperience or poor sex education. Acquired FOD reflects inadequate stimulation, context or physiological change. Risk factors: anxiety, depression, sex guilt, relationship conflict, abuse, DM, SCI, MS, cancer, and drugs (SSRIs, antipsychotics, antiandrogens).
- Treatment: **education is the cornerstone**, plus directed masturbation, sensate focus and mindfulness. If arousal is impaired, treat the arousal problem. If arousal is intact, increase stimulation (vibrators; MCQ the EROS clitoral vacuum is the only FDA-cleared FSD device). Oestrogen or testosterone helps postmenopausal women with deficiency. Clitoral phimosis or adhesions can be corrected surgically.

ALGORITHM

SSRI-induced dysfunction: was it present before treatment (depression itself causes dysfunction)? → wait for tolerance (up to ~4 months) → still present after 3–6 months: lower the dose or switch to or add **bupropion** or mirtazapine; consider sildenafil

9 · GENITO-PELVIC PAIN / PENETRATION DISORDER

- **Dyspareunia**: superficial (introital) or deep (endometriosis, prolapse, pelvic floor). **Vaginismus**: involuntary muscle contraction that prevents penetration.
- **Vulvodynia**: vulval pain, usually burning, for **≥3 months** with no identifiable cause; localised or generalised, provoked or unprovoked.
- **Provoked vestibulodynia (PVD)** is the commonest type (~**10–12%** of women; associated with irritable bowel syndrome and fibromyalgia). Primary PVD starts with the first attempt at penetration (tampon or intercourse). It is associated with nulliparity, family history, enuresis and dysmenorrhoea. Secondary PVD (~two-thirds) follows a period of pain-free sex; the vestibular area involved is larger and the clitoral hood is often the most painful site.
- Mechanism: vestibular nerve proliferation (a peripheral pain generator), amplified by a **hypertonic pelvic floor** and fear–avoidance.
- **Clitorodynia**: hood adhesions trap keratin → pseudocysts and keratin pearls → inflammation and pain. Under-diagnosed — **examine the glans routinely**.

ALGORITHM

Education + vulval care (avoid irritants) → topical lidocaine 5% or lidocaine–prilocaine (EMLA) **10 min before intercourse** → **multimodal pelvic floor physical therapy** (myofascial release, trigger points, stretching, down-training, biofeedback, graded **dilators**, electrical stimulation) + psychological/couples therapy → vestibular oestrogen if hypo-oestrogenic → **vestibulectomy** for refractory localised PVD

- **Multimodal physical therapy is the standard of care** for PVD. Electromyography (EMG) biofeedback helps most when the pelvic floor is overactive.
- Oestrogen deficiency **increases** the density of vulval nerve fibres and oestrogen reduces it. So **topical works better than systemic, and vestibular better than vaginal**.
- †Second-line options: amitriptyline, gabapentin, intralevator botulinum toxin for a hypertonic pelvic floor.
- **Vestibulectomy** removes the vestibule to within 1–2 mm of the meatus. It works better than non-surgical treatment when disease is localised and neuroproliferative.

Treatment of GSM

ALGORITHM
Non-hormonal lubricants and moisturisers + regular sexual activity (first line according to the North American Menopause Society, NAMS) → **low-dose vaginal oestrogen** (†minimal absorption, no progestogen needed) or †vaginal dehydroepiandrosterone (DHEA, prasterone 6.5 mg nightly, converted locally to sex steroids) or †ospemifene 60 mg orally (a selective oestrogen receptor modulator that acts as an oestrogen agonist on the vagina; for women who cannot use vaginal products; risk of venous thromboembolism) → systemic hormone therapy if vasomotor symptoms are also present

KEY POINT †After **breast cancer**, use non-hormonal measures first. Vaginal oestrogen only after discussion with the oncologist (concern with aromatase inhibitors).

10 · PELVIC SURGERY AND SEXUAL FUNCTION

Surgery damages sexual function through three routes: **nerves** (lateral vaginal wall, pelvic plexus), **clitoral blood supply** (via the distal urethra) and **vaginal shape** (length and calibre). Counsel **before and after** surgery; patients want this far more often than they ask for it.

Radical cystectomy

- Problems by frequency: loss of desire > orgasm disorder > dyspareunia > reduced lubrication; only about half can have intercourse afterwards.
- **Preserving technique**: (1) dissect the bladder off the vagina from below upwards (retrograde), saving the neurovascular bundles on the **lateral vaginal walls**; (2) divide the urethra at the **bladder neck** to keep the clitoral blood supply; (3) reconstruct the vagina so its dimensions are kept.
- **Genital-sparing cystectomy** (when oncologically safe) allows sex to resume earlier (**~6 vs 14 weeks**). Type of urinary diversion does not matter. After cystectomy for benign disease, most women were not counselled beforehand and almost none were asked about sex at follow-up; a third of cancer patients wanted sex-therapy or partner referral. **Never trade oncological control for function.** → *Ch 101*

Hysterectomy and rectal surgery

After benign hysterectomy most women are unchanged or better, and abdominal vs vaginal approach makes no difference. After radical hysterectomy, lubrication problems and dyspareunia predominate at 6 months; most previously active women are active again by 12 months, though less often. Rectal cancer surgery injures the pelvic plexus: dysfunction occurs in a substantial proportion after low anterior resection and in **nearly all** after abdominoperineal resection. Independent risks: neoadjuvant chemotherapy, tumour below the peritoneal reflection, intersphincteric resection, Hartmann's procedure. Include these in preoperative counselling.

Prolapse (POP) repair

- Prolapse stage does not predict sexual function; bother and body image do.
- **Native-tissue repair**: most women improve or stay the same, but de novo dyspareunia can follow any technique.
- Dyspareunia is more common after vaginal than abdominal repair, and after anterior mesh than **sacrocolpopexy** (gold standard for apical prolapse). Biological grafts have higher recurrence and have fallen out of use. Mesh complications (erosion, pain, dyspareunia) can appear years later.
- **Combined anterior + posterior repair consistently increases dyspareunia** — consider staging the repair in sexually active women. **Keep vaginal length, avoid excessive skin excision, preserve introital calibre.** → *Ch 119*

Stress urinary incontinence (SUI) surgery and lower urinary tract dysfunction (LUTD)

- After SUI surgery, about a third improve and most are unchanged. **Curing coital incontinence** predicts improvement.
- Retropubic **tension-free vaginal tape (TVT)** can denervate the vaginal wall (↓ lubrication). **Transobturator tape (TOT)** spares the dorsal nerve of the clitoris. PISQ-12 scores slightly favour TOT, but FSFI shows no difference. Dyspareunia is a strong reason to remove a sling.
- **Coital incontinence** (~1 in 10 women with incontinence): leakage at penetration = stress mechanism; leakage at orgasm = detrusor overactivity (detrusor contraction with urethral relaxation on urodynamics).
- Severity of FSD by type of LUTD: bladder pain syndrome > detrusor overactivity > urgency or mixed incontinence > SUI > dry overactive bladder (OAB) > voiding LUTS. Pairings: HSDD with SUI; orgasm problems with OAB.
- **Bladder pain syndrome** has the worst sexual impact of any LUTD (pain → fear → PFM overactivity → vulvodynia). Treating incontinence improves sexual function: tolterodine improved all FSFI domains in OAB. **Sacral neuromodulation** improves sexual function whatever the cause, which suggests a shared neural pathway.

Genital cosmetic surgery: the evidence is poor. Screen for **body dysmorphic disorder** and disclose the risks (altered sensation, dyspareunia).

11 · SPECIAL POPULATIONS

Ask about orientation and gender identity in every history; a **trans male** was assigned female at birth. Discuss fertility preservation before transition. Main risk of oestrogen: thromboembolism; of testosterone: erythrocytosis (haematocrit **>50%**) — check haematocrit and testosterone every 3 months in year 1, then 1–2 times a year. On spironolactone, check potassium. Women with SCI have impaired sensation, lubrication and orgasm, with genital pain and spasticity, so ask about sexual function. → *Ch 126*

12 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Flibanserin phase III	RCTs vs placebo, premenopausal	↑ Satisfying events, ↑ desire, ↓ distress	First FDA-approved drug for HSDD
RECONNECT (bremelanotide)	2 phase III RCTs	↑ Desire, ↓ distress	On-demand option for premenopausal HSDD
Bupropion for SSRI-induced dysfunction (Cochrane review)	Meta-analysis	Benefit at 150 mg twice daily	Add or switch to bupropion

13 · MCQ ONE-LINERS

1. Female sexual reflex: pudendal afferent, parasympathetic (S2–S4) efferent.
2. Orgasm and resolution are sympathetic (T12–L1).
3. The glans clitoris is non-erectile; dorsal nerves lie superficial to the tunica.
4. Skene glands are the prostate homologue (PSA-positive).
5. Basson model: responsive desire; spontaneous desire not required.
6. Masters–Johnson has no desire phase; Kaplan added it.
7. DSM-5 FSIAD needs ≥3 of 6 criteria, ≥6 months, with distress.
8. DSM-5 GPPPD merges dyspareunia and vaginismus.
9. DSM-5 FOD adds reduced intensity.
10. Commonest FSD is HSDD; second is FOD.
11. PLISSIT: intensive therapy means referral.
12. BSSC-W is not validated; DSDS is validated.
13. FSFI: 19 items, 6 domains.
14. PISQ is the only validated prolapse/incontinence sexual questionnaire.
15. Random testosterone measurement is useless in women.
16. Vaginal pH ≥4.5 without infection suggests atrophy.
17. Wickham striae are pathognomonic of lichen planus.

8. Flibanserin: 5-HT1A agonist/5-HT2A antagonist, 100 mg at bedtime; alcohol warning.
9. Bremelanotide: melanocortin agonist, 1.75 mg subcutaneously, ≤8/month; hyperpigmentation.
10. Both HSDD drugs are approved only before menopause.
11. Thyroxine and oral contraceptives raise SHBG.
12. Bupropion is the choice for SSRI-induced dysfunction.
13. Coital anorgasmia without clitoral stimulation is normal.
14. EROS clitoral vacuum is the only FDA-cleared FSD device.
15. PGAD: arousal not relieved by orgasm; associated with Tarlov cysts.
16. Vulvodynia requires ≥3 months of pain without a cause.
17. Multimodal pelvic floor physical therapy is standard for PVD.
18. Vestibulectomy extends to within 1–2 mm of the meatus.
19. Female cystectomy: preserve lateral vaginal walls and the distal urethra.
20. Combined anterior + posterior colporrhaphy increases dyspareunia.
21. TOT spares the dorsal clitoral nerve; TVT may denervate the vaginal wall.
22. Bladder pain syndrome has the worst sexual impact of any LUTD.
23. Genital-sparing cystectomy: sexual activity resumes at ~6 vs 14 weeks.
24. Coital leakage at orgasm suggests detrusor overactivity; at penetration, SUI.
25. Topical vestibular oestrogen works better than systemic because it reduces vulval nerve density.

★ Putting it together: female pelvic medicine decisions

Reading time ≈ 7 min (slow pace) · Integrates Ch 77–122 · use these as the skeleton of long-essay answers

A · WOMAN WITH URINARY INCONTINENCE: THE PATHWAY

ALGORITHM

1. Classify by history and bladder diary: stress (SUI), urgency (UUI), mixed (MUI), overflow, functional. Examine: cough stress test, urethral mobility, pelvic organ prolapse (POP) staging, post-void residual (PVR), urinalysis (→ *Ch 105*).

2. Urodynamics are not needed before surgery for uncomplicated, demonstrable SUI (VALUE trial). Use them for mixed or unclear symptoms, voiding dysfunction, previous surgery, neurological disease or high-grade prolapse.

3. Conservative first: weight loss, fluid and caffeine adjustment, supervised pelvic floor muscle training (PFMT), bladder training; pessary or vaginal device; vaginal oestrogen in postmenopausal women (→ *Ch 114*).

4. SUI surgery if conservative care fails (→ *Ch 116, 118*); UUI → drugs, botulinum toxin, neuromodulation (voiding book).

B · CHOOSING AN SUI OPERATION

Operation	Best for	Main drawbacks
Retropubic midurethral sling (MUS)	Standard first choice; better for intrinsic sphincter deficiency	Bladder perforation (cystoscopy mandatory), voiding dysfunction, rare vascular/bowel injury
Transobturator MUS	Avoids retropubic space (previous pelvic surgery, obesity)	Groin/thigh pain; slightly less effective for severe SUI
Autologous fascial pubovaginal sling	Recurrent SUI, after mesh complications, concomitant urethral repair (diverticulum, fistula), patient declines mesh	Larger operation, more voiding dysfunction
Burch colposuspension	Patient declines mesh; concurrent abdominal surgery	Posterior compartment prolapse, de novo urgency
Bulking agent	Frail patient, minimal surgery wanted	Lower, short-lived cure

C · PELVIC ORGAN PROLAPSE: WHICH REPAIR?

ALGORITHM

Stage with POP-Q; leading edge sets the stage; stage II = within 1 cm of the hymen (→ *Ch 105*).

Asymptomatic → observe. **Symptomatic** → pessary or surgery.

Apex is the key: anterior and posterior defects often follow apical loss, so support the apex — vaginal uterosacral or sacrospinous suspension, or abdominal/robotic sacrocolpopexy (most durable) (→ *Ch 117*).

Not sexually active and frail → colpocleisis (high success, low morbidity).

Occult SUI: discuss a concomitant sling (OPUS: fewer post-op leaks, more complications).

D · REPAIRING A VESICOVAGINAL FISTULA

KEY POINT Confirm the type (double-dye test), image the upper tracts, and biopsy any fistula after radiation or cancer. Most post-hysterectomy fistulae are supratrigonal and repaired vaginally (Latzko or layered); choose the abdominal route when the ureter is involved, the fistula is large or high, or vaginal repair has failed. Principles: excise or freshen the tract, tension-free watertight closure in layers, non-overlapping suture lines, a vascularised flap (Martius vaginally, omentum abdominally) when tissue is poor or irradiated, and continuous bladder drainage before a cystogram (→ *Ch 121*).

E · MESH COMPLICATIONS IN FOUR RULES

ALGORITHM

1. Examine (speculum) and do cystoscopy for any suspicion; complications can appear years later.
2. Small vaginal exposure → vaginal oestrogen ± trimming; larger or symptomatic → excision with closure.
3. Mesh in the bladder or urethra → remove it (endoscopic or open/transvaginal) and repair; no new mesh at the same time.
4. Remove enough mesh to solve the problem; never remove asymptomatic mesh (→ *Ch 119*).